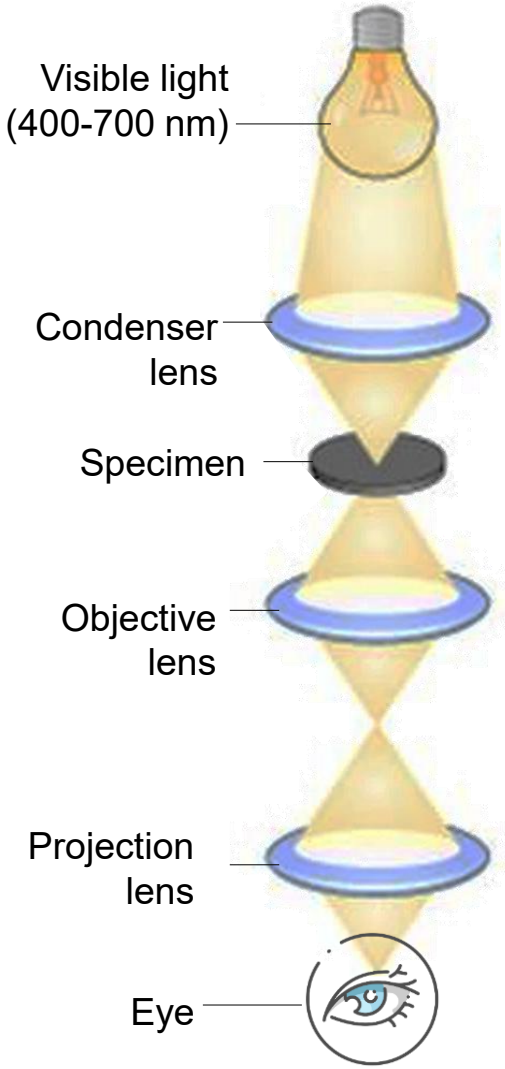
MATE 6301

Transition Electron Microscopy (TEM)

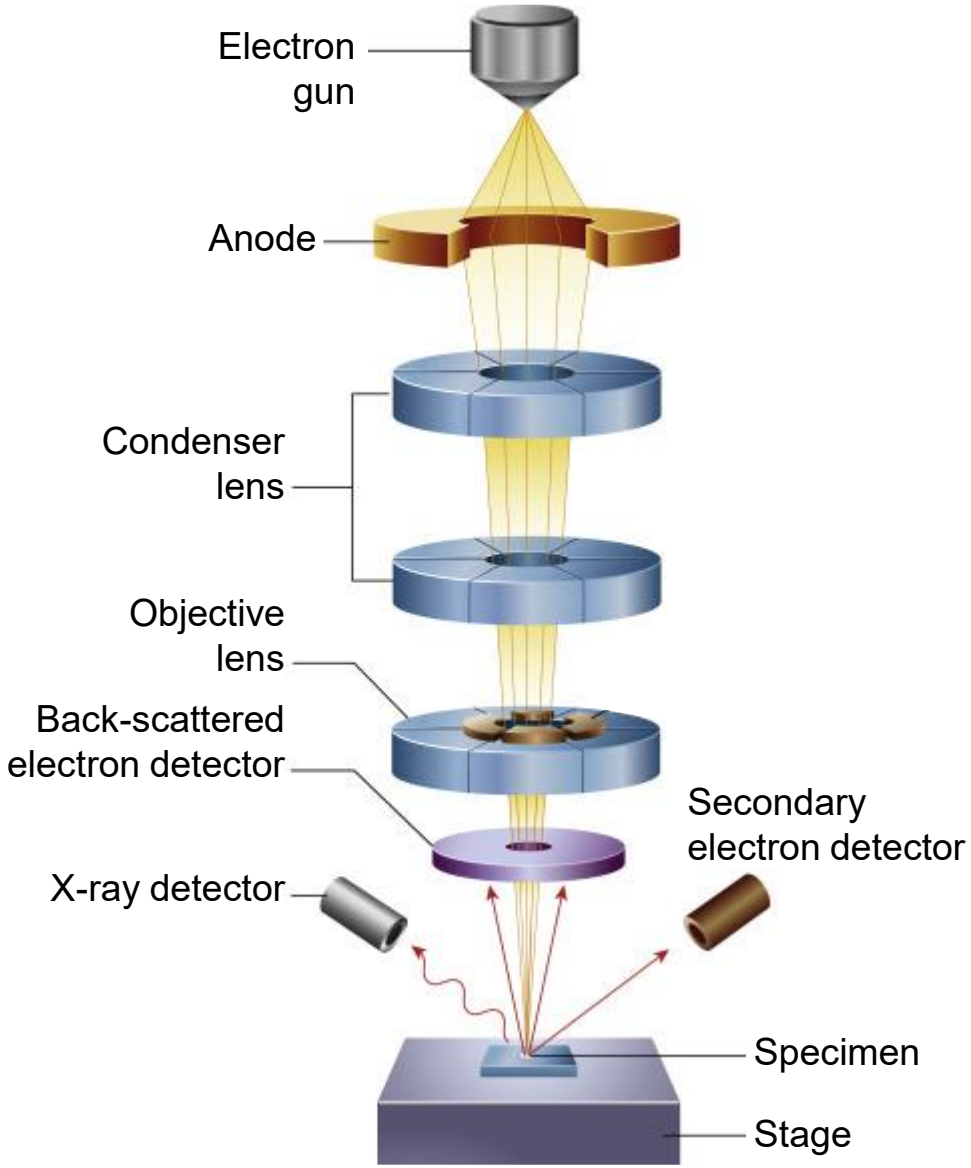
MATE 6301 | Light Microscope *vs* Electron Microscope

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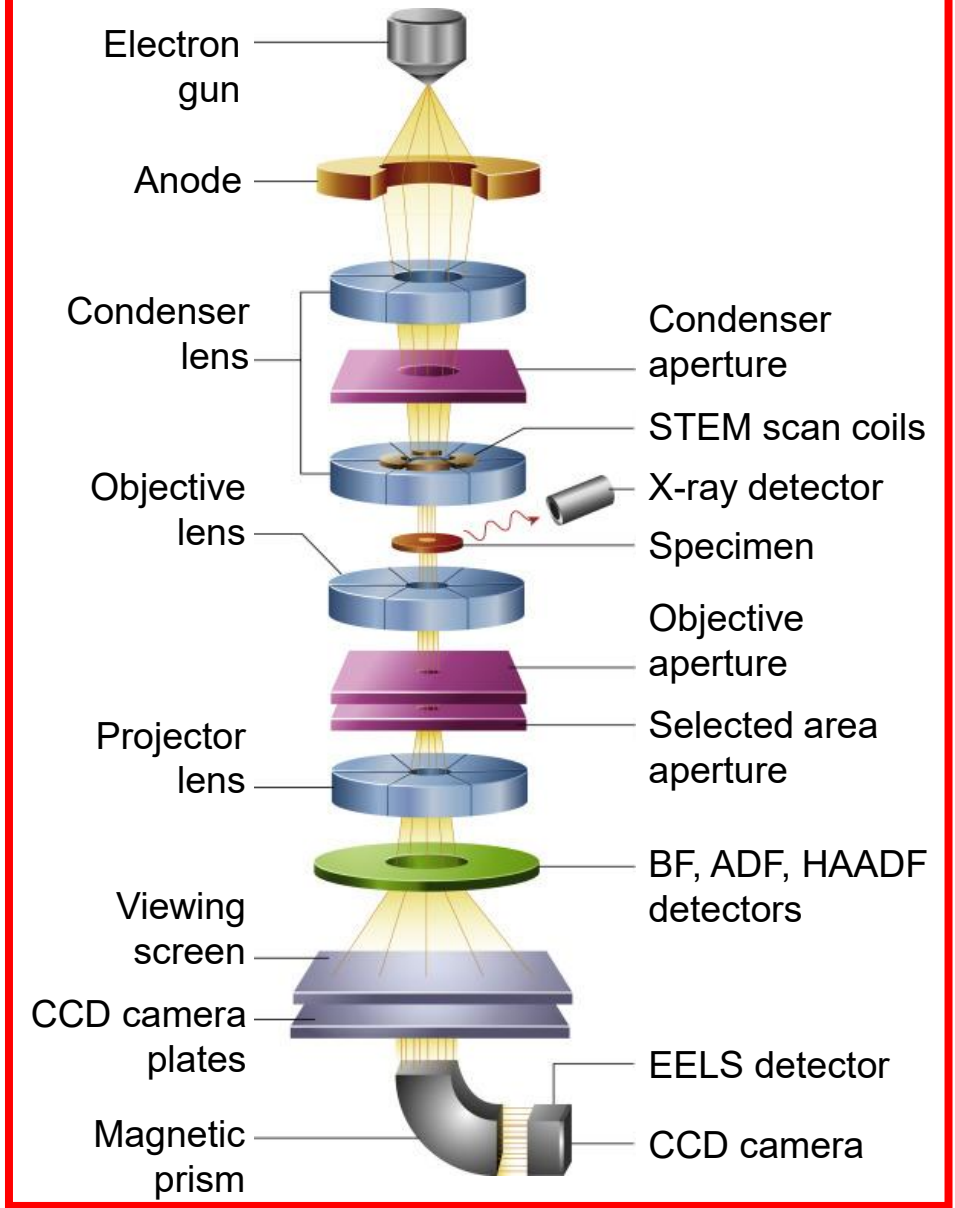
Light microscope



Scanning EM



Transmission EM

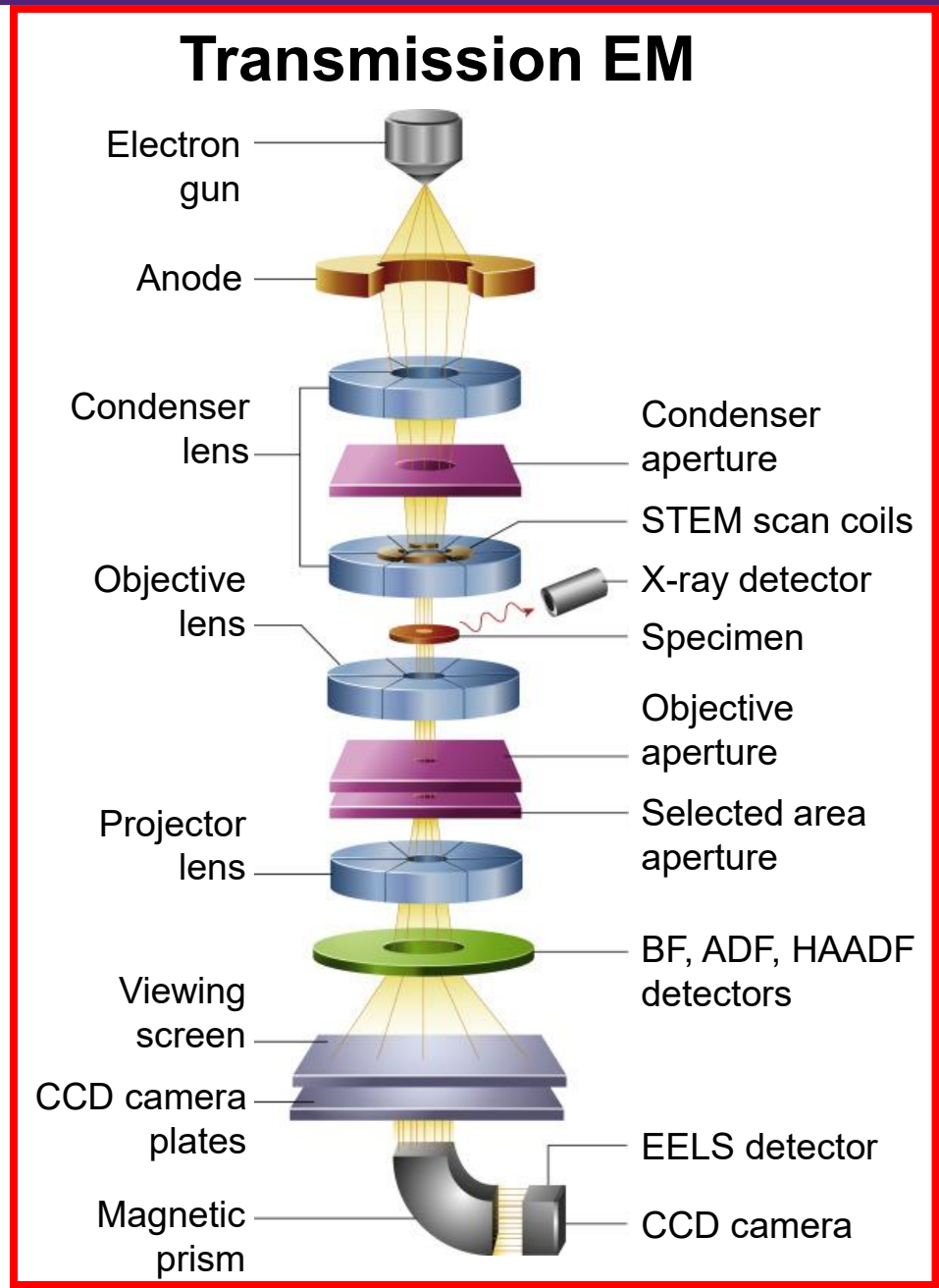


MATE 6301 | Transmission Electron Microscope (TEM)



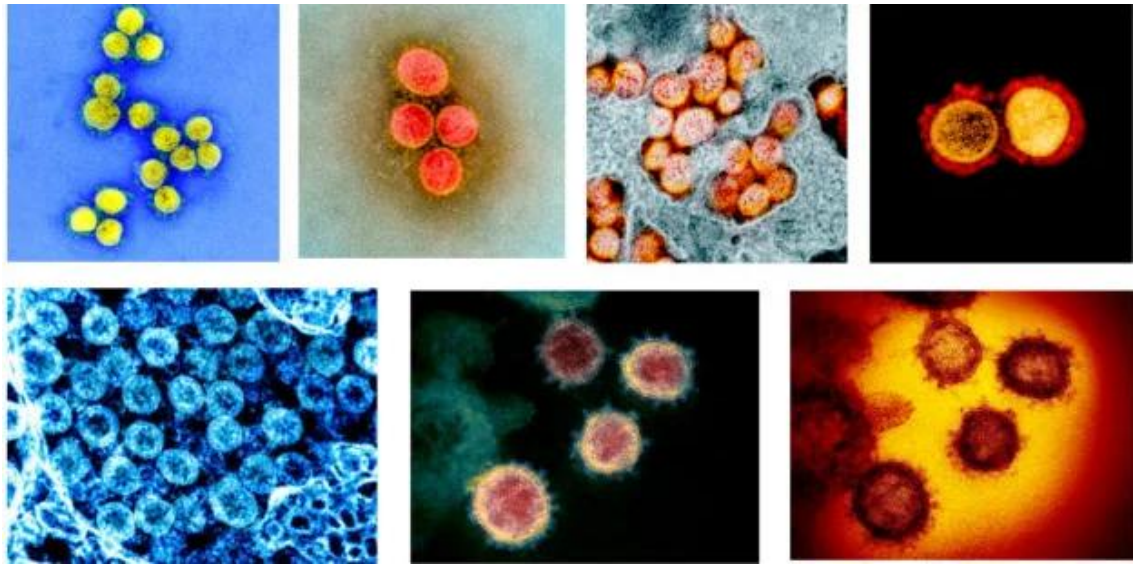
The father of transmission electron microscopy (TEM).

In 1932-33, he came up with the concept and developed a prototype TEM for his Ph.D. He worked further and commercialised the first TEM on the market in 1939.



What is TEM?

Transmission Electron Microscope (TEM) is a powerful electron microscope that uses a beam of electrons to focus on a specimen. TEM has the potential to magnify an image up to 10-50 million times.



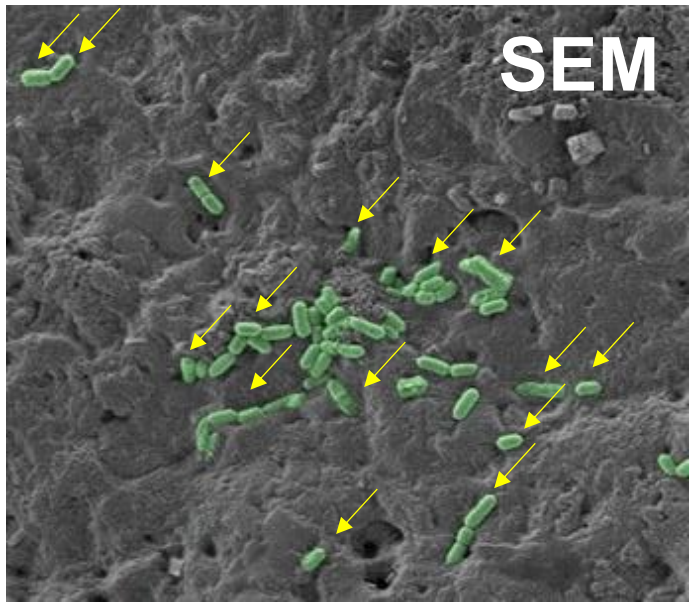
[1]

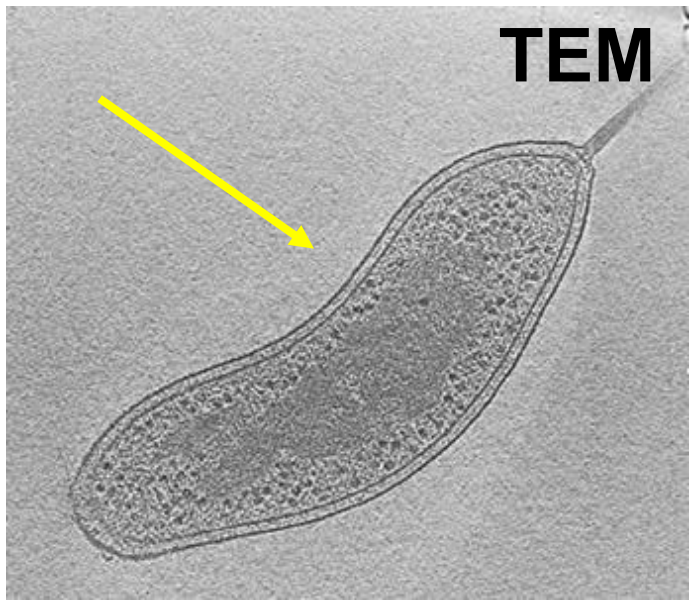
What material information can be obtained?

Applying TEM to the real world has fuelled many discoveries and advancements. Scientists can acquire a basic understanding of the internal structures of materials and biological systems down to the atomic level.

Information such as the morphological features to high-resolution lattice images, compositions and crystal information of the specimen.

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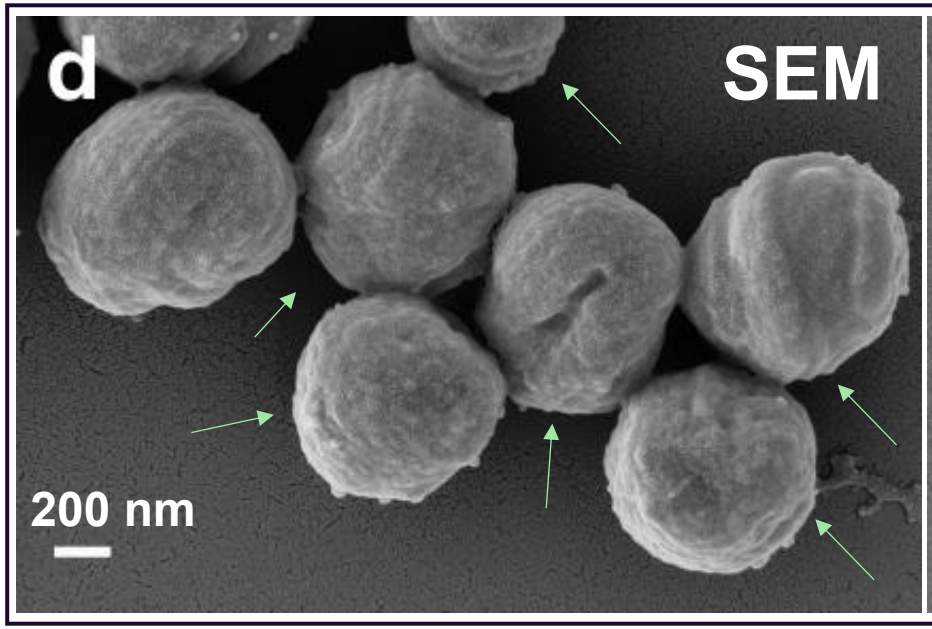


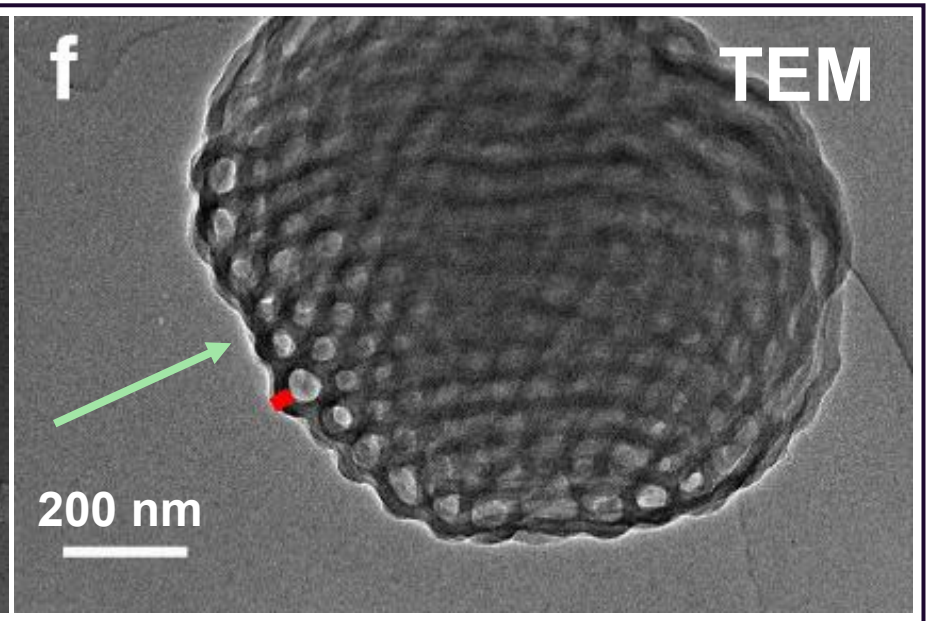
TEM offers…

- 2D projection of 3D objects.
- Finer details of internal structures.
- Much higher resolution & magnification.



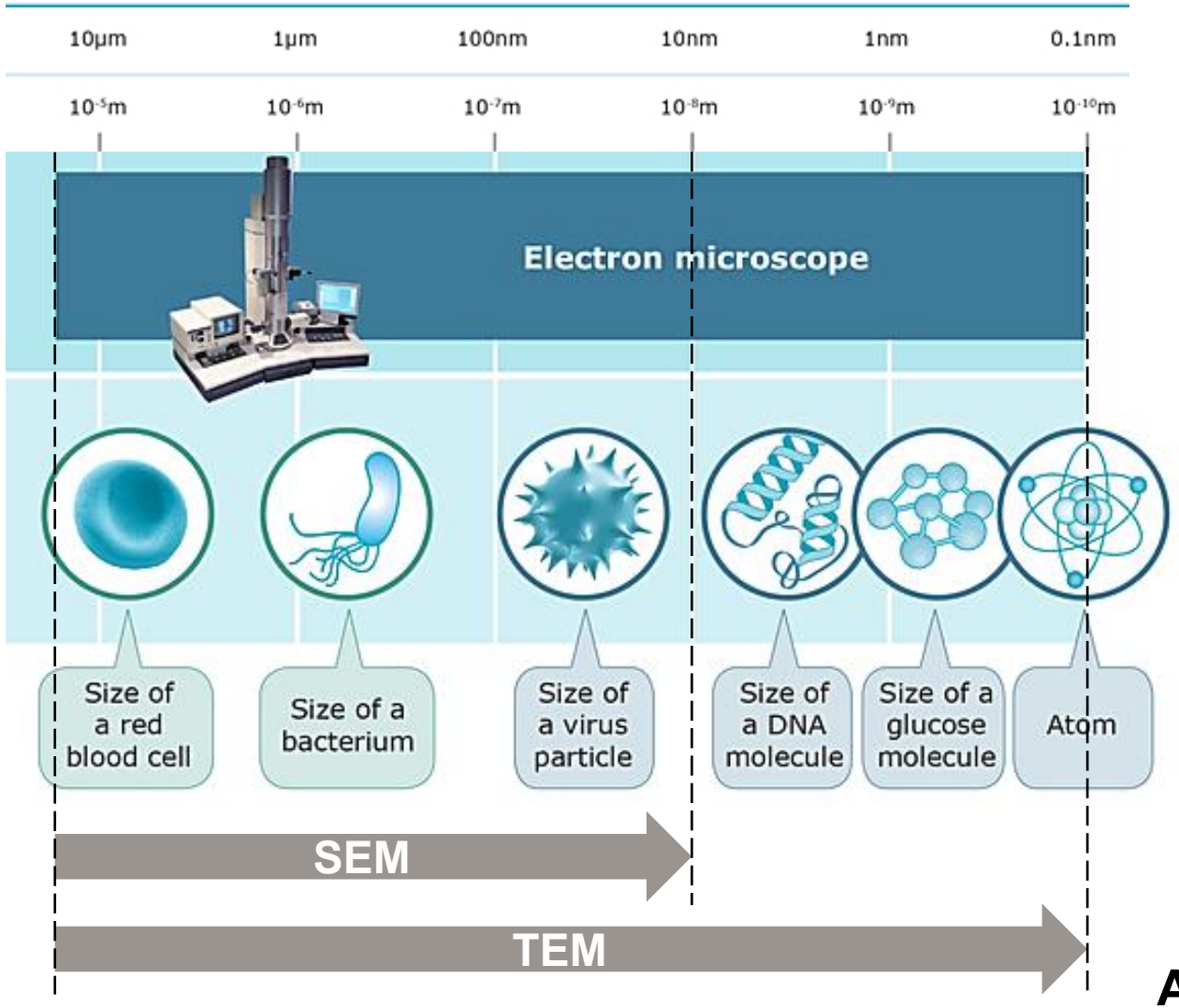
Why?





	SEM	TEM
Electronic type	Scattered scanned electrons	Transmission electrons
Image formation	Electrons are captured and counted by the detector Display on PC computer screen	Imaging directly on a fluorescent screen on a PC computer screen using a charge-coupled device (CCD)
Image information	Surface 3D images	2D internal structure projection image
Maximum multiplier	About 1 – 2 million times	More than 50 million times
Maximum field of view	Large	Limited
Sample thickness	Unrestricted	Usually, < 150 nm.
Acceleration Voltage	1 – 30 kV	60 – 300 kV
Difficulty of operation	Easy to use, little or no sample preparation required	Not easy to use, require sample preparation and training prior to use
Cost	Cheaper	More expensive
Speed	Quick	Slow

Limit of resolution (resolving power): The limit or ability of a microscope to distinguish details of a specimen.





$$\lambda = \frac{h}{p} \longrightarrow \lambda = \frac{h}{\sqrt{2meV}}$$

TEM operates at much higher accelerating voltage (80 – 200 kV) than SEM (1 – 30 kV).

at V = 100 kV (100,000 V), $\lambda_{electron}$ = 0.038 Å

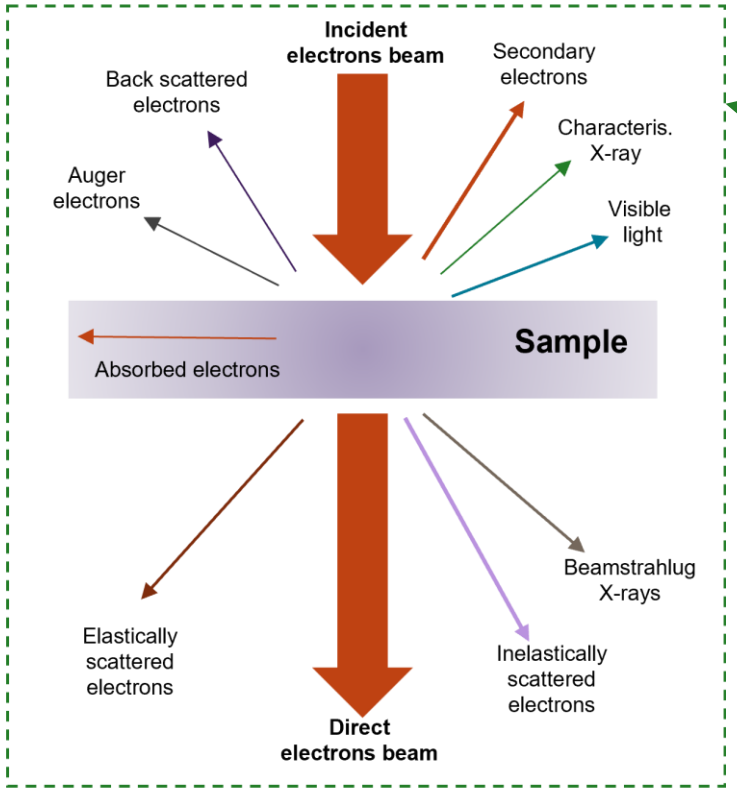
at V = 200 kV (200,000 V), $\lambda_{electron}$ = 0.027 Å

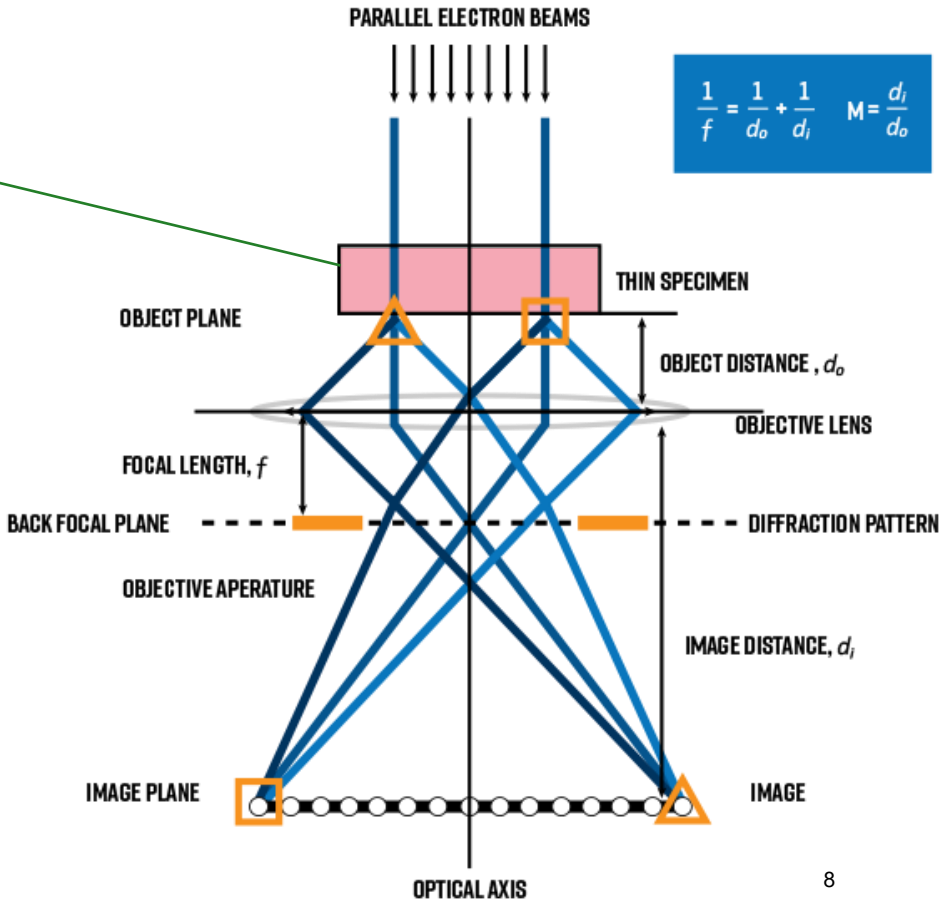
at V = 300 kV (300,000 V), $\lambda_{electron}$ = 0.022 Å

Atomically resolved image collection is possible !

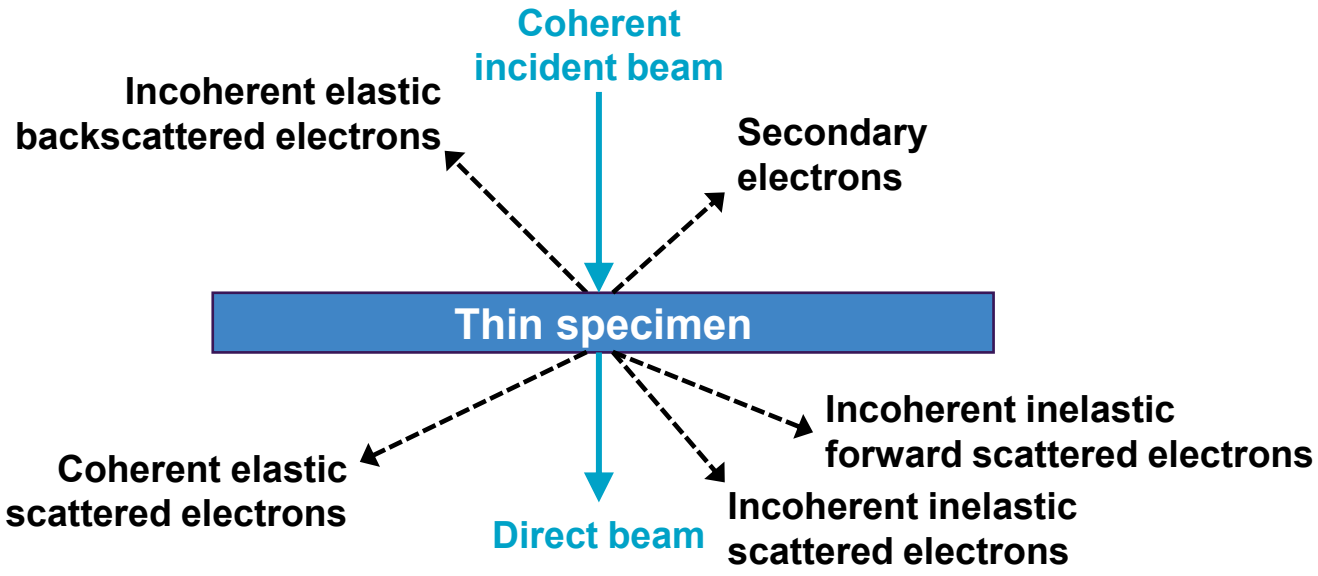
Operating Principle:

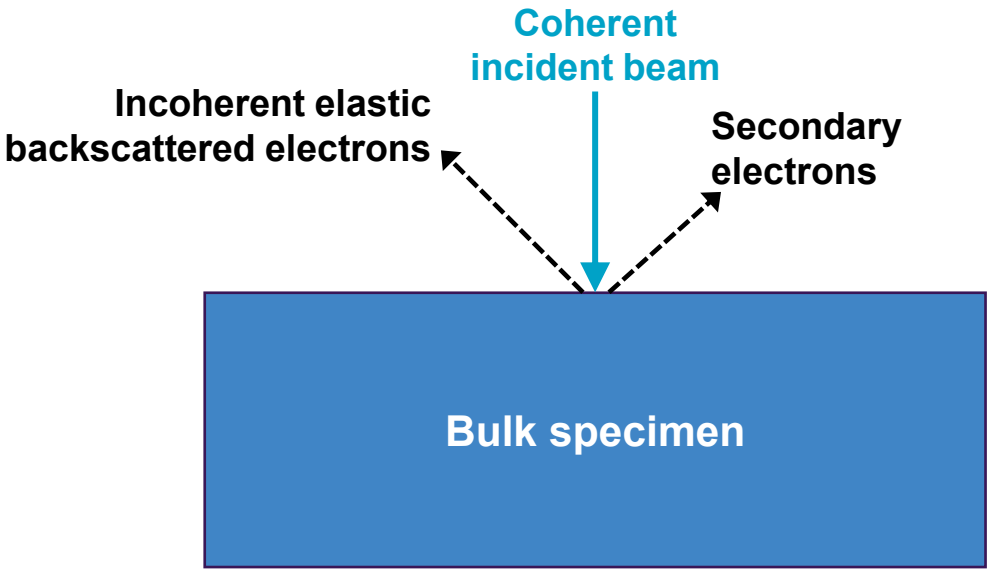
- de-Broglie's equation: $\lambda = h/mv$, When velocity (v) increases, wavelength (λ) decreases.
- Wavelength of electrons is based on the transmission of electrons through the object.
- Bright field imaging relies on incident electrons being scattered and disappearing from the beam depending upon the compositional density and crystal orientation of the sample.



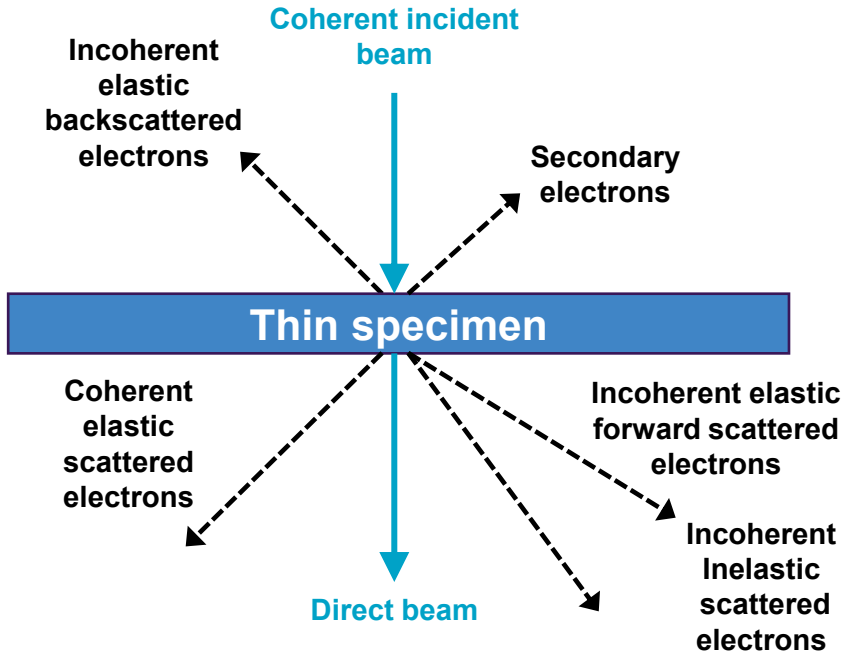


- High accelerating voltage is used in TEM to allow E-beam to penetrate through the specimen.
- This fact restricts the TEM observation to sufficiently thin samples.
- If the sample is too thick, effective penetration of the sample by E-beam is hard to achieve, thus generating dark object in the image without useful information on the internal structure.





Scattering in TEM occurs as both forward and back scattering, though we almost always use the forward scattered signals (Scattering in the SEM is always backscattering).



Coherent scattering has the strong relationship between phase of waves.

Incoherent scattering has the weak or non-existing phase relationship.

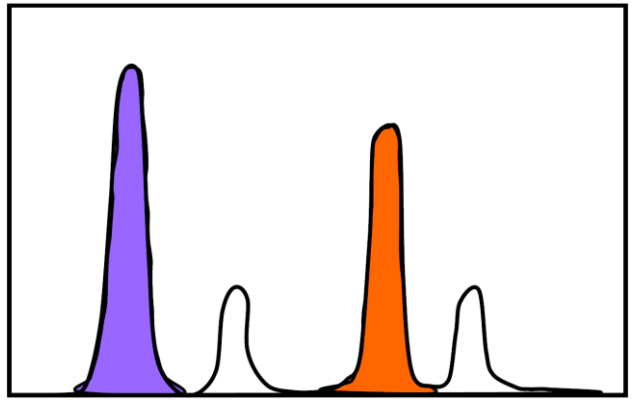
- Elastic scattered electrons conserve the energy from the incident electron beam.
- Inelastic scattered electrons lose the energy form the indecent electron beam.

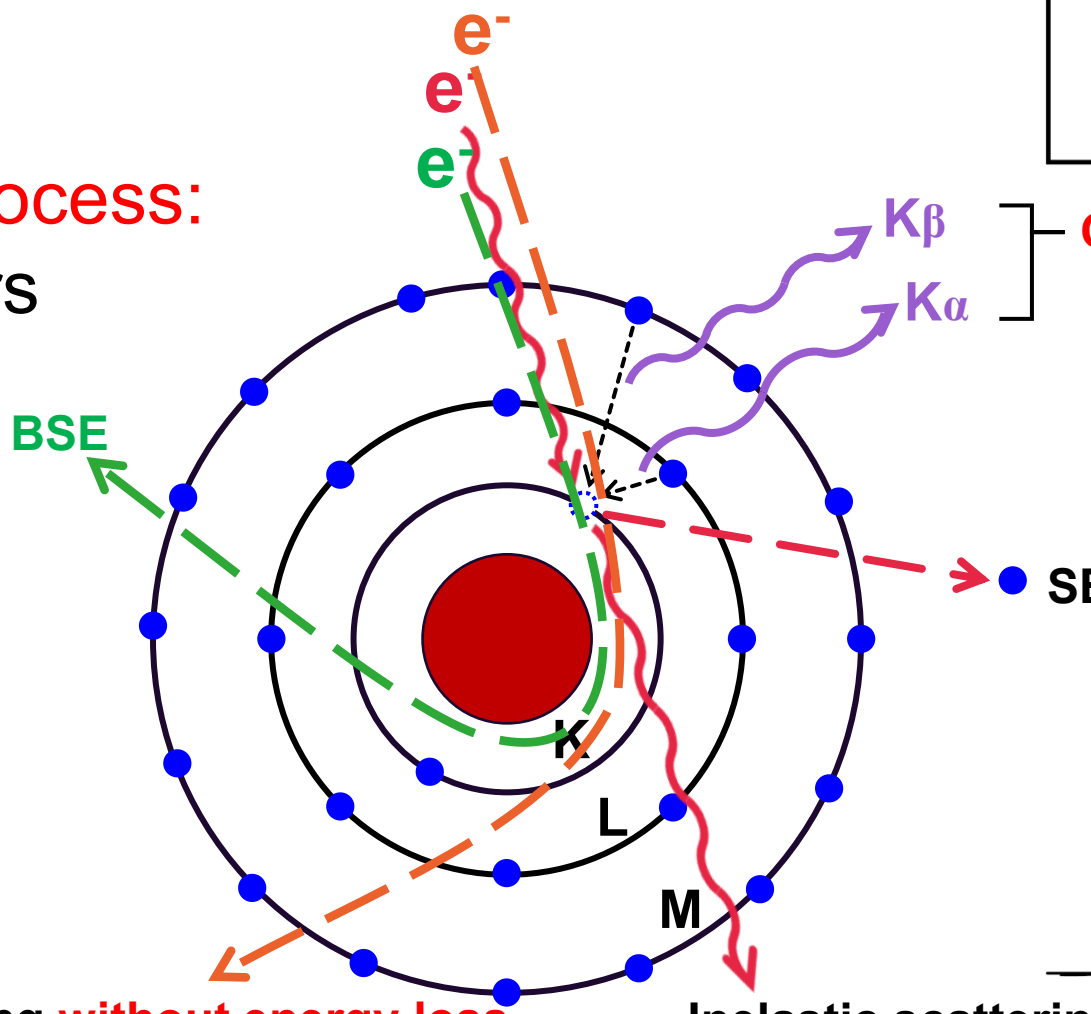
ELASTIC scattering process:

- Incidental beam does not suffer energy loss

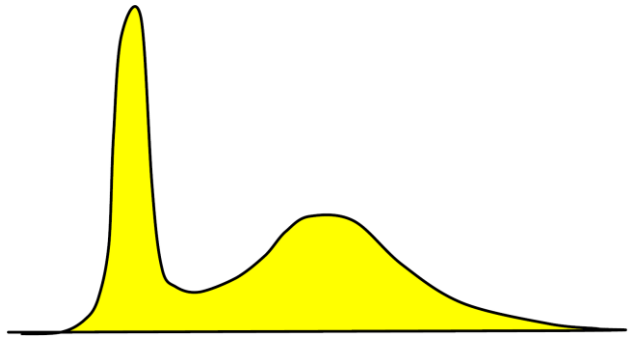
INELASTIC scattering process:

- Incidental beam suffers energy loss





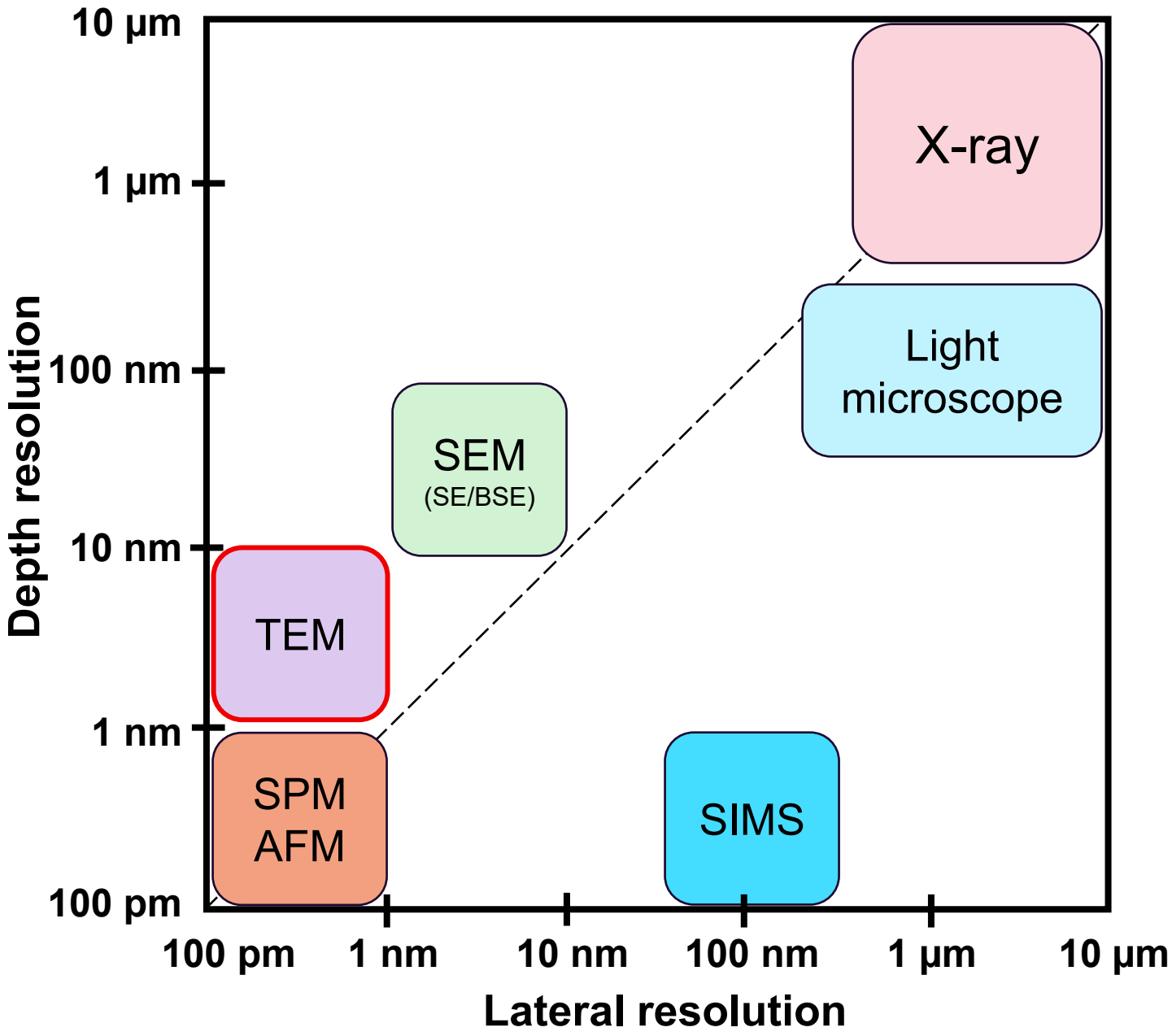
Characteristic X-rays
E = hv



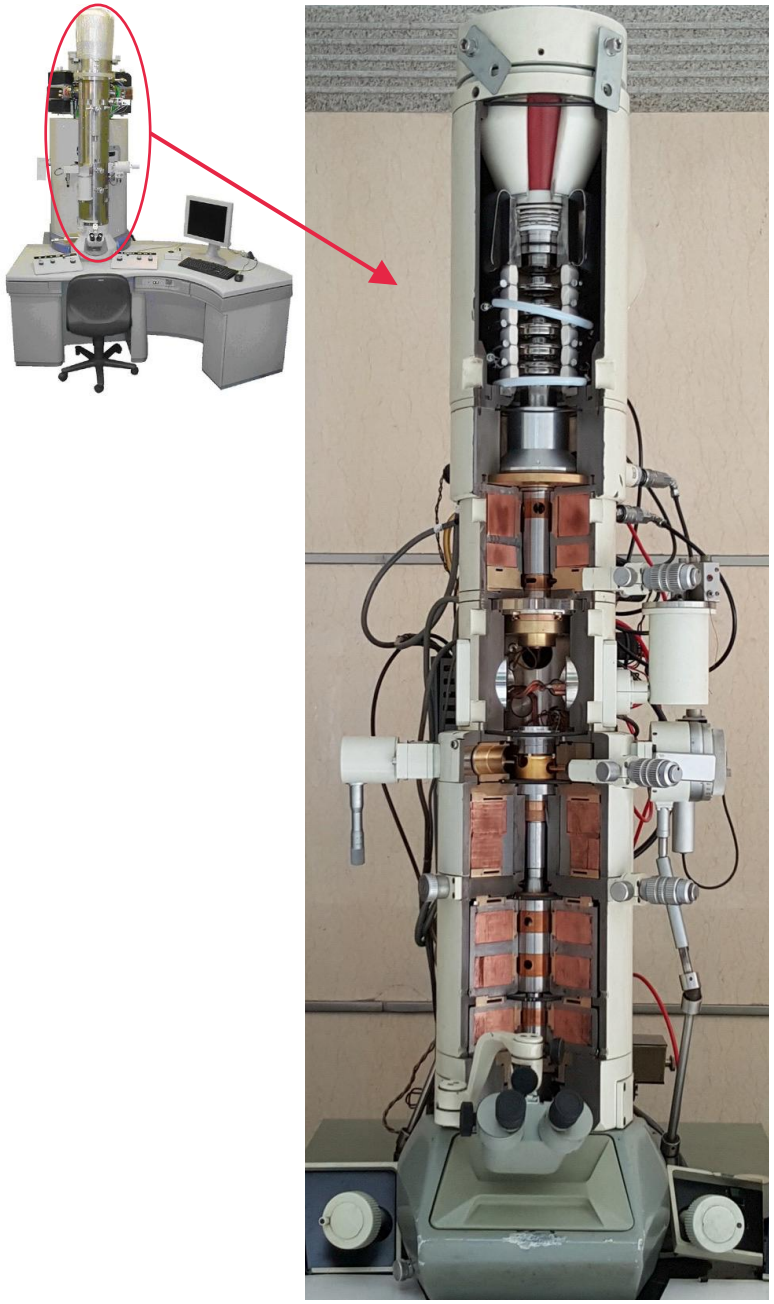
Elastic scattering without energy loss

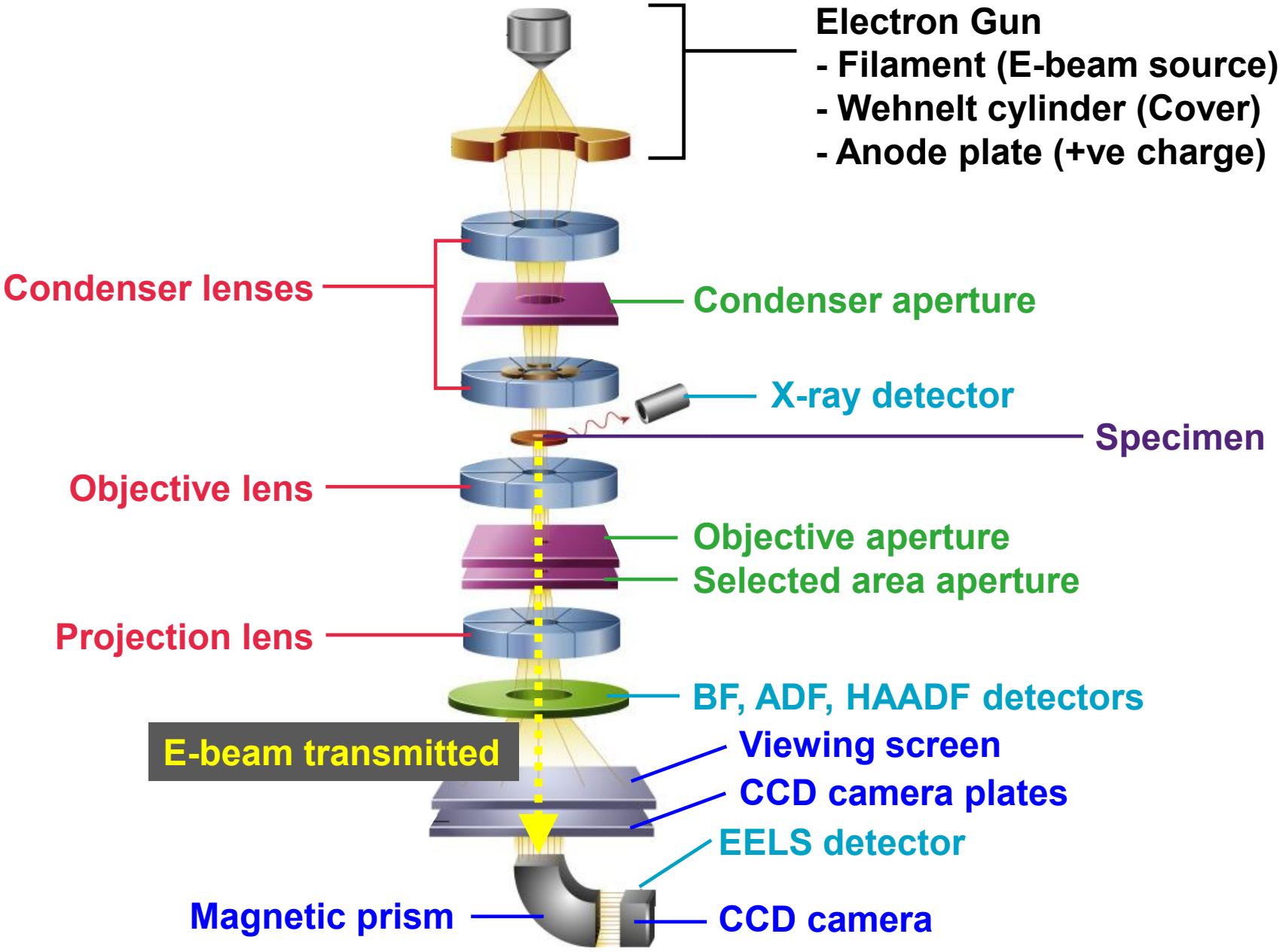
Inelastic scattering with energy loss

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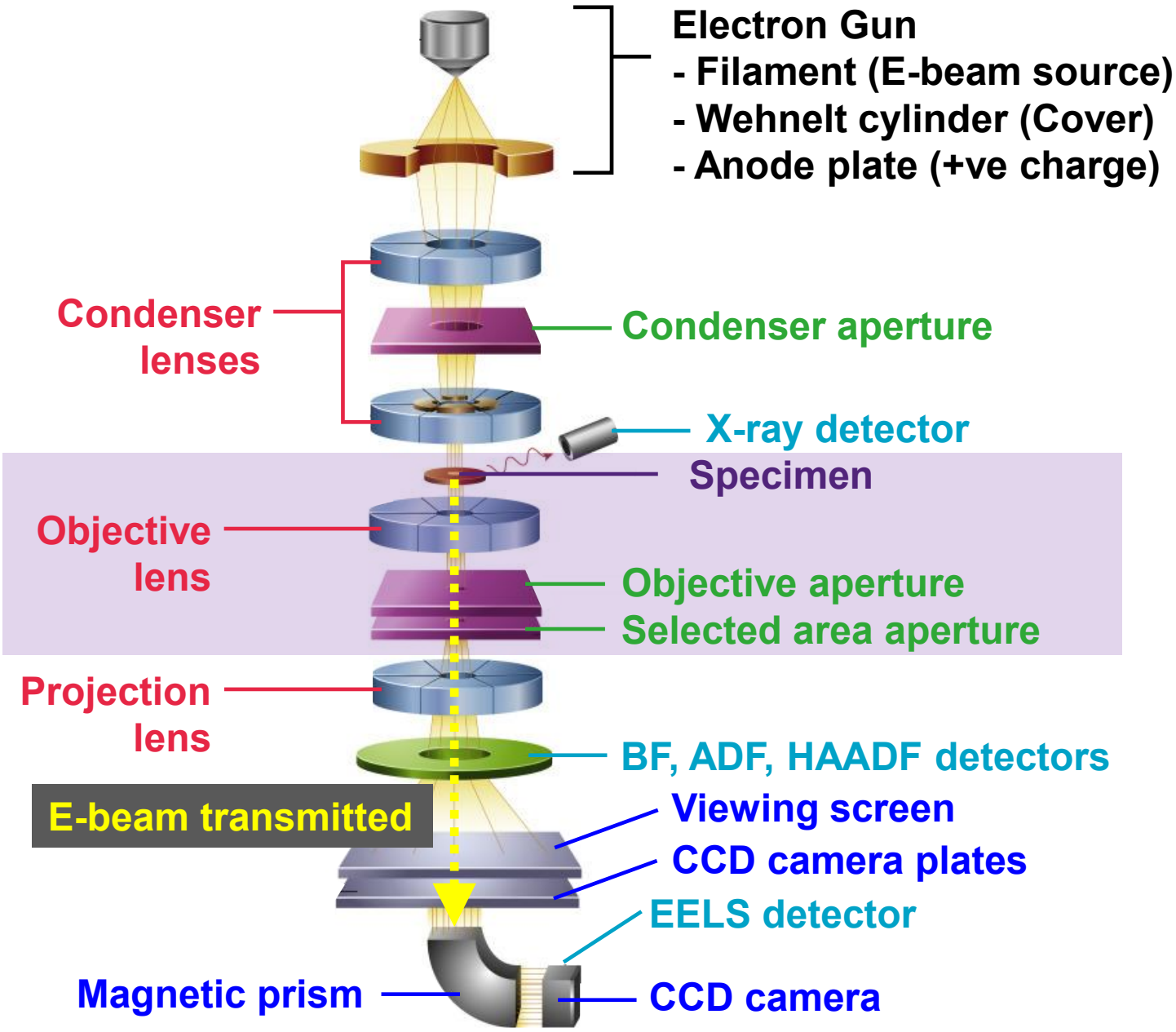


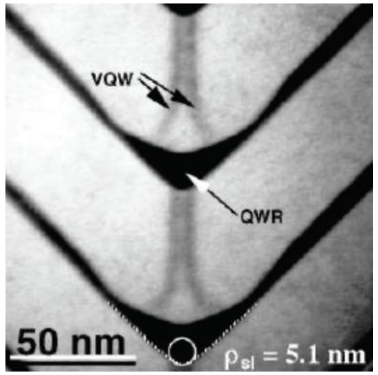
MATE 6301 | How does TEM work?



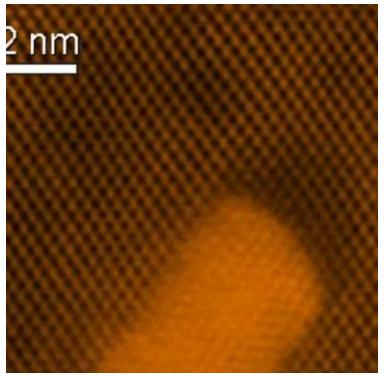


MATE 6301 | How does TEM work?

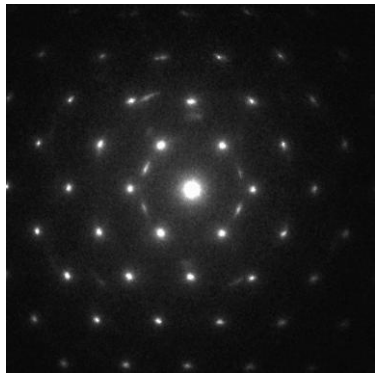


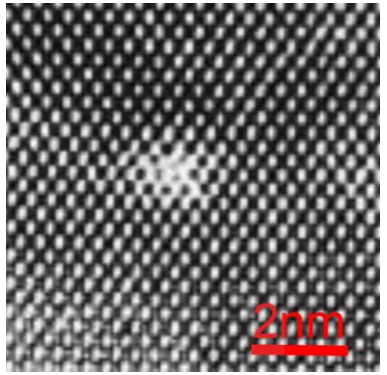


Diffraction contrast

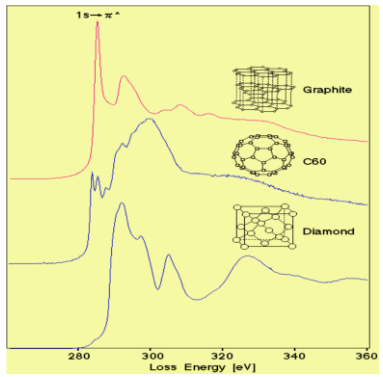


STEM imaging

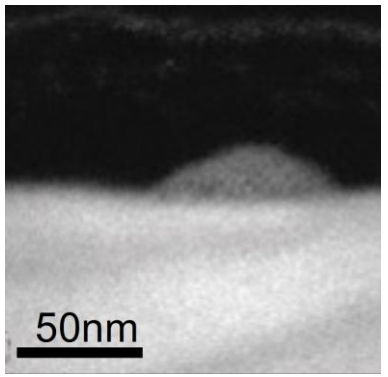

Electron diffraction



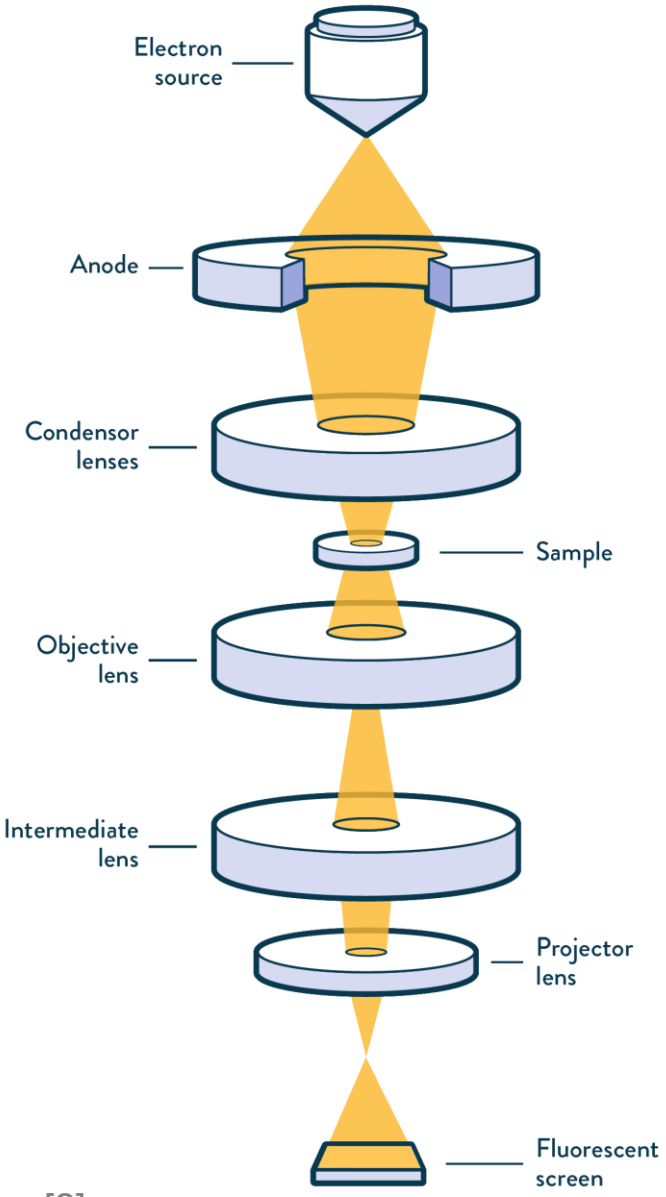
HRTEM imaging



EELS spectroscopy



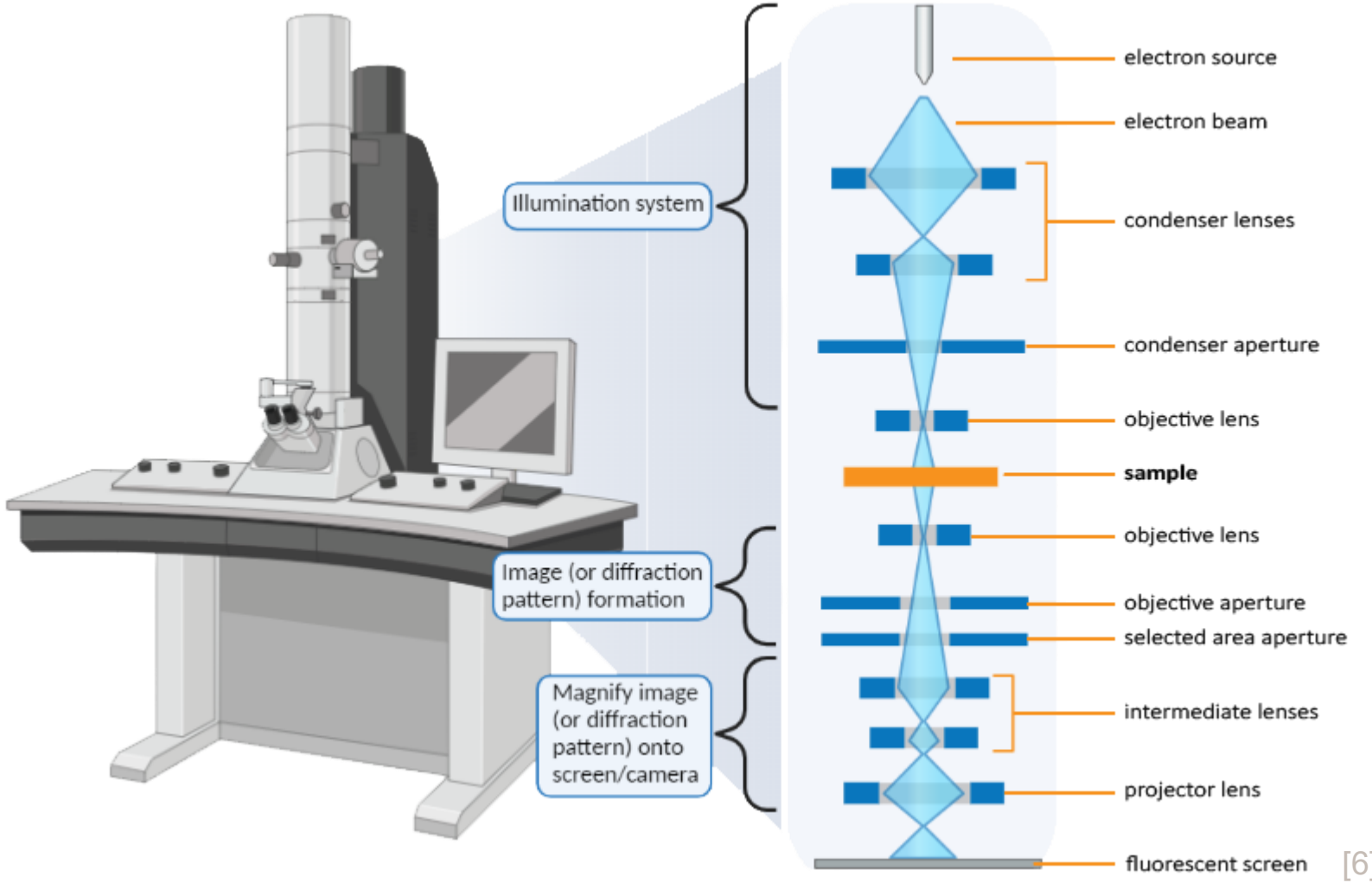
Energy filtered TEM



[2]

Fundamentals of TEM

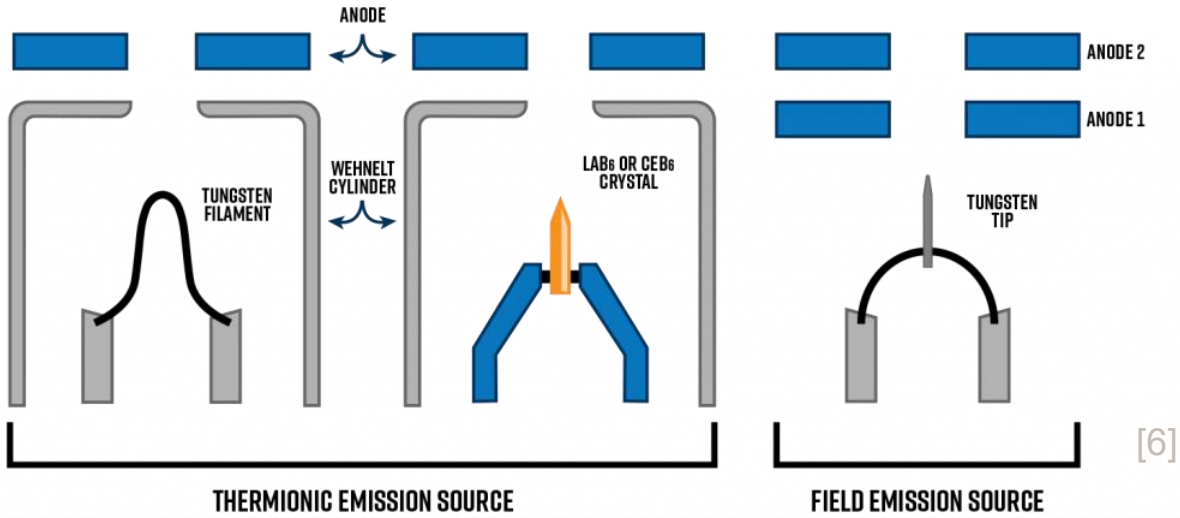
The five major components in TEM are the electron gun, vacuum system, electromagnetic lenses, specimen stage and imaging devices.



[6]

1. Electron Gun:

The TEM consists of an electron emission source, which may be a **tungsten filament or a lanthanum hexaboride (LaB_6) source**, known as an electron gun. By connecting this gun to a high-voltage source of about **100 – 300 kV**, the gun begins to emit electrons by either **thermionic or field electron emission** into the vacuum.



[6]

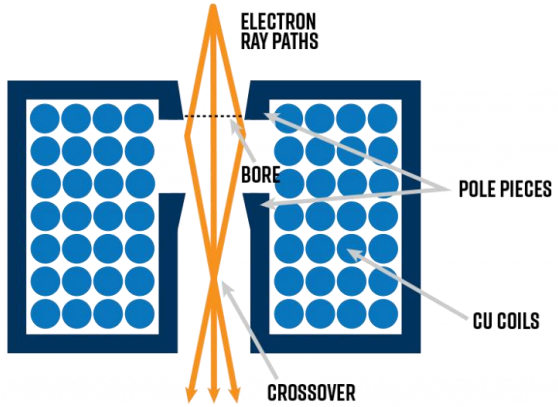
2. Vacuum System:

Electrons behave like light only when they are manipulated in a vacuum. Different vacuum pumps are used to obtain and maintain **high to ultra-high vacuum pressure as low as 10^{-8} Pa**. A number of airlocks and separation valves are used to avoid the need to evacuate the whole column every time a specimen or photographic material or filament is exchanged.

[6]

3. Electromagnetic Lenses:

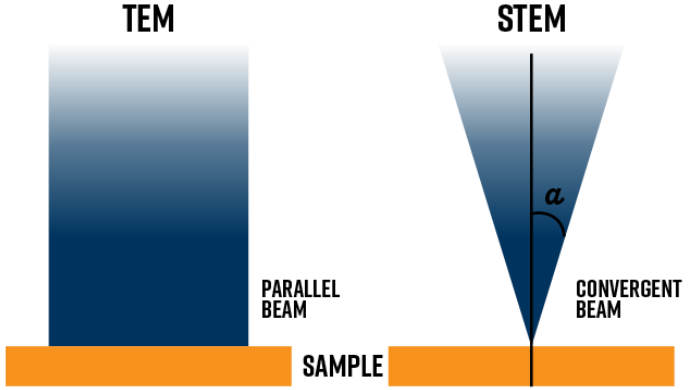
The **electromagnetic lenses typically consist of Cu coils**, and when an electric current is passed through these coils, an **electromagnetic field is created between the pole pieces creating a gap in the magnetic circuit**. By varying the current through the coils, the strength of the field, and thereby the **power of the lens, can be varied**.



Typically, a TEM consists of three stages of lensing – **condenser lenses, objective lenses,** and **projector lenses**.

- The **condenser lenses** are responsible for **primary beam formation**.
- The **objective lenses focus on the beam that comes through the sample itself**.
- The **projector lenses** are used **to expand the beam onto the phosphor screen** or **other imaging devices such as film**.

The magnification of the TEM is due to the ratio of the distances between the specimen and the objective lens' image plane.



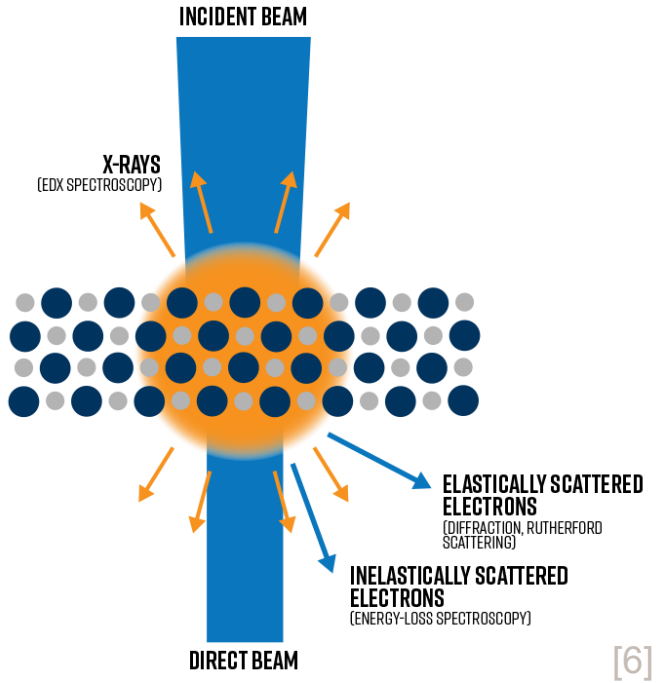
[6]

Comparison of TEM and STEM illumination which are used to form a parallel beam or convergent probe, respectively, at the sample entrance surface.

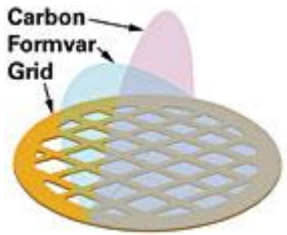
4. Specimen Stage:

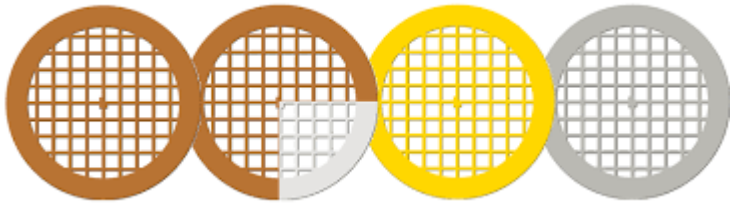
The specimen is placed on a mesh having a diameter of about 2.5 mm and is held by a specimen holder. The **mesh or grid is made of copper, molybdenum, gold,** or **platinum**. The specimen stage also includes airlocks to allow for the insertion of the specimen holder into the vacuum with a minimal increase in pressure in other areas of the microscope.

Sample preparation includes the preparation of thin sections of a maximum thickness of a few tens or hundreds of nanometers. As TEM specimen holders permit the rotation of a sample by a desired angle, multiple views of the same specimen can be obtained by rotating the angle of the sample along an axis perpendicular to the beam. By taking multiple images of a single TEM sample at differing angles, typically in 1° increments, a set of images known as a "**tilt series**" can be collected.



[6]

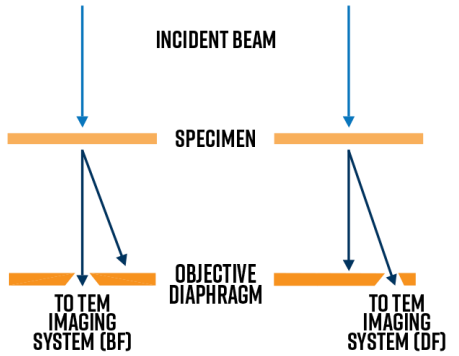




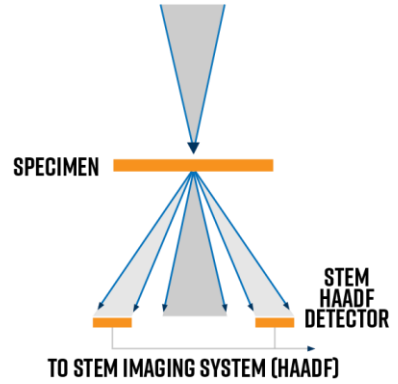
[7]

5. Imaging Devices:

Originally, TEMs used a fluorescent screen that emitted light when impacted by the transmitted electrons for real-time imaging and adjustments; and a film camera to record permanent, high-resolution images. Modern instruments rely primarily on solid-state imaging devices, such as a **CCD (charge-coupled device) camera**, for image capture. They may still include a fluorescent screen, but it may be observed by a video camera. [3]



[6]



[6]

Note:

Samples must be thin (less than 150 nm) to allow an electron beam to pass through the sample, to form an image.

Risk Involved:

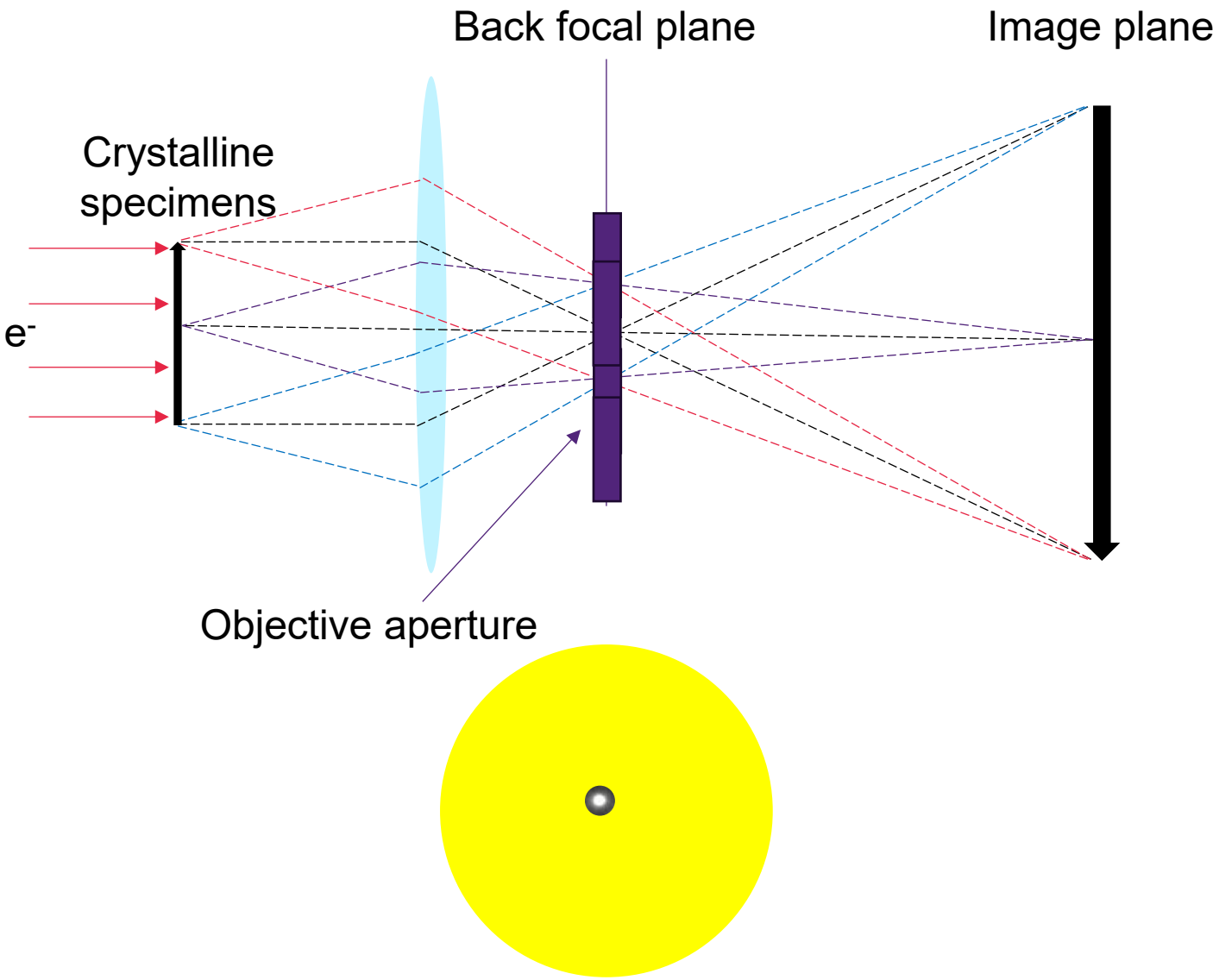
- Longer exposure of organic or biological samples to the electron beam can affect the sample.
- Free from solvents & moisture (can react to the electron beam and affect the vacuum).

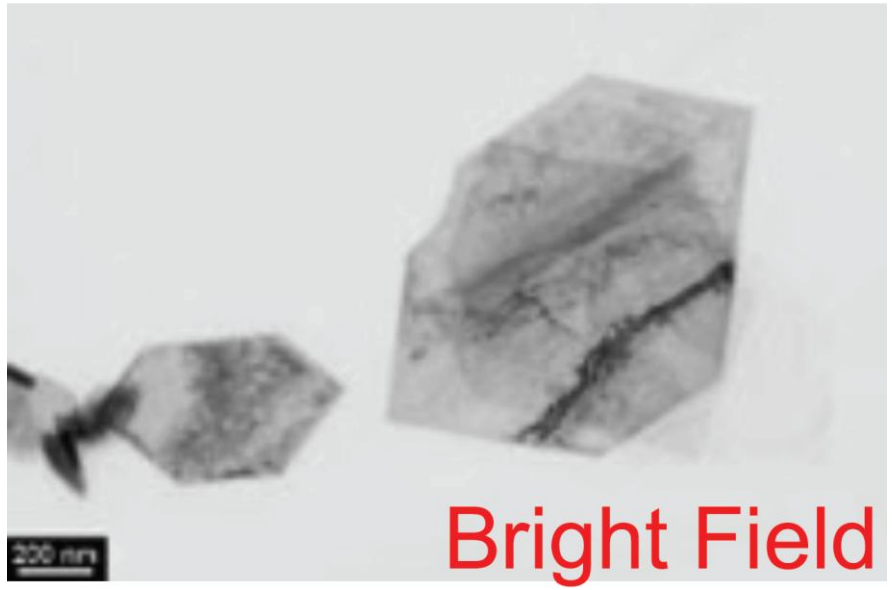
Imaging Techniques

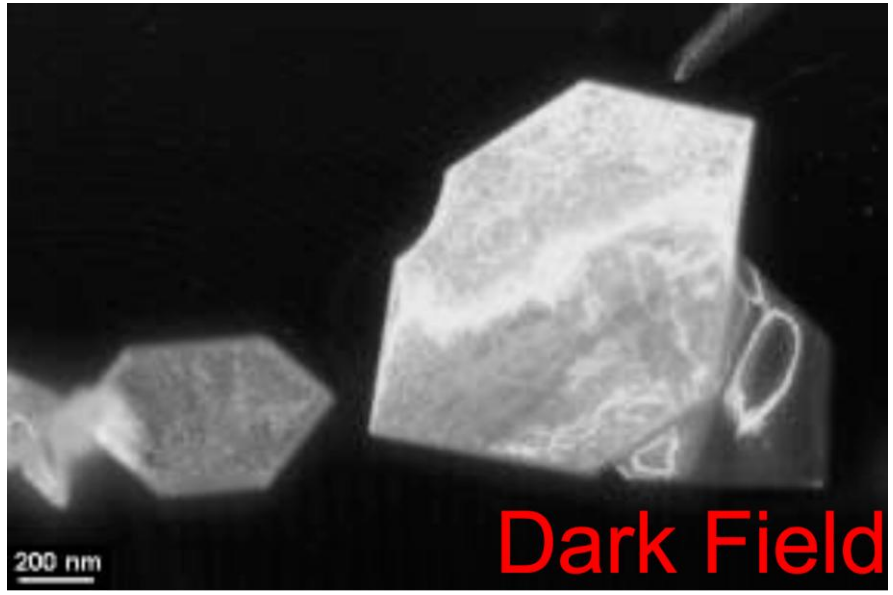
- **Conventional Imaging Modes:**
  - Diffraction Contrast
  - Phase Contrast
  - Mass/Thickness Contrast
- **Diffraction Modes:**
  - Electron Diffraction
- **Inelastic Signals:**
  - Characteristic X-ray (also called Energy Dispersive Spectroscopy – EDS)
  - Electron Energy Loss Spectroscopy (EELS + EFTEM)
- **Scanning Transmission Electron Microscopy (STEM)**

Diffraction contrast

- Elastic scattered electrons are typically used to form images and diffraction patterns

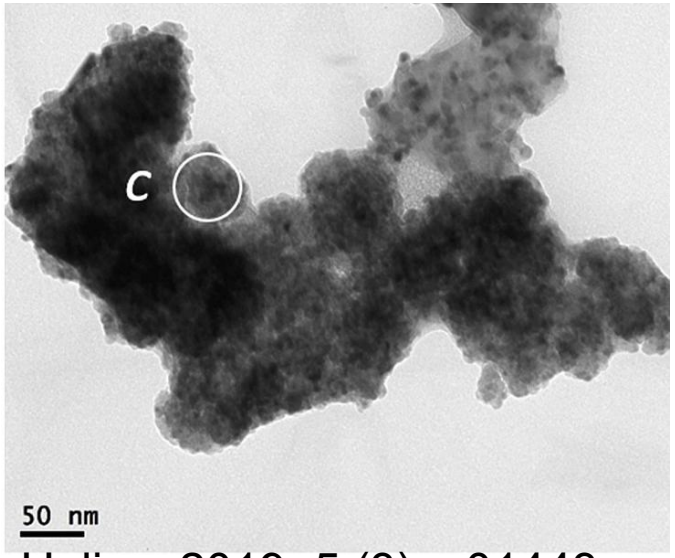






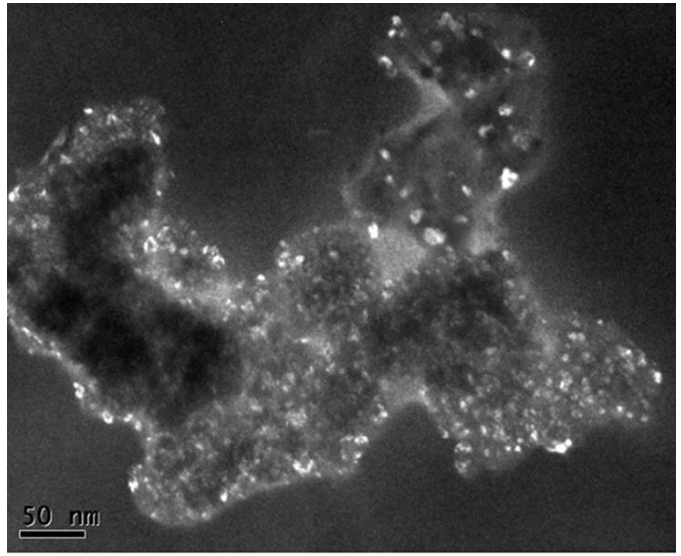
- Diffraction contrast is used for crystalline specimens
- Diffraction contrast is formed **with a single diffracted (can be transmitted) electron beam** and realised with an aperture at the back focal plane of the objective lens
- Diffraction contrast is mainly used for study of
  - Defects
  - Precipitates
  - Grain sizes and distributions
  - ……

Bright-field TEM image

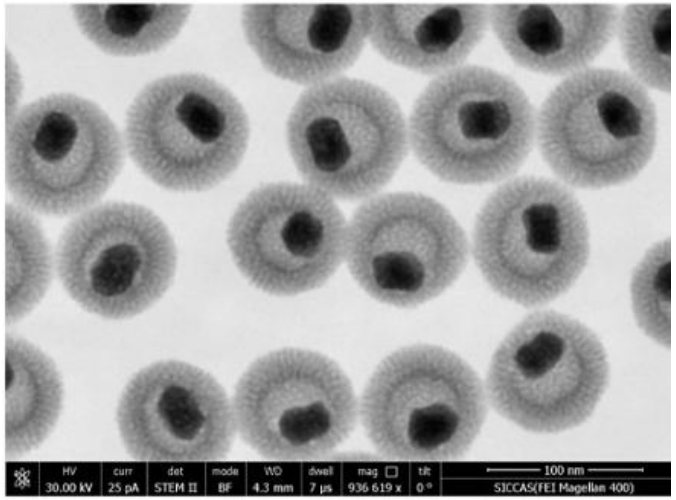


Heliyon2019, 5 (3), e01443

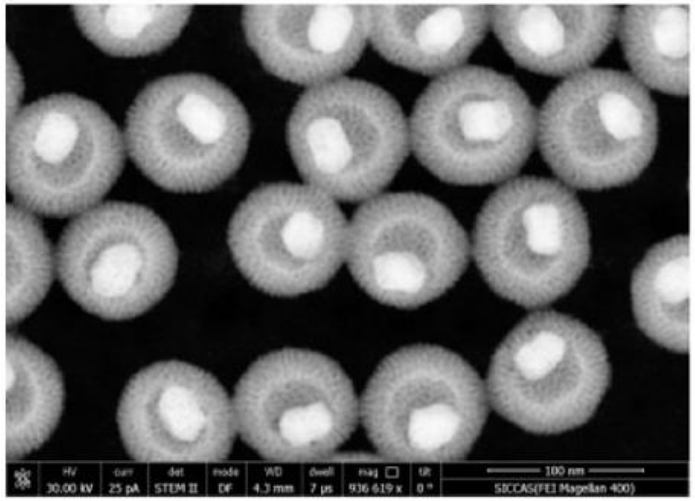
Dark-field TEM image



Fe_2O_3 on SiO_2

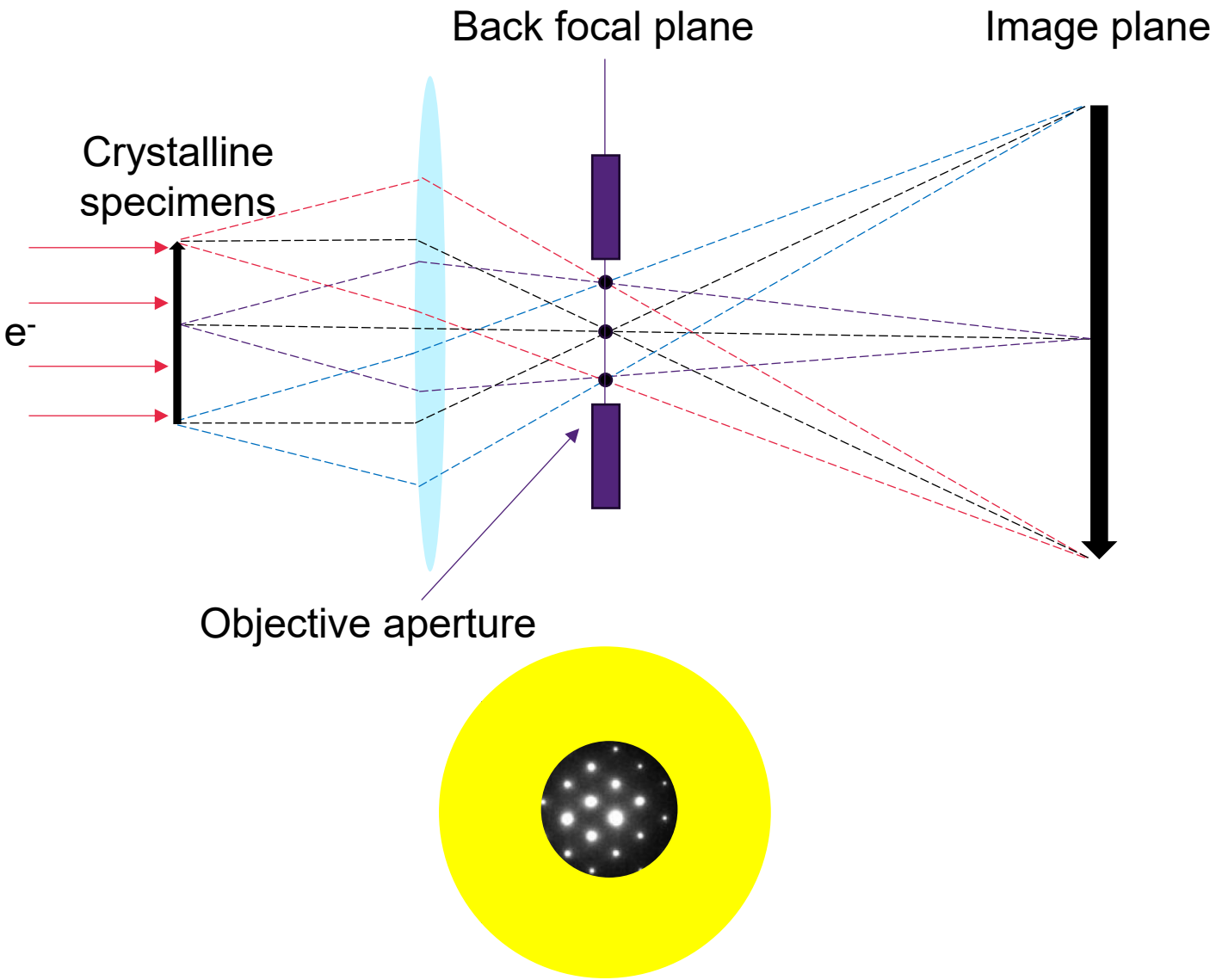


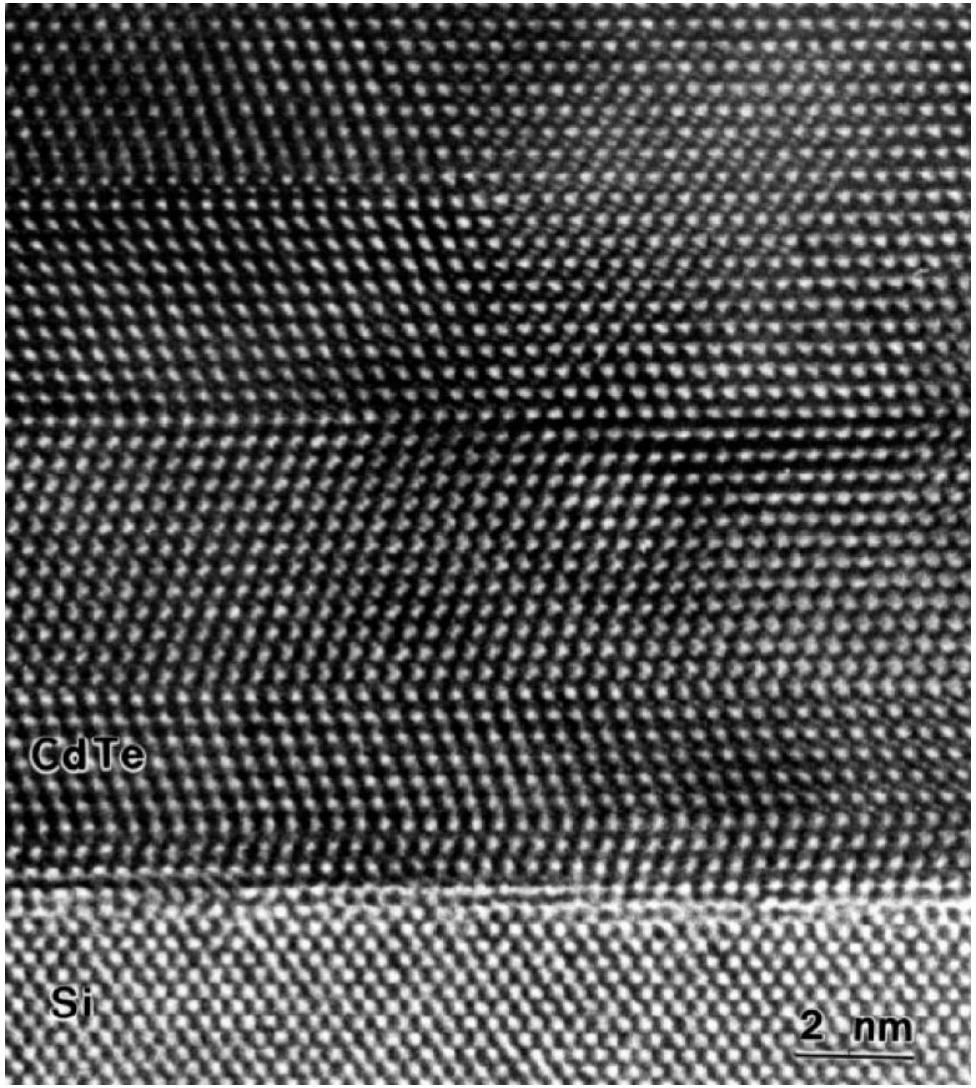
Ruthenium nanoparticle core in SiO_2 shell



J. Am. Chem. Soc. 2014, 136, 27, 9701–9709

- Elastic scattered electrons are typically used to form images and diffraction patterns





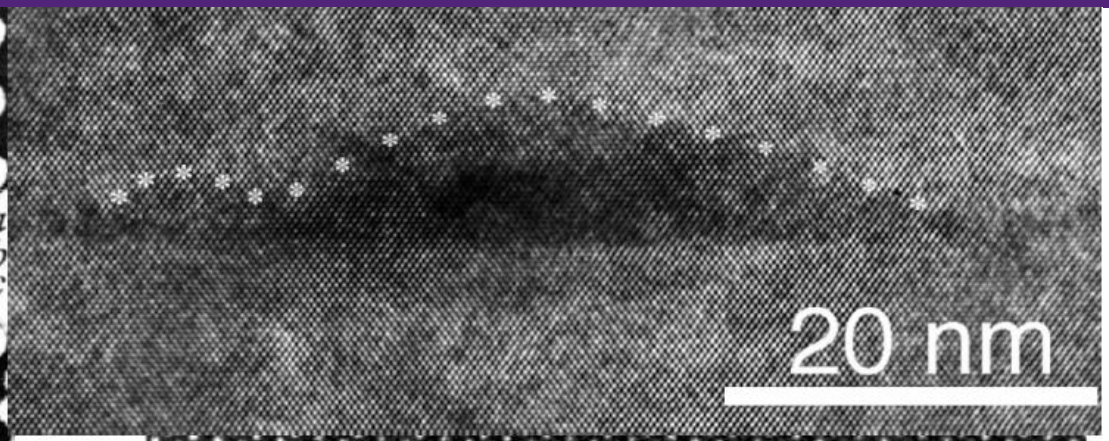
- Phase contrast deals also with crystalline specimens
- Phase contrast is formed **with at least two diffracted electron beams through their interference**.
- Phase contrast is realised with (or without) an aperture at the back focal plane of the objective lens
- Phase contrast is mainly used for study of
  - Defects
  - Precipitates
  - Interfacial structures
  - Crystal growth
  - ……

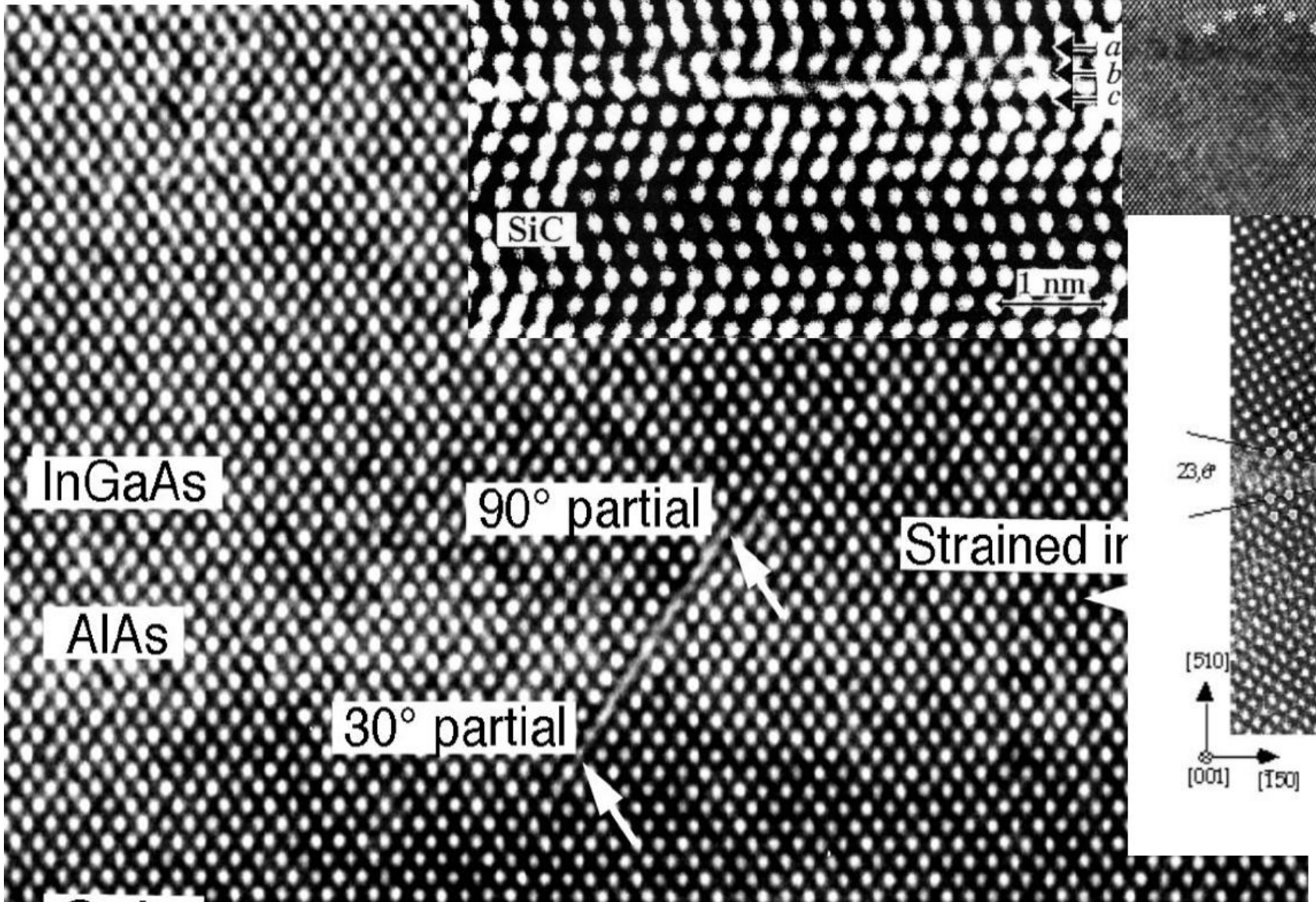
- Phase contrast requires **very thin specimens** (< 20 nm typically) for the Interference

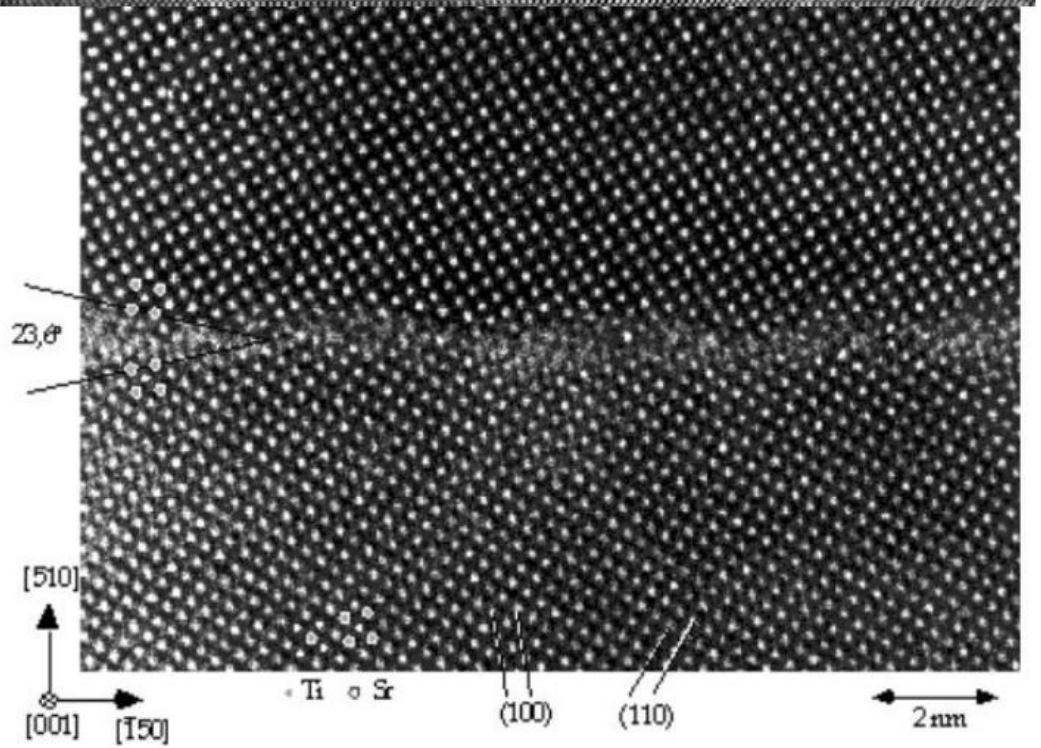
- Crystals should be **orientated** to the directions where the **zone axes** should be parallel to the incidental electron beam

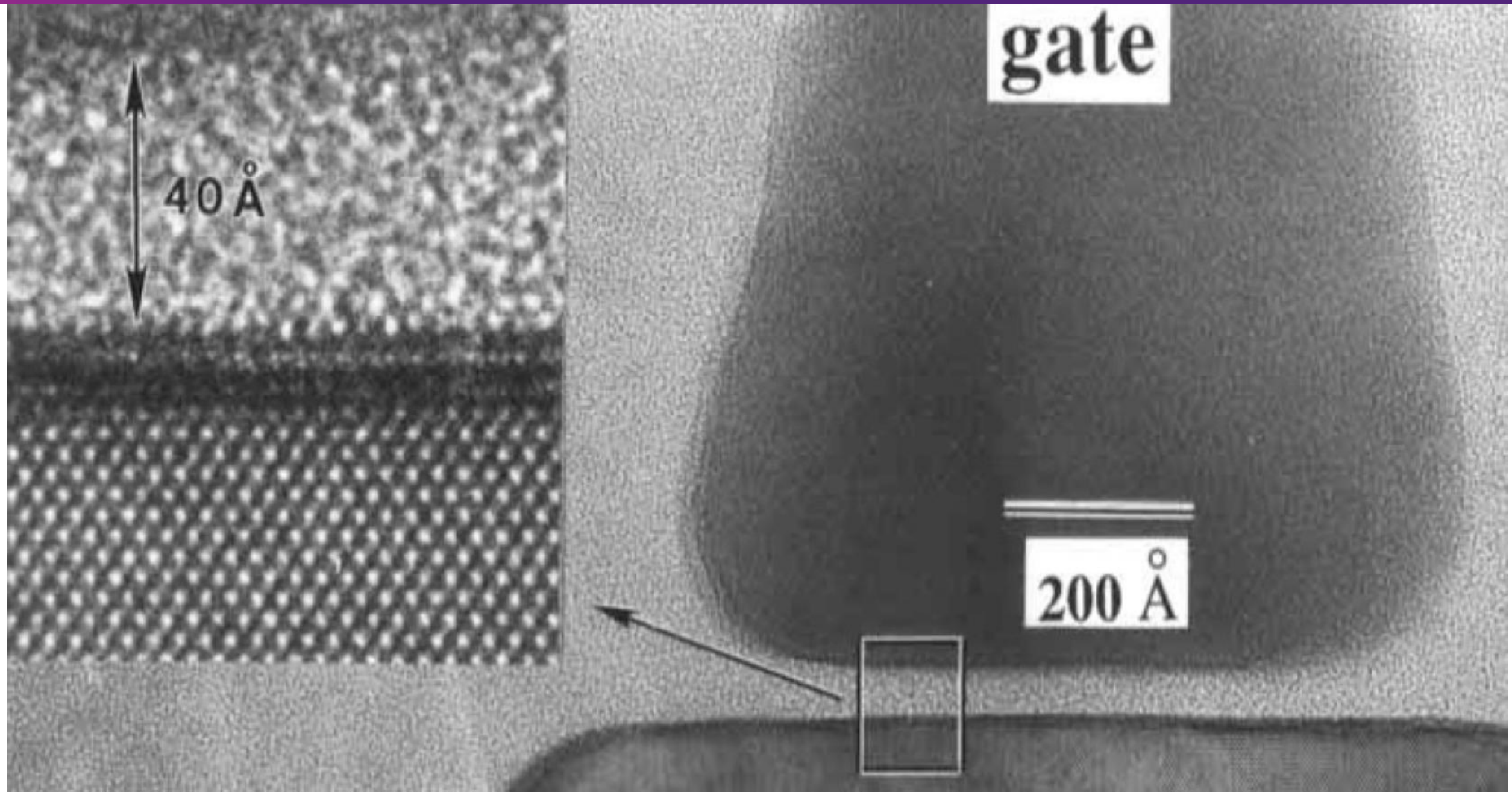
- Sometimes, image simulations are needed for gain correct interpretation as atoms can be black or white in the image


GaN
SiC
1 nm


20 nm


InGaAs
AlAs
90° partial
30° partial
Strained ir

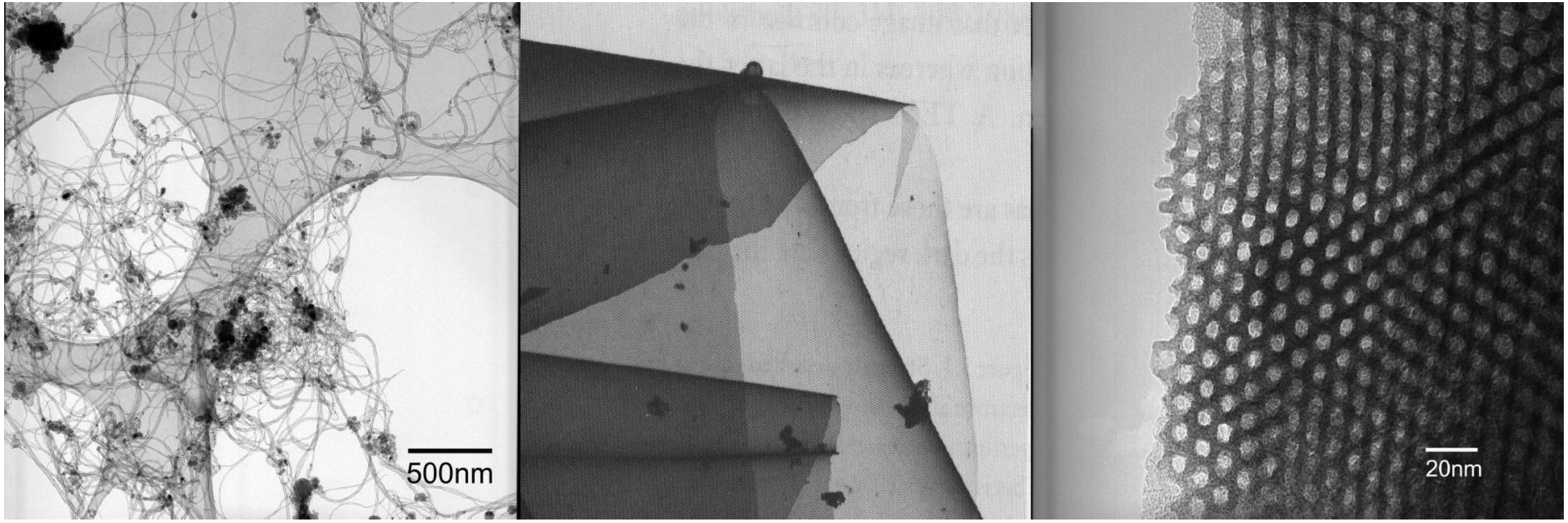

[510]
[001]
[1̄50]
Ti
Sr
(100)
(110)
2 nm

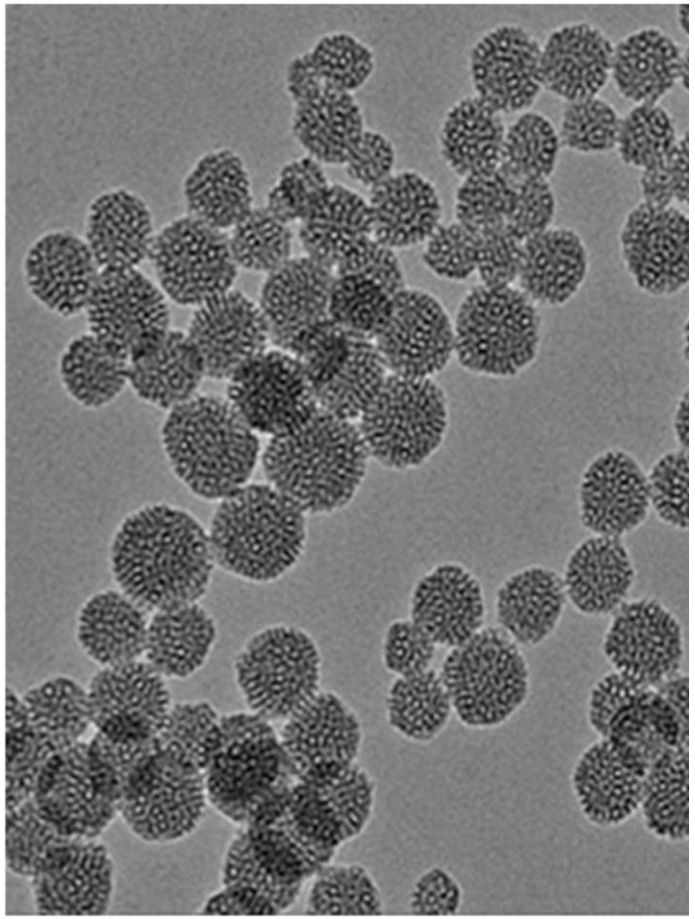

40 Å
gate
200 Å

- Mass/thickness contrast arises from incoherent elastic scattering at the nucleus (Rutherford cross section)
- The interaction between electrons and heavy atoms is stronger than that between electrons and light atoms . If the thickness is homogeneous, areas in which heavy atoms are concentrated appear with darker contrast than such with light atoms (mass contrast). Of course, more electrons are scattered in thick than in thin areas; thus, thick areas appear dark (thickness contrast). However, a thick area with light elements might have the same contrast as a thinner area with heavier atoms.

- The contrast depends upon atomic number Z and sample thickness t.

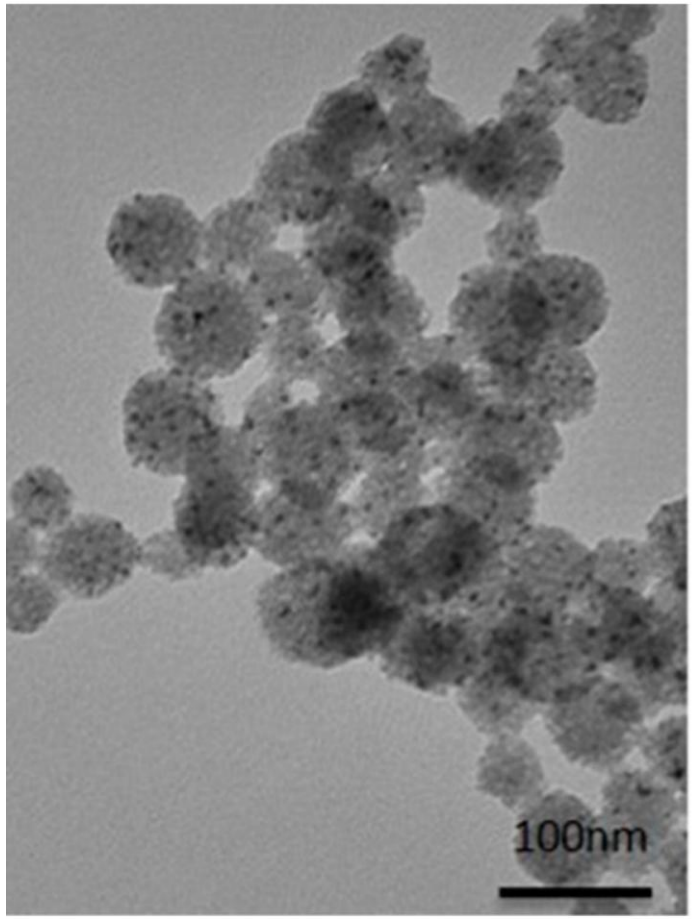
- Amorphous materials, such as polymers and amorphous porous
- Crystalline materials, but with all electron beam participating the formation of the images.
- Biological materials



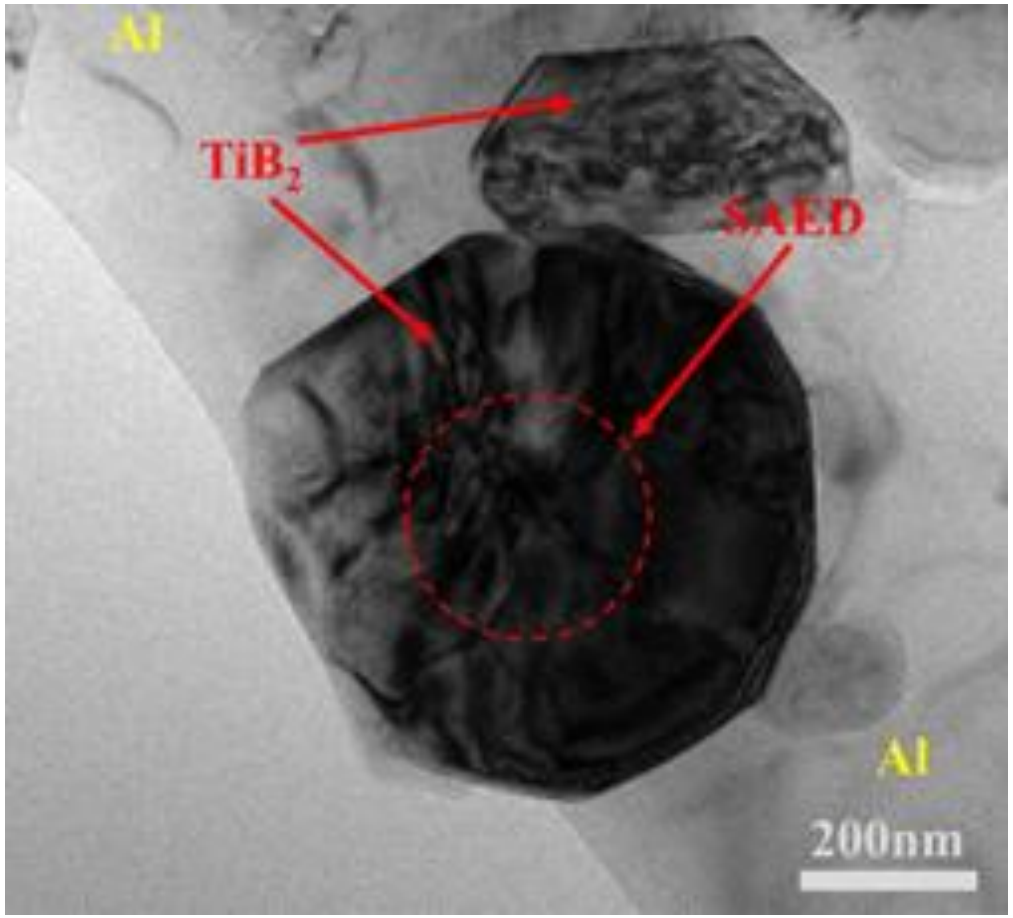


Mesoporous silica

Int. J. Mol. Sci. **20**(7), 1560 (2019).



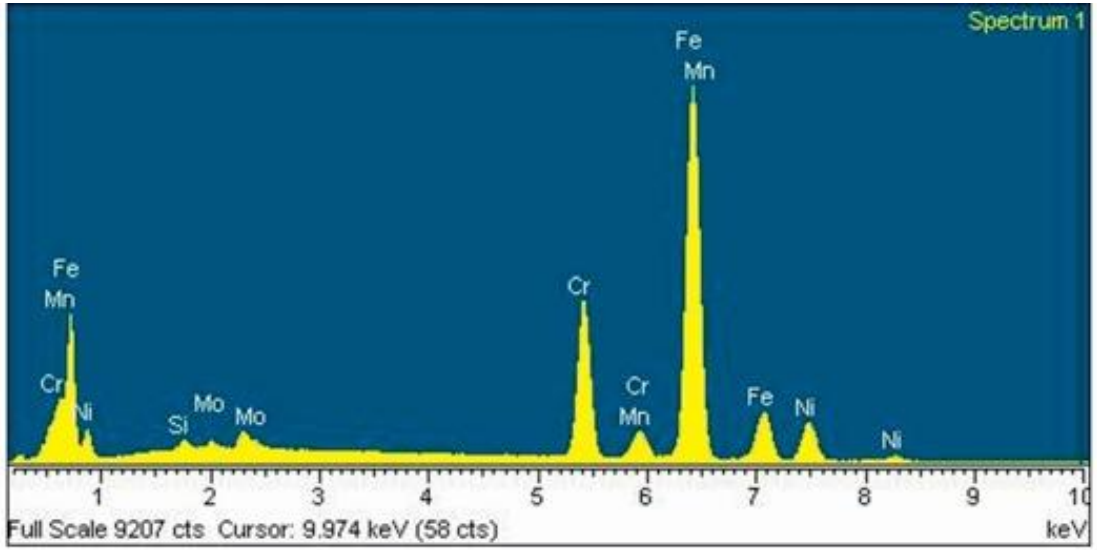
Pt-loaded mesoporous silica



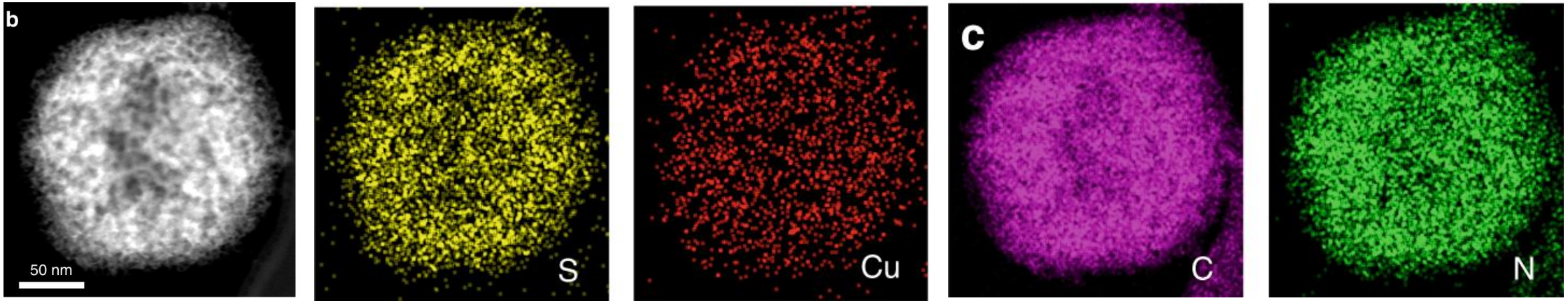
Mater. Res. Express **8** 036514 (2021).

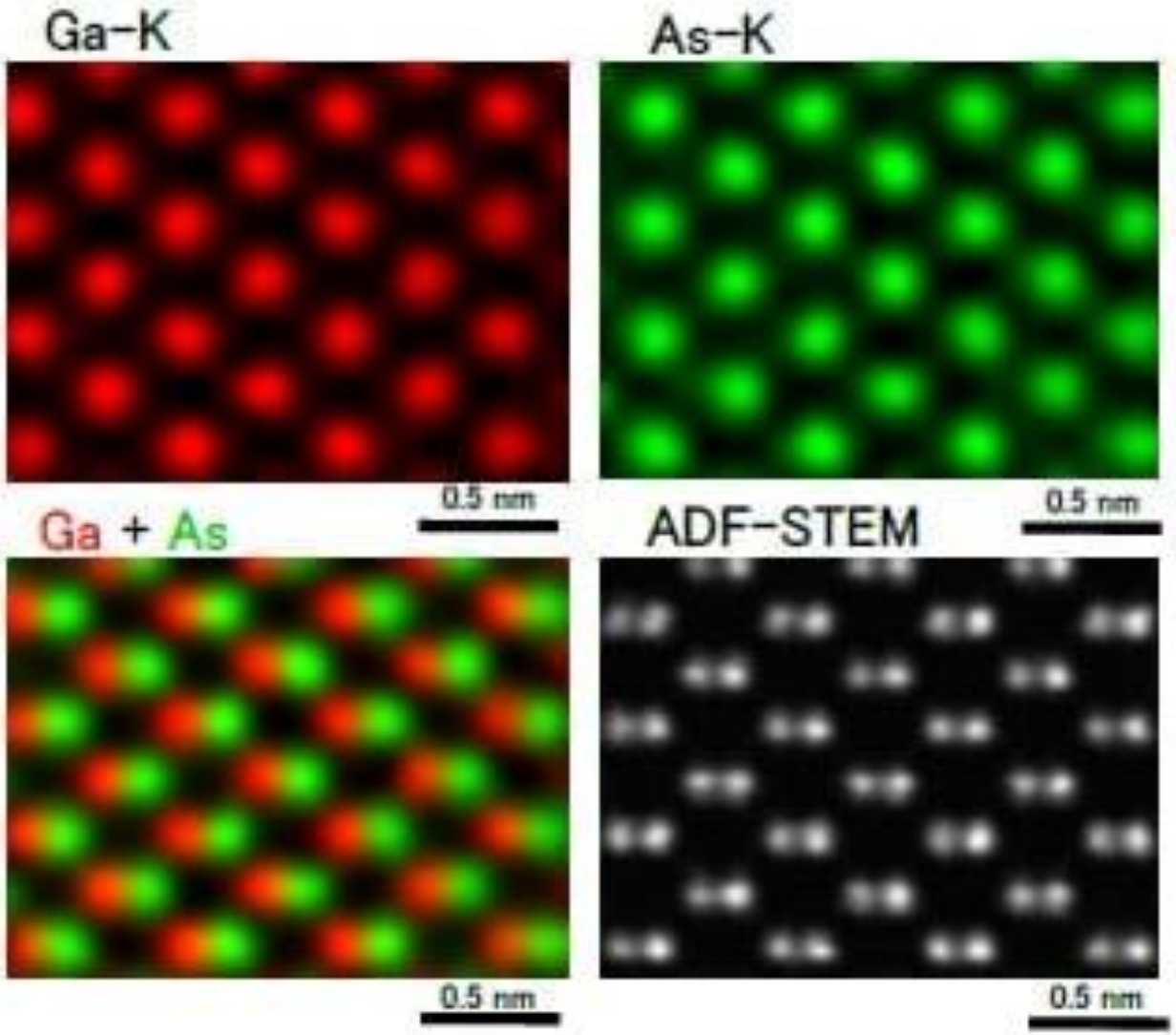
- Inelastic scattered electrons are used to form the basis of spectroscopy inside the TEM.

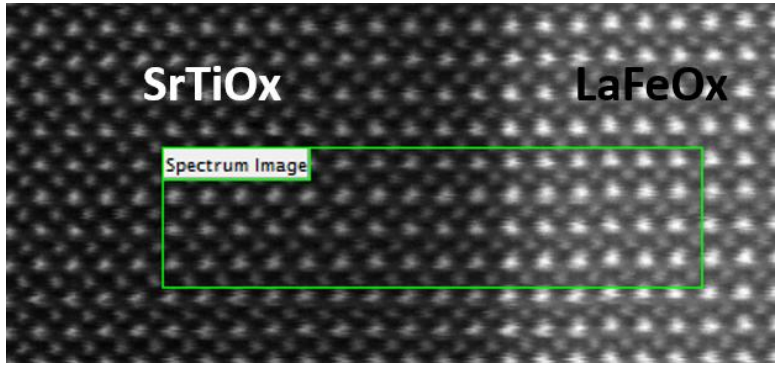
  - X-ray spectroscopy / Energy dispersive spectroscopy

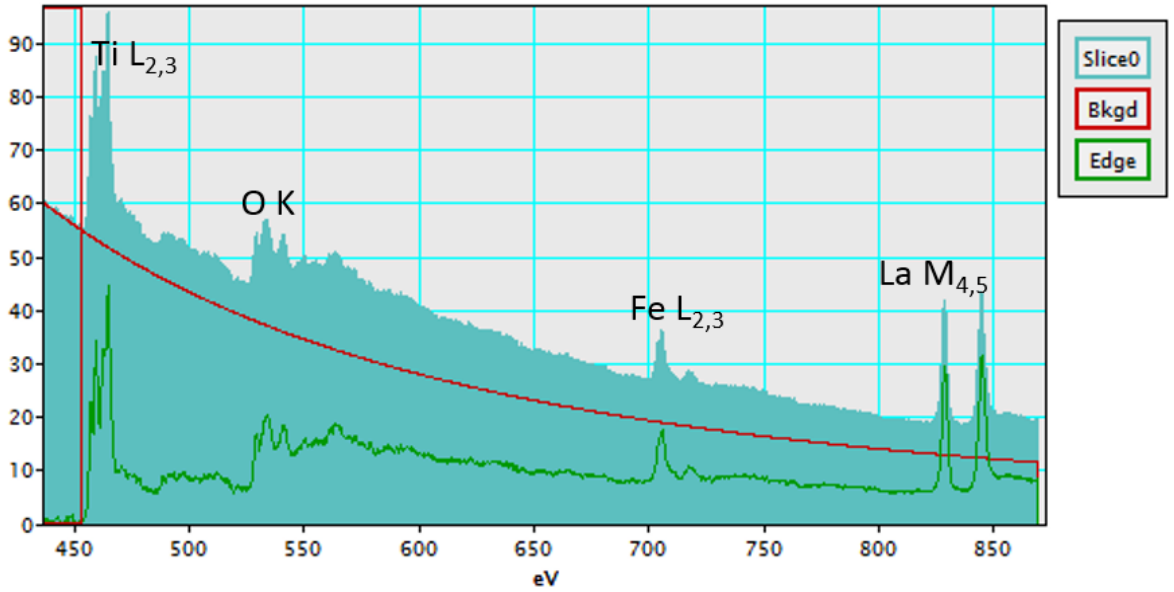


Element	Weight %	Atomic %
Si	0.54	1.08
Cr	17.53	18.80
Mn	1.45	1.47
Fe	66.61	66.50
Ni	11.11	10.55
Mo	2.76	1.60
Totals	100	100









Atomic-level-resolution EDX elemental map of a gallium arsenide monocrystal <011>.

https://www.hitachi-hightech.com/global/en/sinews/topics/0672/

Scanning Transmission Electron Microscope (STEM)

Conventional TEM mode (**T**ransmission **E**lectron **M**icroscope) mode

- Static/fixed beam
- Incident beam is relatively broad, unfocused
- Most relevant subtype: high-resolution (lattice, HRTEM) imaging

STEM mode (**S**canning **T**ransmission **E**lectron **M**icroscope) mode

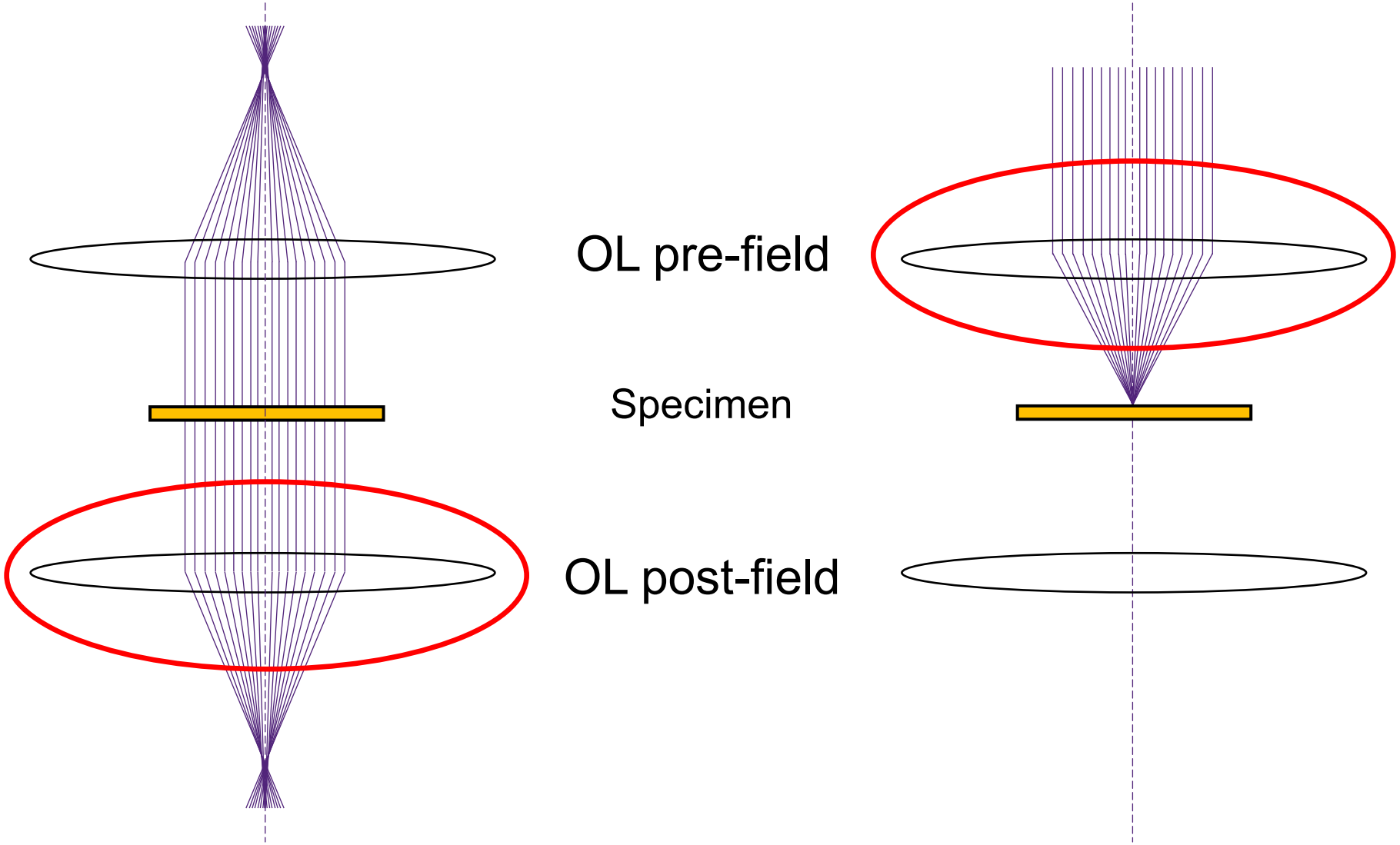
- Rastered/scanning beam
- Incident beam is focused
- Most relevant subtype: HAADF-STEM imaging

S/TEM is instrument that can operate in TEM or STEM mode.

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TEM mode

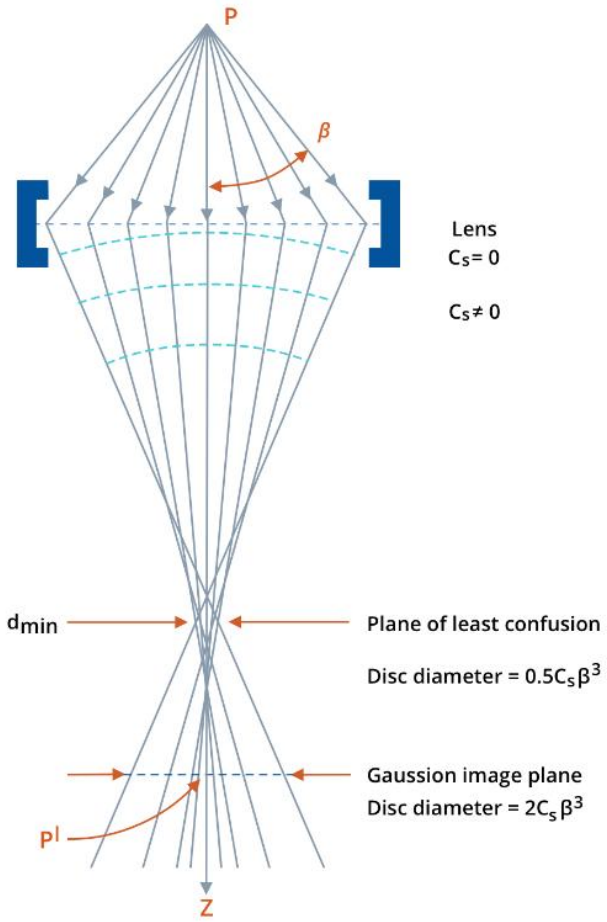
STEM mode



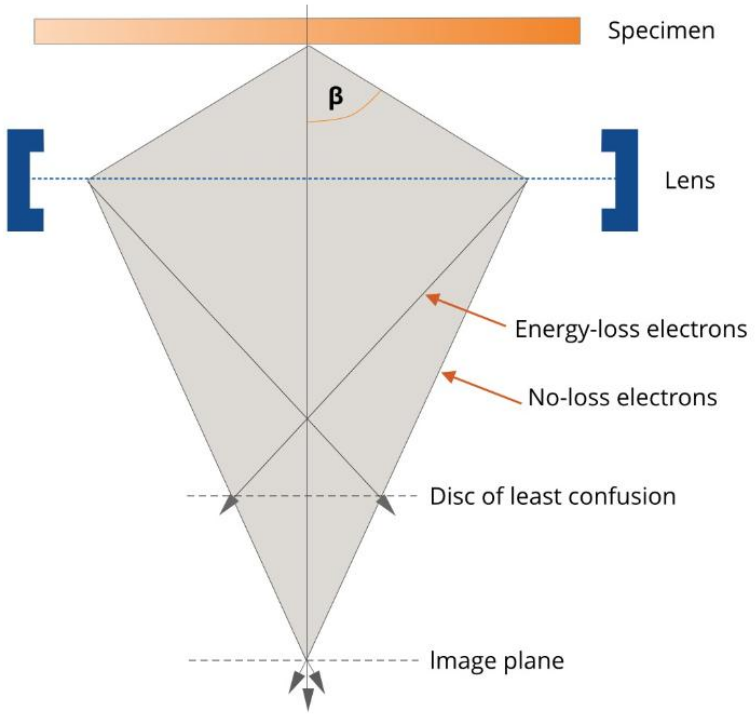
0.2 nm the limit of resolution for typical TEM.

To further improve on the limit of resolution, imperfection in the lenses (aberrations) must be corrected.

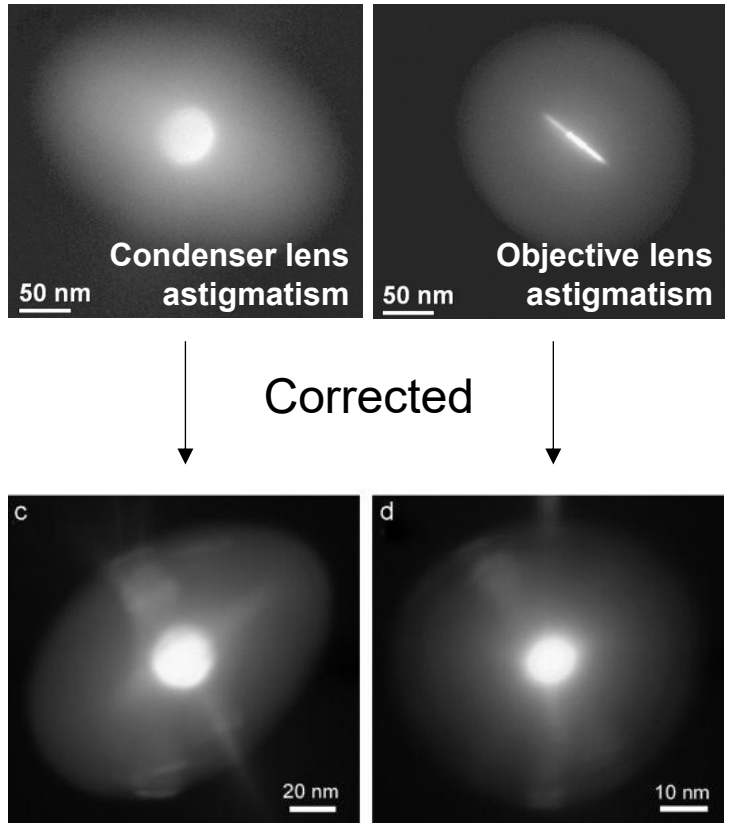
1. **Spherical aberration**



2. **Chromatic aberration**



3. **Astigmatism**

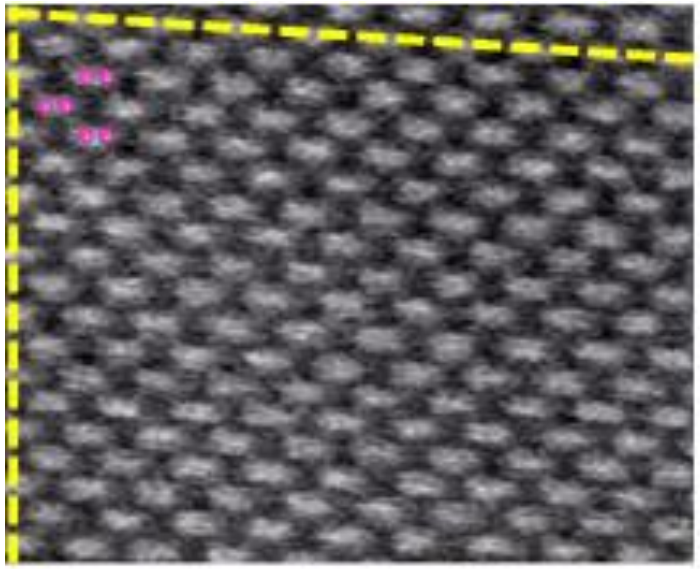


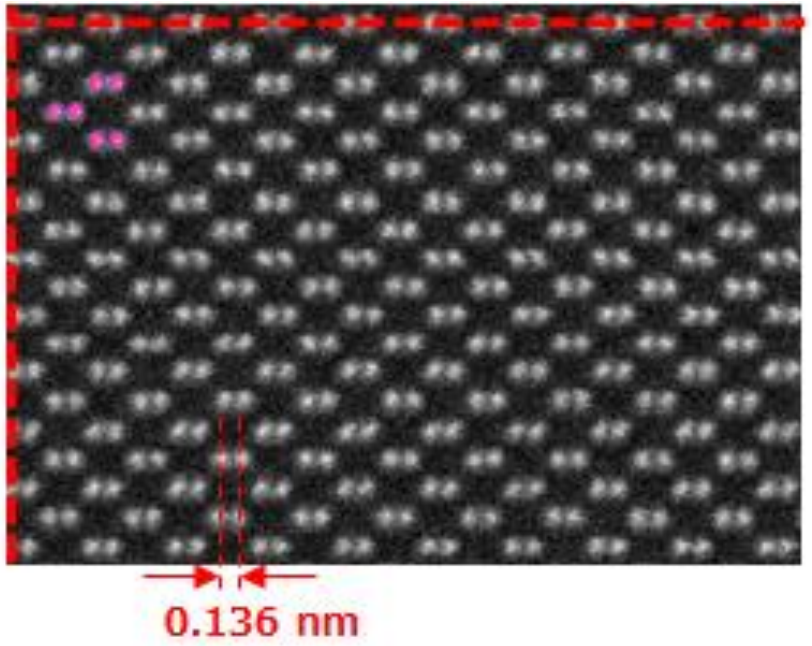
Aberration correction

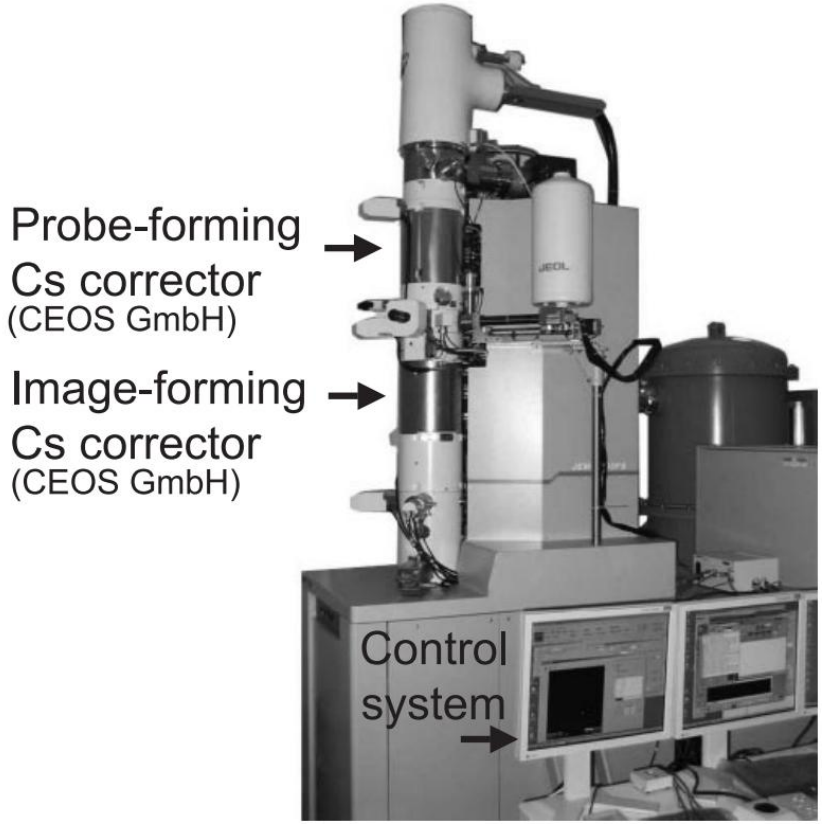
- Spherical aberration
- Chromatic aberration

Aberrations (other than astigmatism) cannot be corrected by TEM operators because it is inherently caused by the imperfection in magnetic lenses.

Therefore, they need to be corrected by additional hardware.

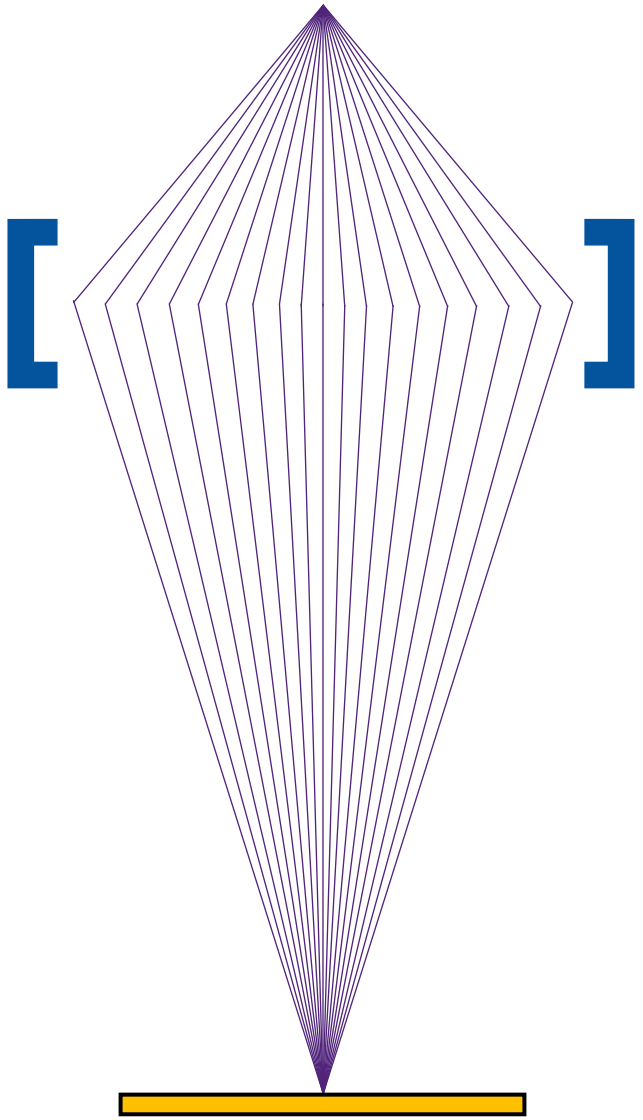




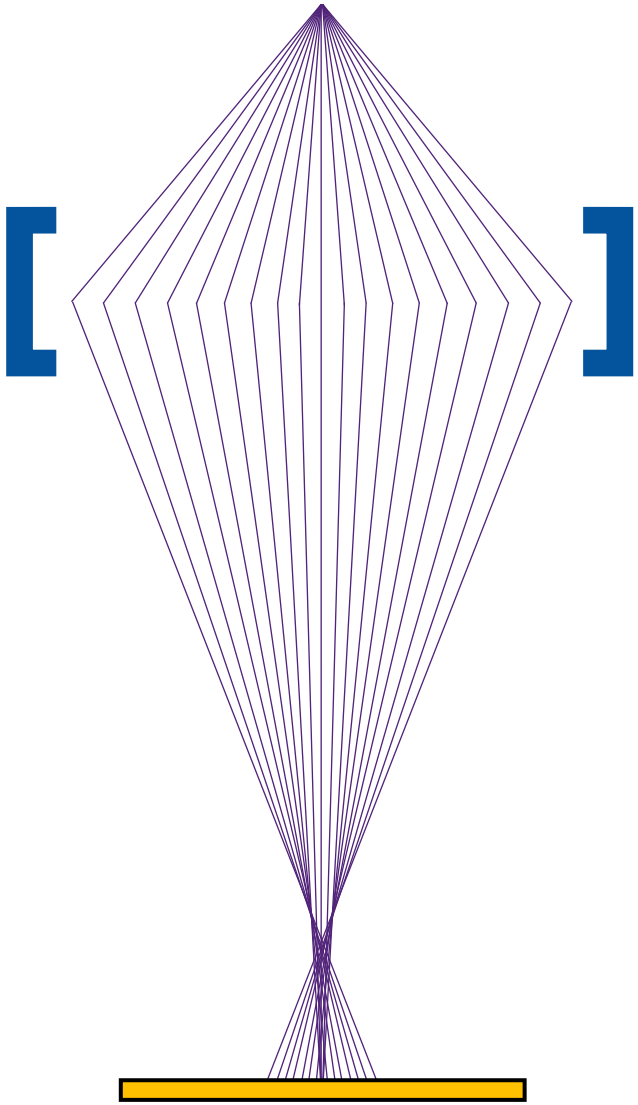


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Expectation



Reality



HRTEM (lattice) imaging resolution:

$$\delta_{HRTEM} = 0.64(C_s\lambda^3)^{1/4}$$

STEM imaging resolution:

$$\delta_{STEM} = 0.43(C_s\lambda^3)^{1/4}$$

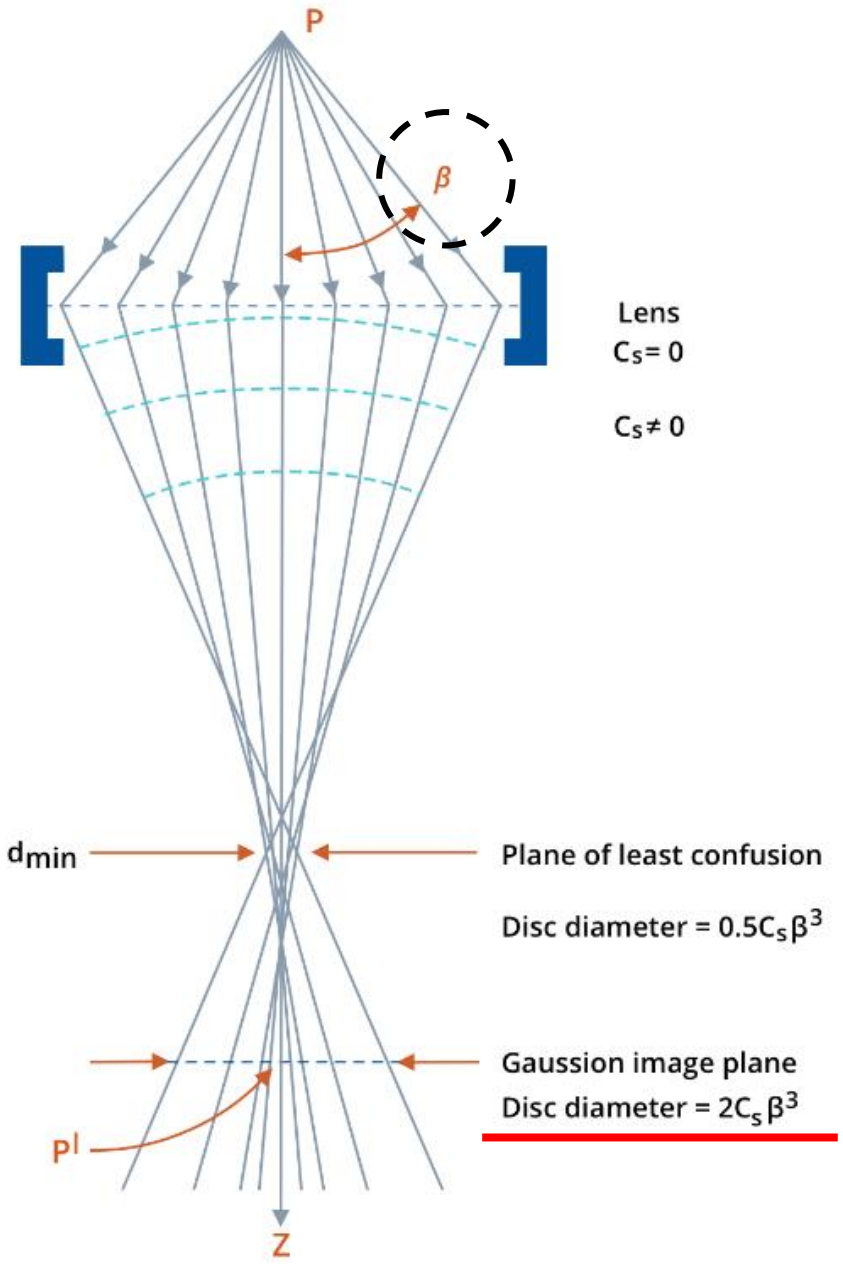
To get high resolution image, δ value needs to be made smaller.
Based on the equations, lower δ can be obtained by

1. *Decreasing C_s.*
2. *Decreasing λ.*

In practice, there is a limit on λ to avoid beam damage.

- λ = 1.97 pm (300 kV) for physical sciences (roughly).

→ Remaining option: Make C_s *smaller*.



The spherical aberration (C_s) is caused by the imperfection of objective lens.

Electron rays entering at large angles or further from optic axis ***bent excessively*** upon exiting.

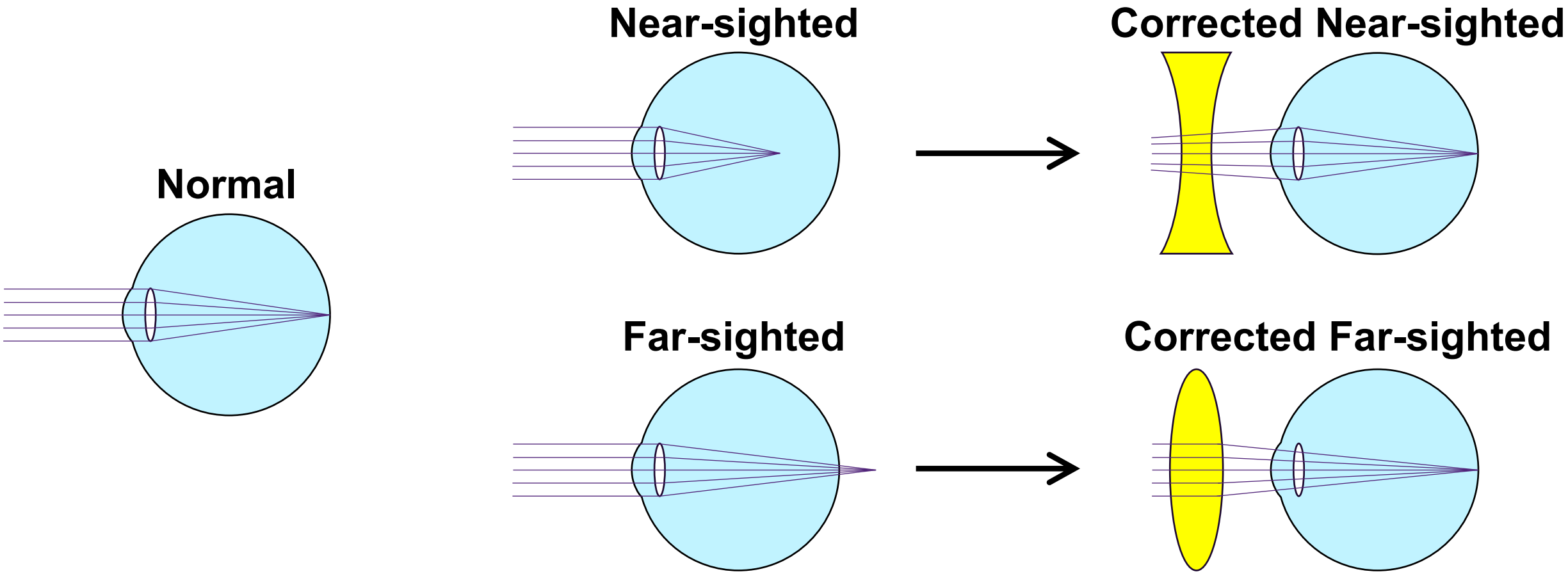
With C_s, off-axis rays are inhomogeneously generated.

The radius of the spherical aberration disk (r_{sph}) in the image plane can be calculated by the below equation.

$$r_{sph} = C_s \beta^3$$

C_s = the spherical aberration coefficient (a constant for a particular lens, units of mm).
β = the maximum semi-angle of collection of the lens.

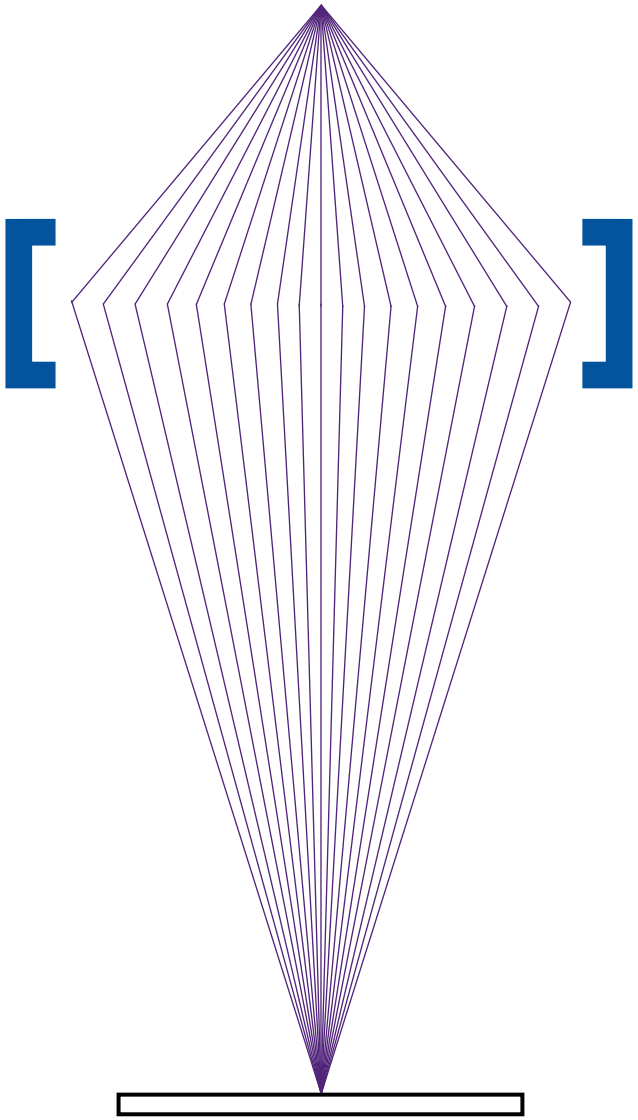
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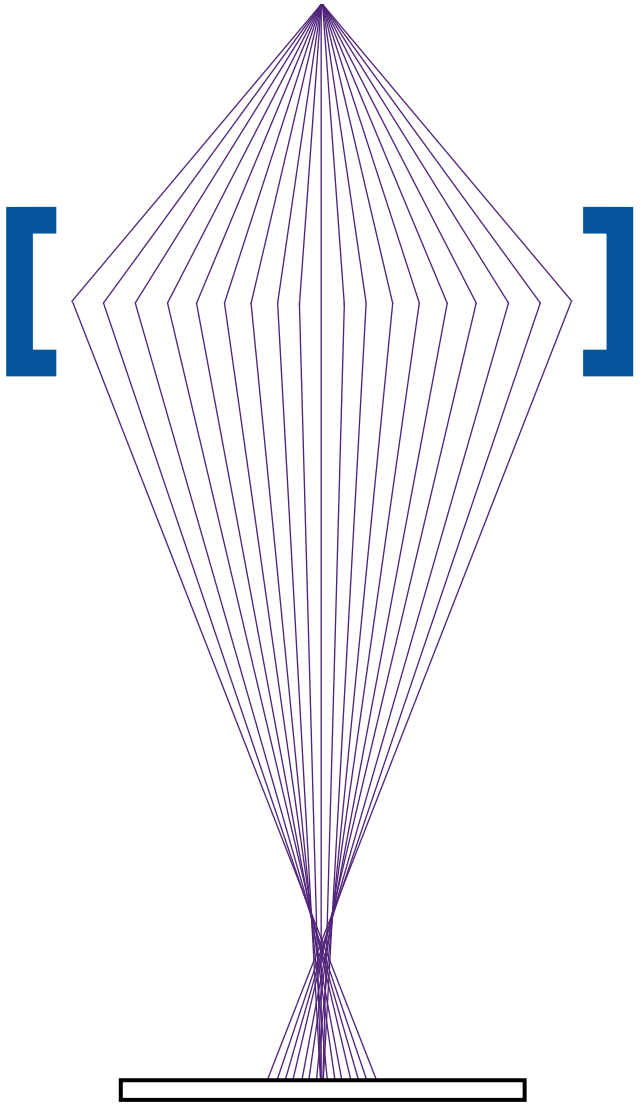
Glass lenses do not make you correct your eye but alters the path of light rays to get to the right focal plane.

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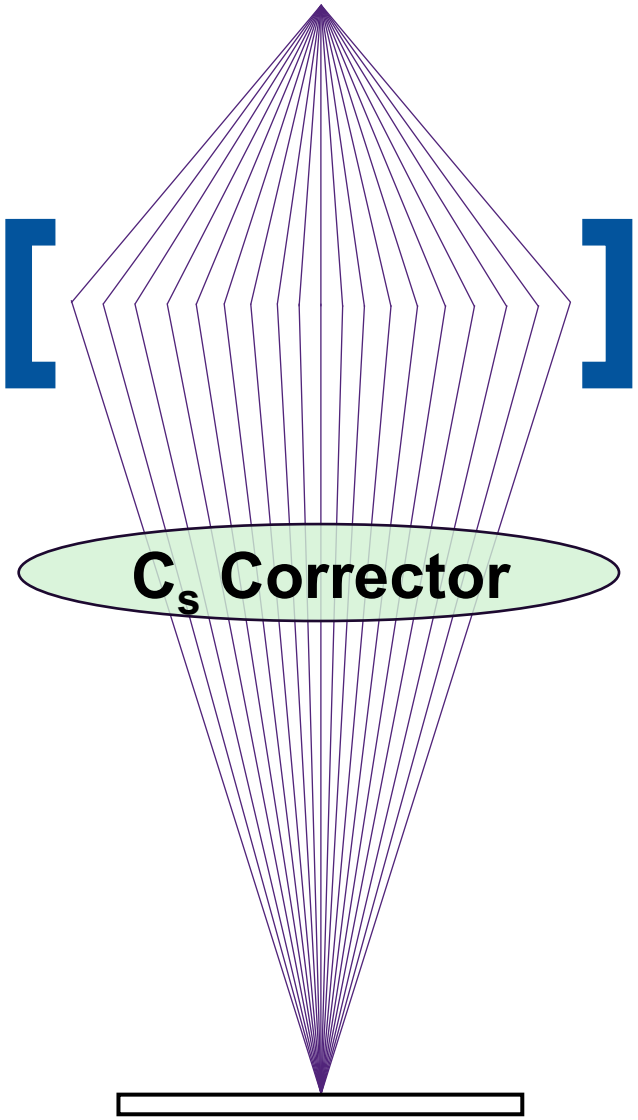
Expectation



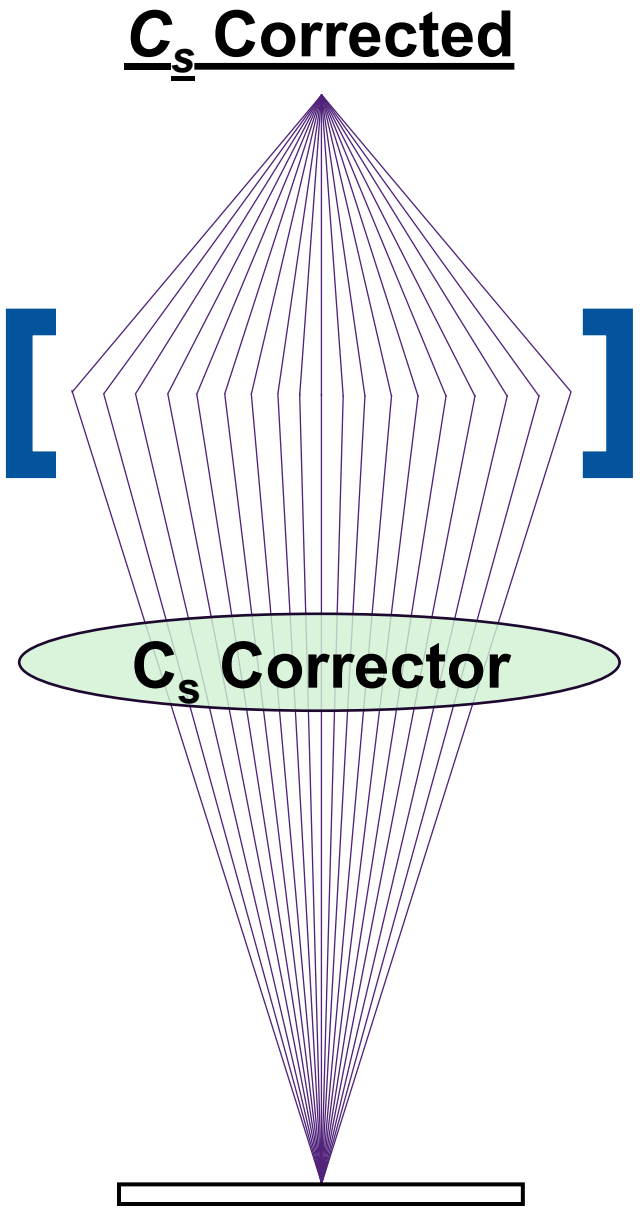
Reality



C_s Corrected

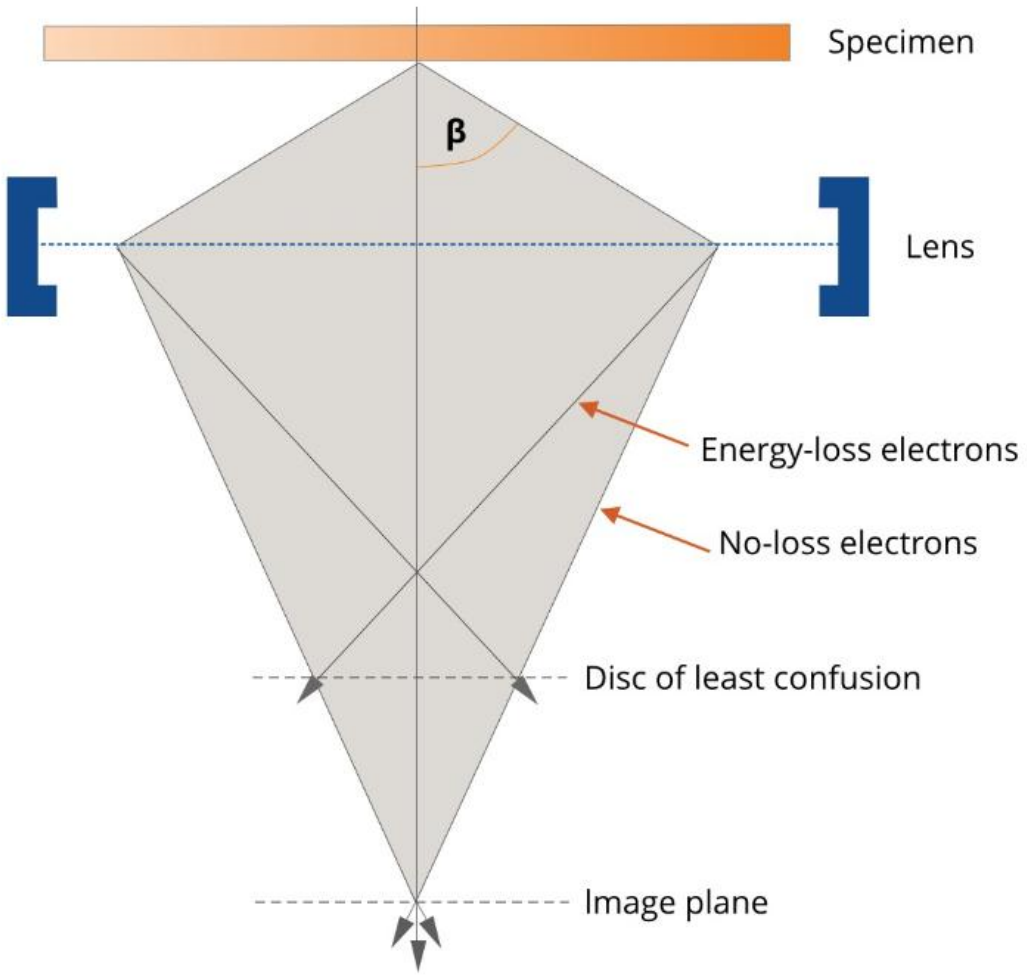


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- Imaging
  - ➢ Ability of resolve ≤ 0.1 nm features at 30 – 300 kV
    - What is currently commercially available:
      - ➢ < 0.06 nm (300 kV)
      - ➢ ~ 0.1 nm (30 kV)

- Analytical
  - ➢ < 0.1 nm probes with high current (up to ~1 nA)
    Ability of resolve ≤ 0.1 nm features at 30 – 300 kV
    - Fast(er) atomic-resolution chemical analysis
      - ➢ Energy dispersive spectroscopy (EDS)
        - ○ Elemental mapping
      - ➢ Electron energy loss spectroscopy (EELS)
        - ○ Elemental mapping
        - ○ Oxidation state mapping

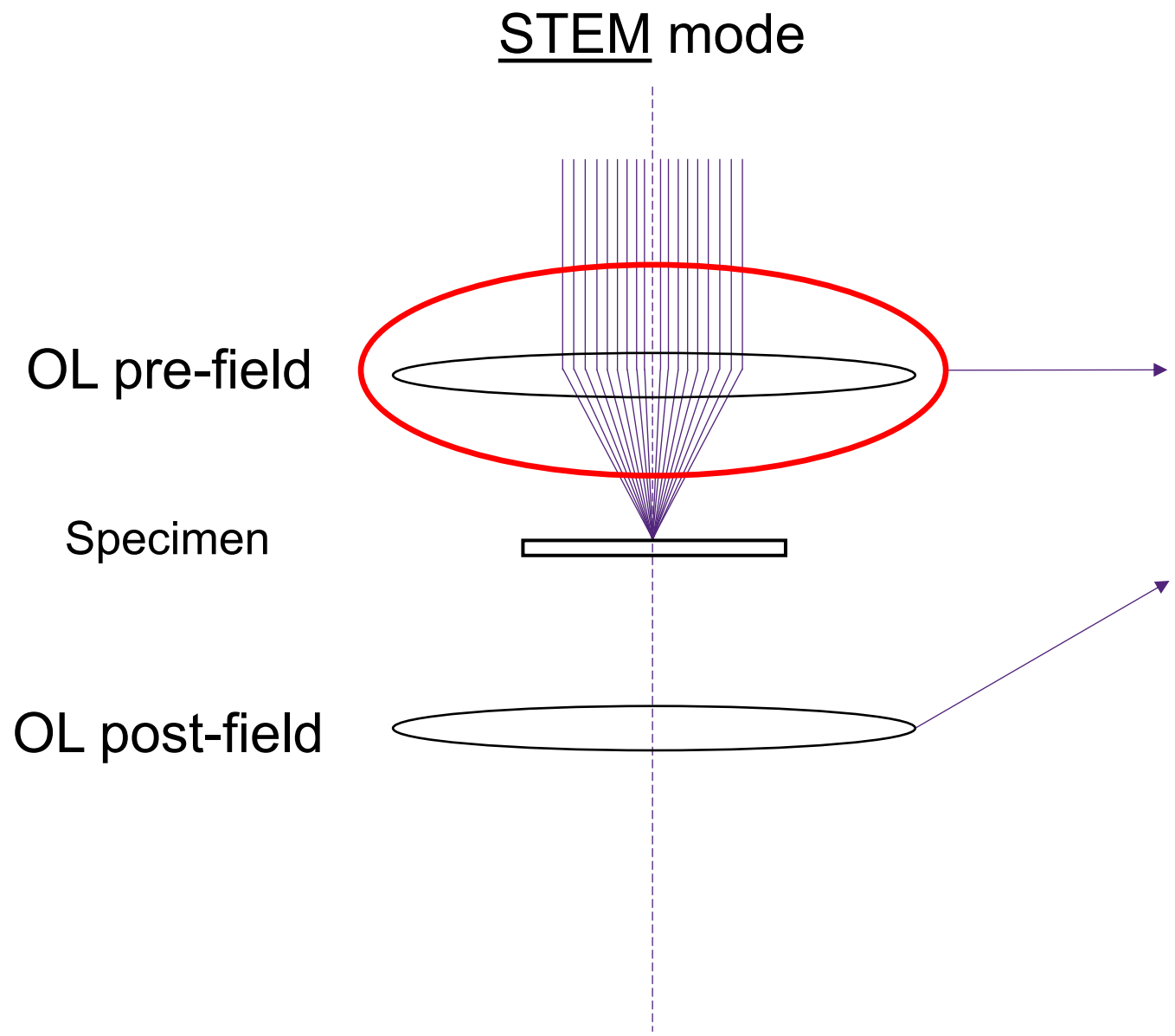


The term chromatic aberration is related to the energy of the electrons.

Electrons are not monochromatic. Electrons emerge from the gun at a whole range of energies and are bent by the objective lens to different degrees (electrons that have lost energy are bent more strongly). Thus, once again, electrons from a point on the specimen form a disk image, as for spherical aberration. The radius (r_{chr}) of the disk is given by:

$$r_{chr} = C_c \frac{\Delta E}{E_0} \beta$$

C_c = the chromatic aberration coefficient of the lens (length).
ΔE = the energy loss of the electron.
E_o = the initial beam energy.
β = the maximum semi-angle of collection of the lens.

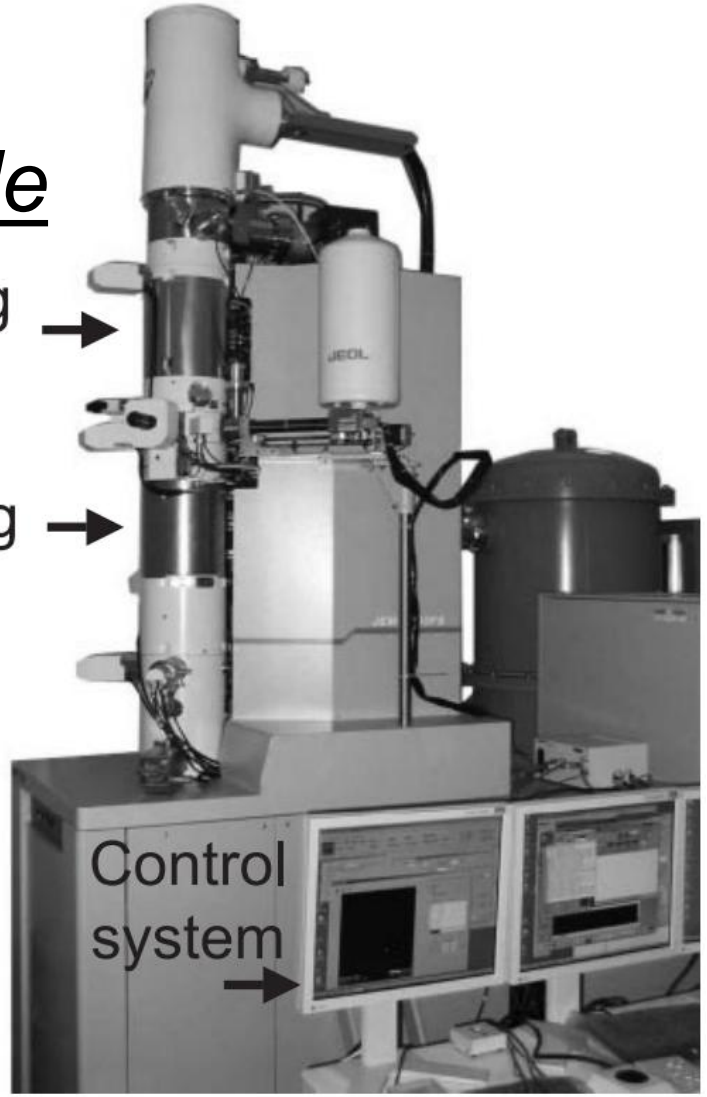


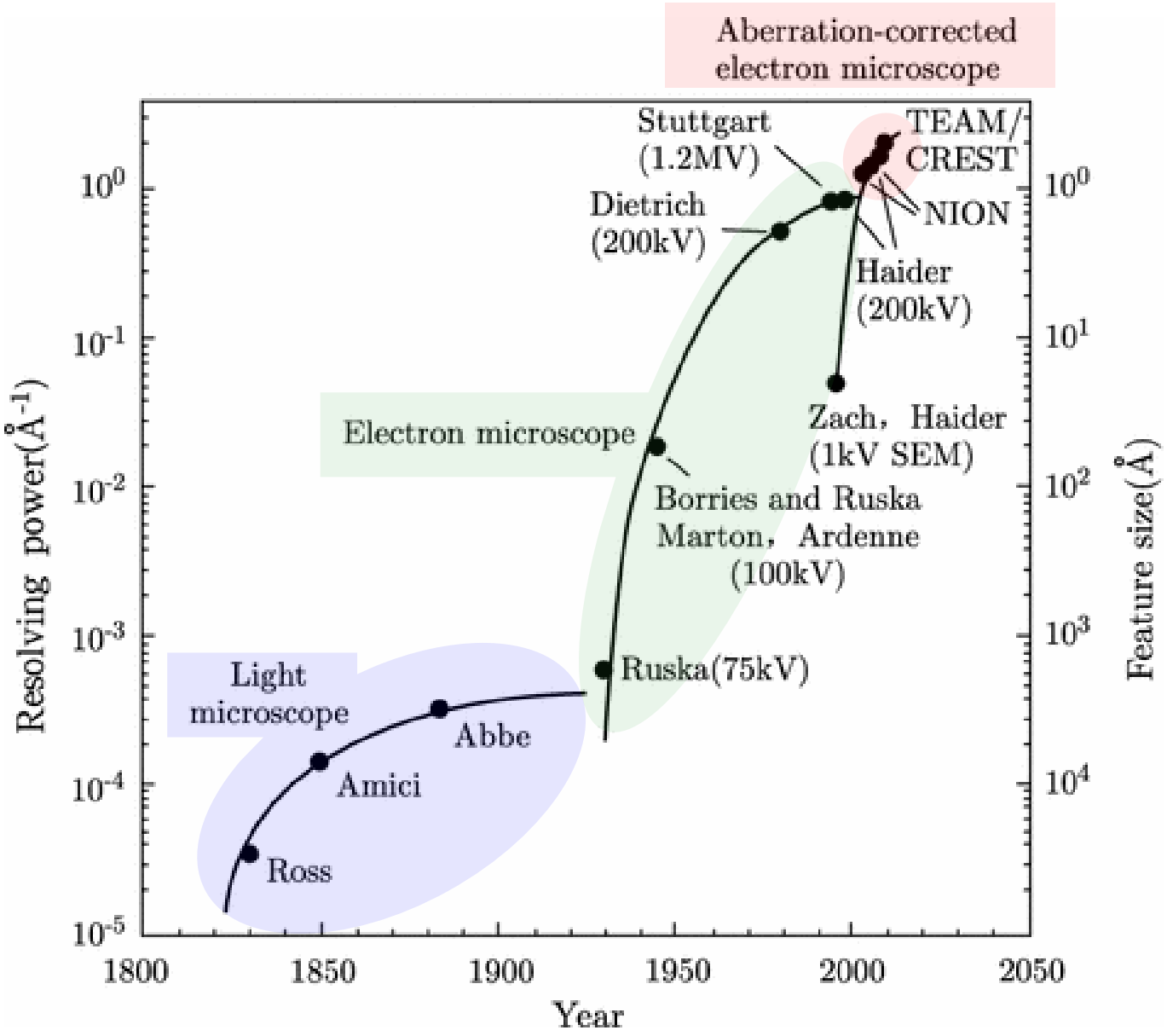
STEM mode

Probe-forming
Cs corrector
(CEOS GmbH)

Image-forming
Cs corrector
(CEOS GmbH)

TEM mode





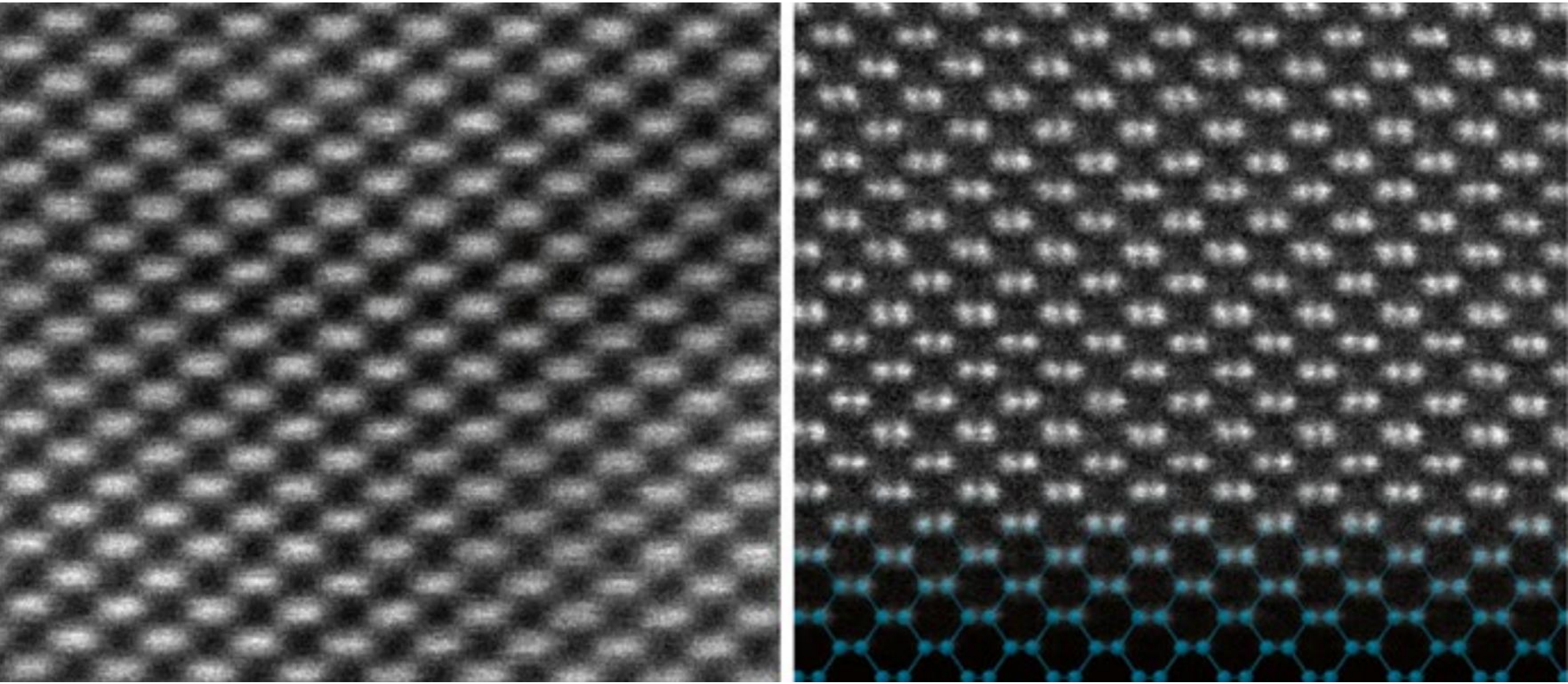
With aberration correction, the limit of resolution of 0.05 nm can be achieved.

MATE 6301 | Cs-corrected STEM image

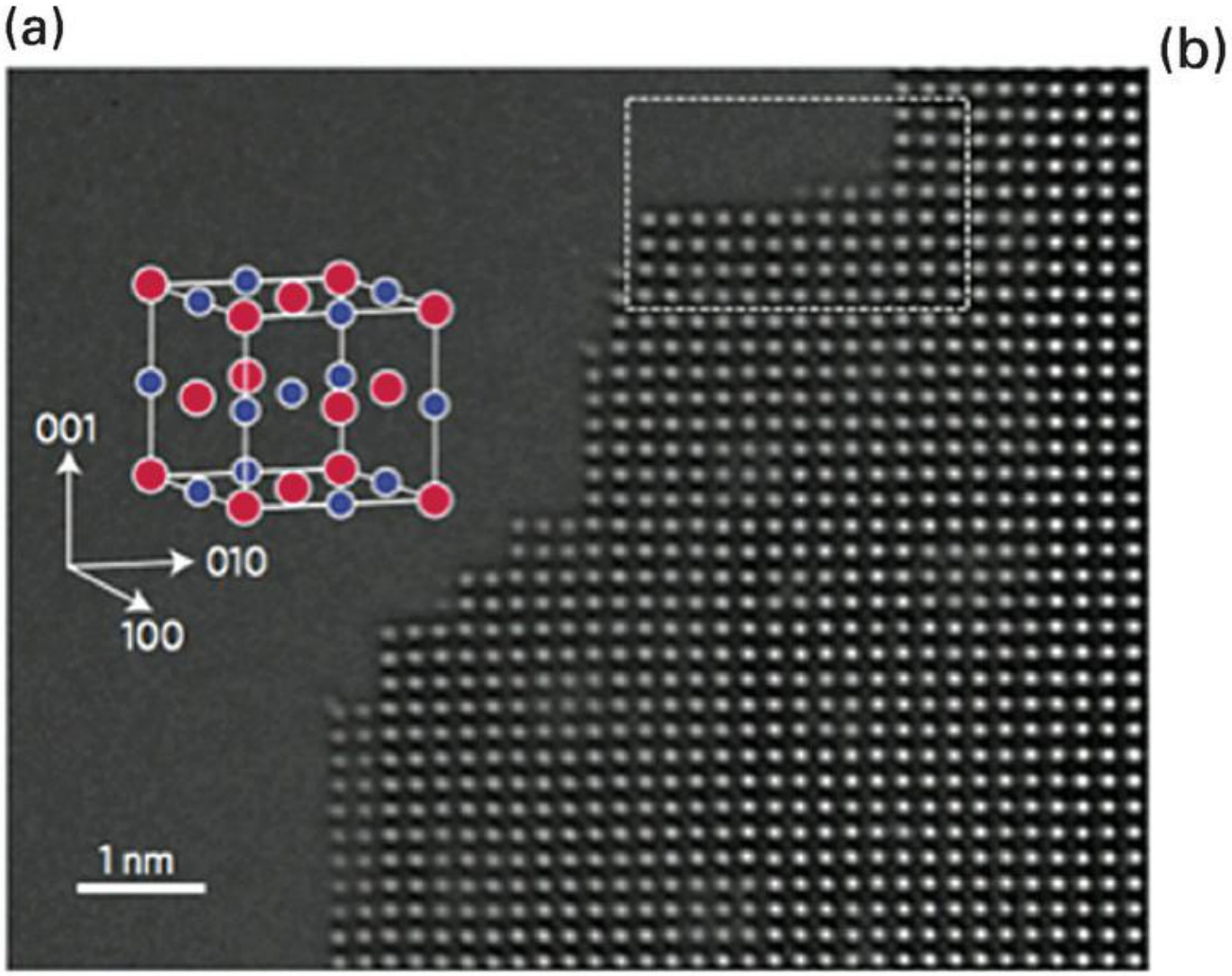
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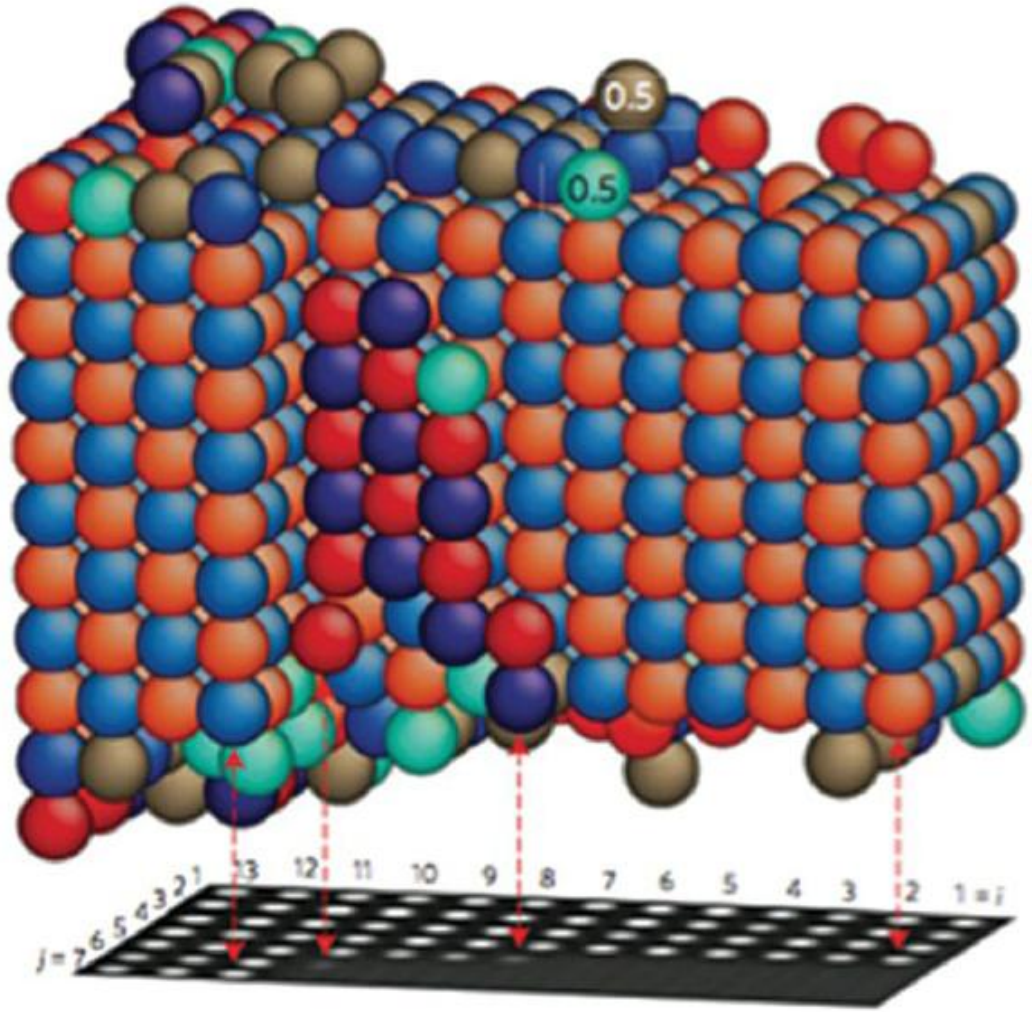
Uncorrected

C_s-corrected

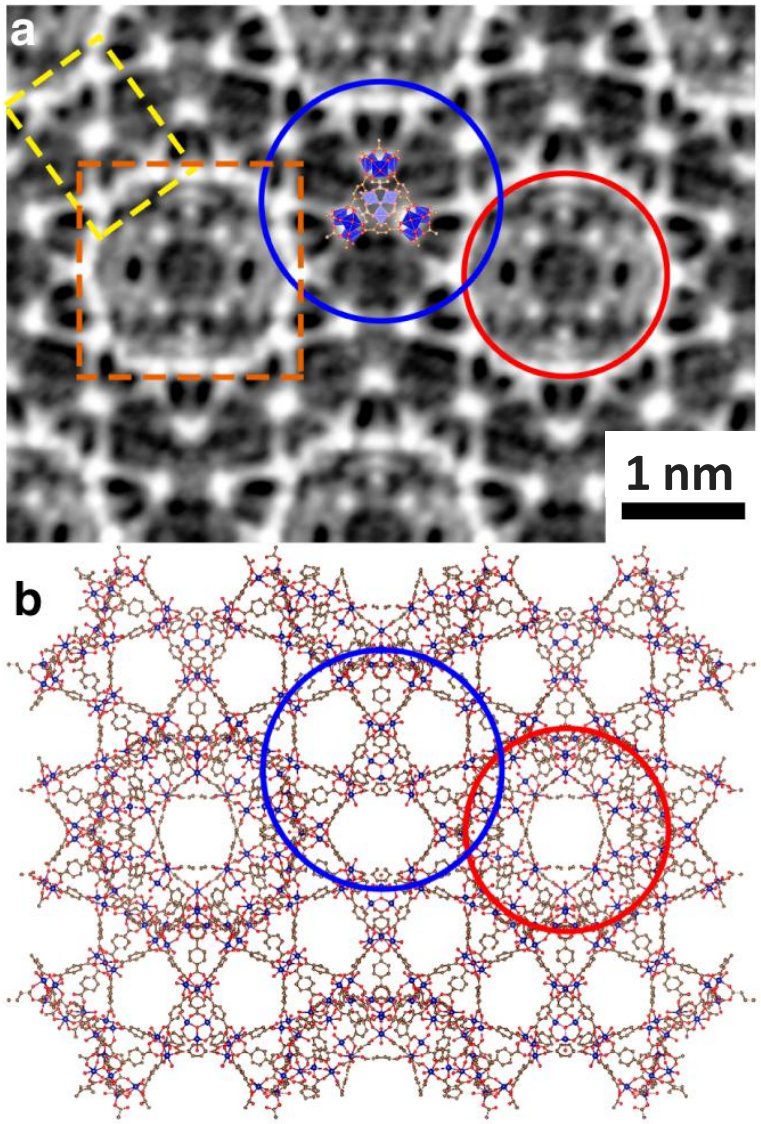


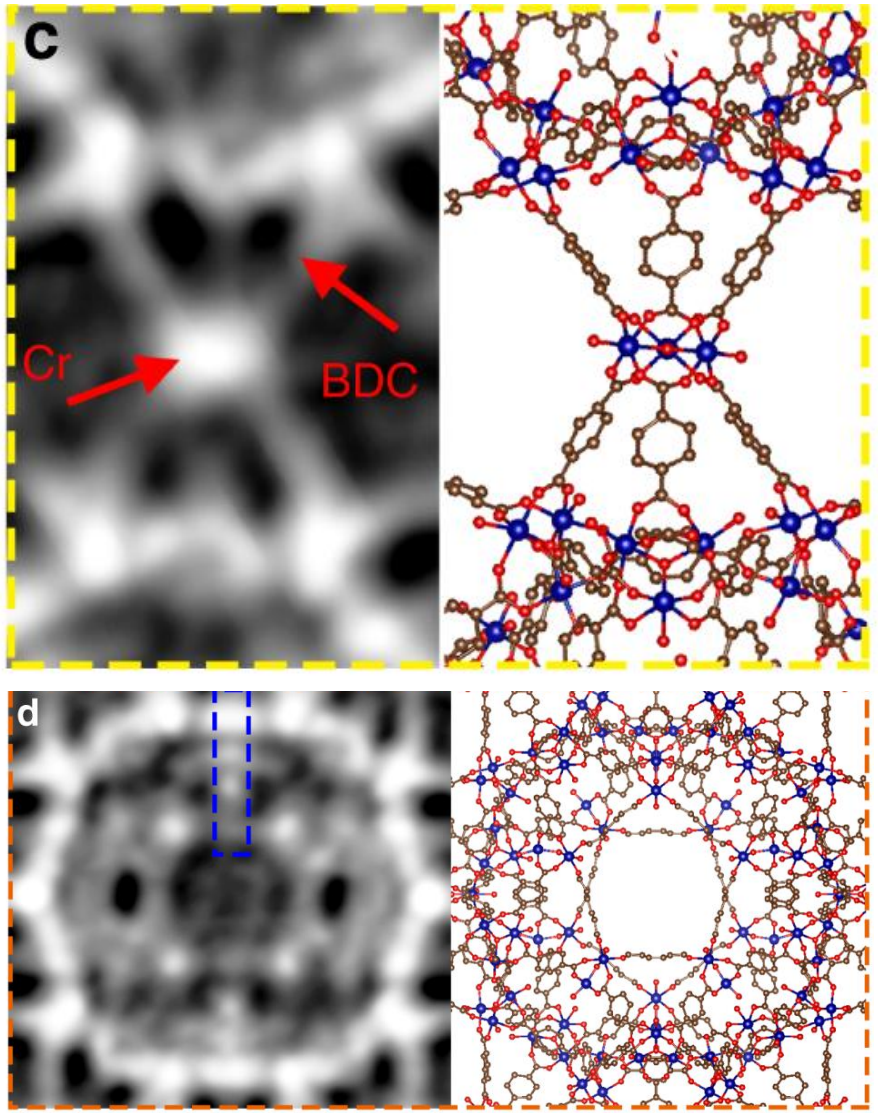
It’s like swtiching from standard definition to 4k UHD TV!


(a)
001
010
100
1 nm

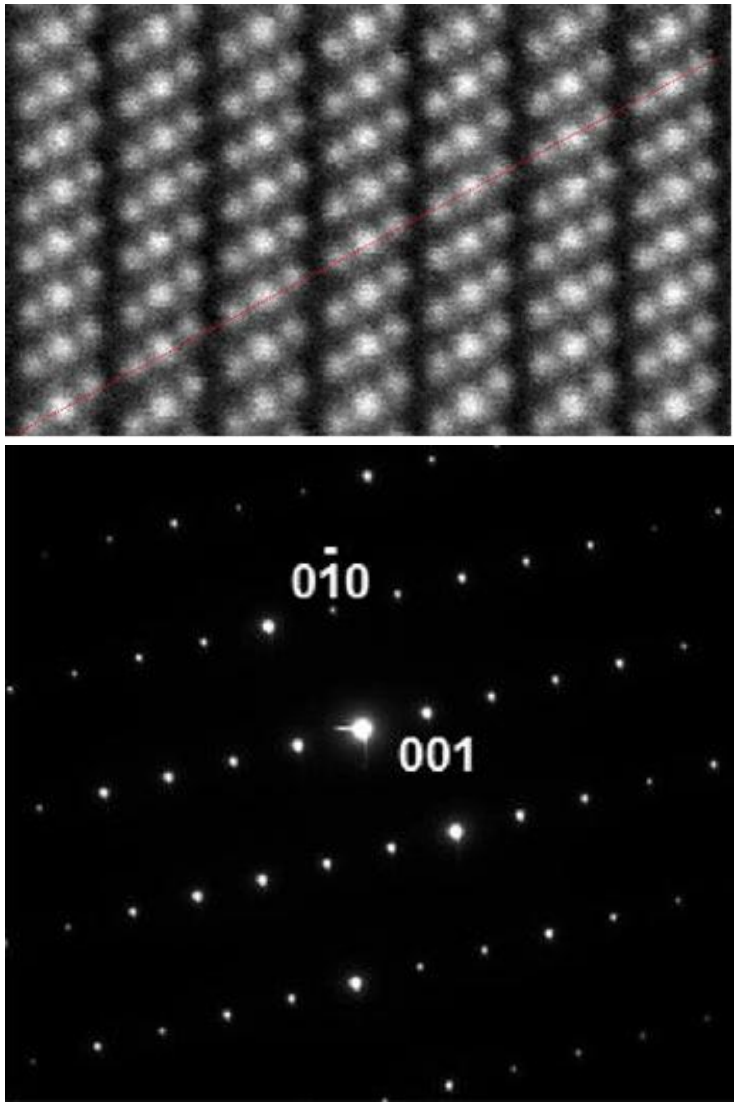

(b)
0.5
0.5

MATE 6301 | Cs-corrected STEM image





Nat Commun **11**, 2692 (2020).



CrystEngComm **16**, 8678-8683 (2014).

- Microstructural analysis
- Defects identification
- Interfacial analysis
- Crystal structure
- Seeing atomic arrangement of crystals
- Elemental analysis at nanometre scale

- **Advantages**
  - ✓ Superior resolution
  - ✓ Ability to obtain compositional and crystallographic information from specimens with high spatial resolution

- **Limitations**
  - ➢ Very small volume of material studied – is it representative?
  - ➢ Limitations on samples
  - ➢ Sample preparation can be quite lengthy and involved
  - ➢ Electrons can damage sample and introduce artifacts

- **Structure of TEM**
  - Principles of TEM contrasts
  - TEM imaging modes include:
  - Diffraction contrast (bright field/dark field)
  - Phase contrast
  - Mass/thickness contrast

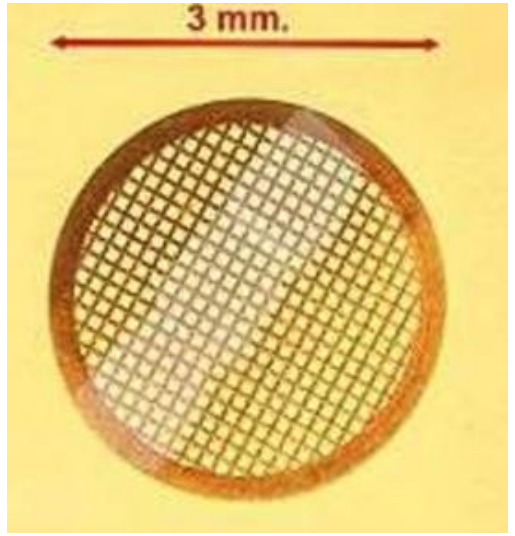
- **Compositional information obtained through:**
  - Characteristic X-rays
  - STEM

Sample Preparation for TEM

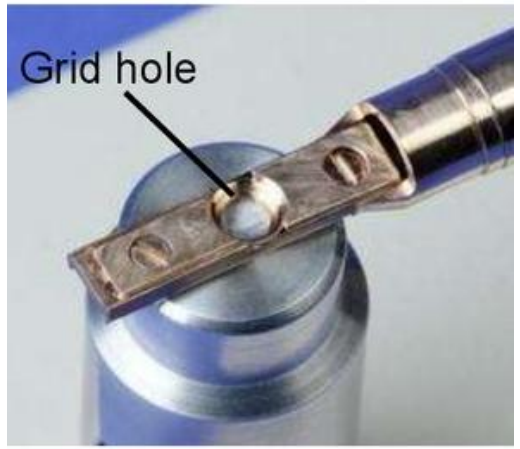
Sample Preparation

Powdered samples/Nanoparticles:

- Small amount of powdered samples or in the case of a thin film sample, we scrape the film deposited on the substrate and transfer it to a volatile solution (*i.e.*, isopropyl alcohol, ethanol).
- Then the solution containing the nanoparticle is allowed to sonicate for 10 minutes (this short duration is just to maintain the pore structure without being collapsed and to avoid heating of samples).
- This solution is then drop-cast on the carbon-coated side of the TEM grid, and the solvent is allowed to evaporate completely before the analysis.

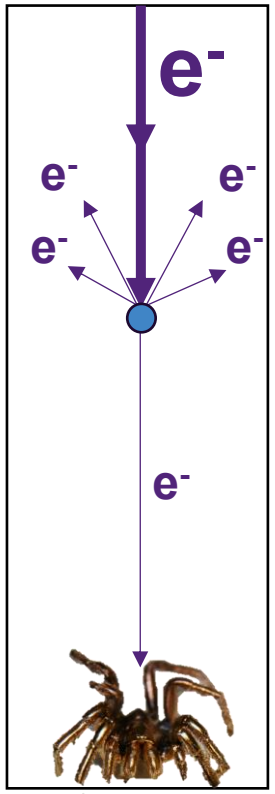


[4]

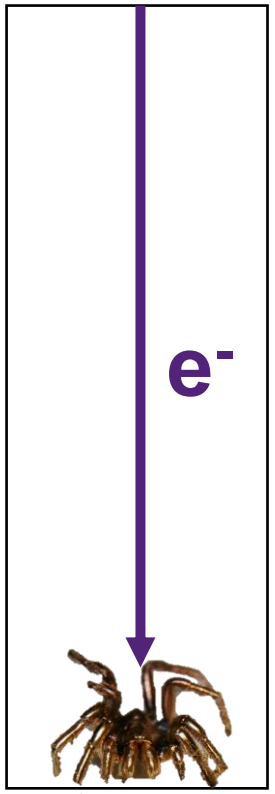


[5]

- An electron beam wouldn't have a very long mean free path in air. This is why the TEM is under vacuum, so the electrons can make it from the source to the sample.



In air



In vacuum

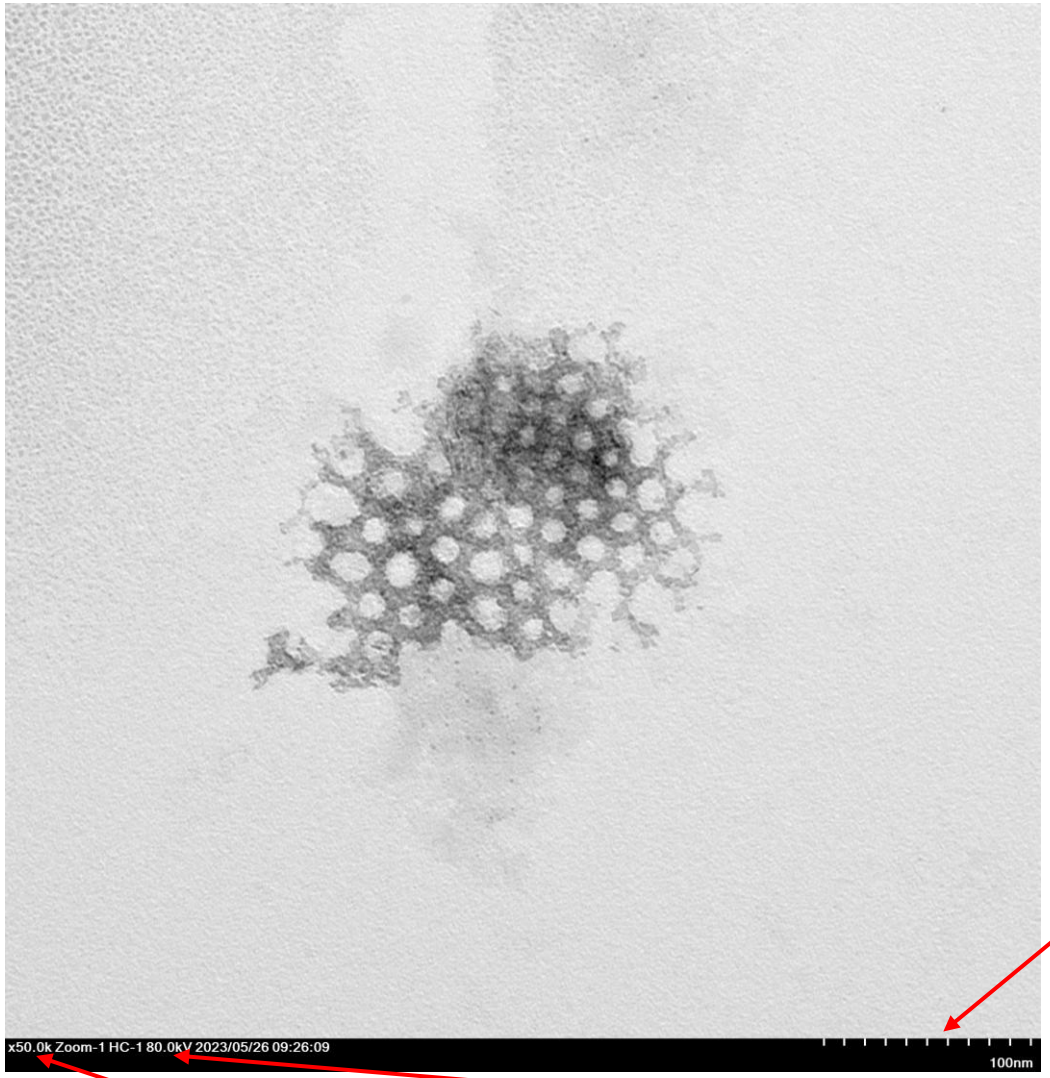


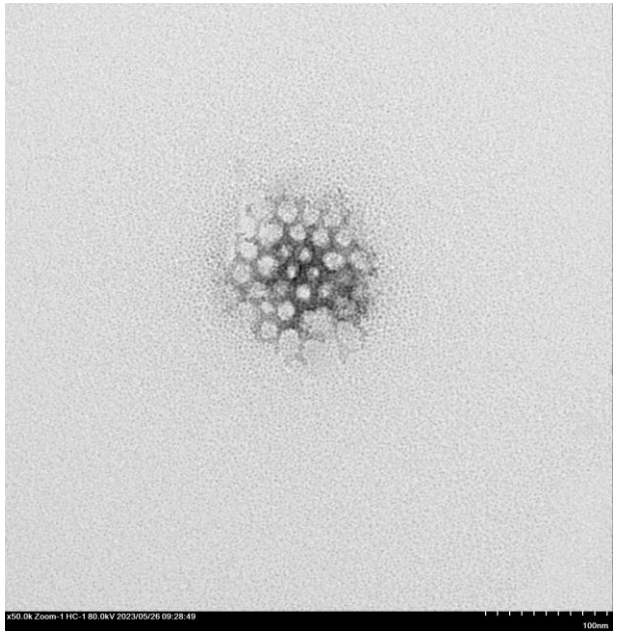
Powder-free gloves

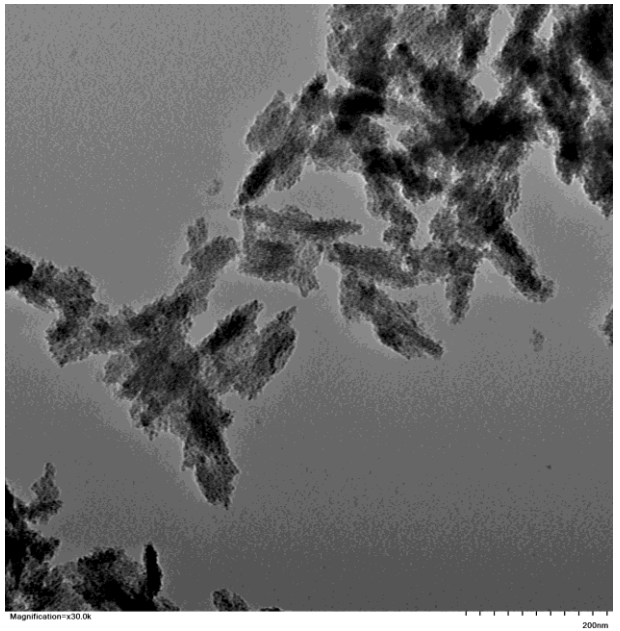




Results:


x50.0k Zoom-1 HC-1 80.0kV 2023/05/26 09:26:09
100nm
Scale bar
Magnification
Accelerating Voltage


x50.0k Zoom-1 HC-1 80.0kV 2023/05/26 09:28:49
100nm


Magnification=x30.0k
200nm

Key-Points

The University Of Queensland AUSTRALIA

1. Sample preparation
   - Conductive sample
   - Non-conductive sample

2. Beam alignment
   - Focal plane (or working distance) adjustment
   - Column (aperture & lens) alignment
   - Astigmatism correction

3. Aberration correction
   - Spherical aberration
   - Chromatic aberration

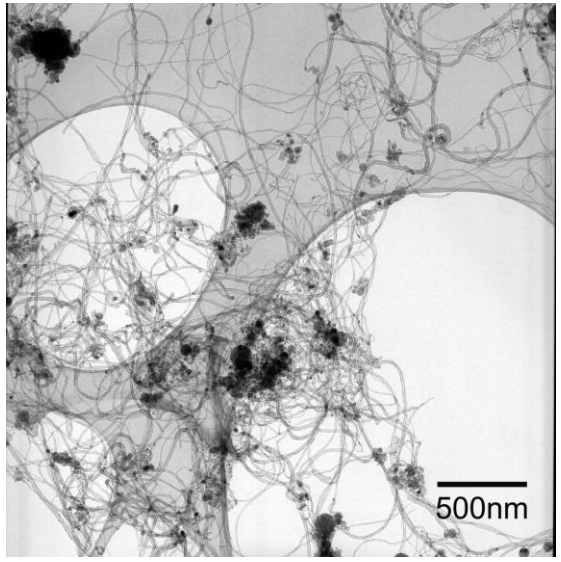
4. Imaging parameters
   - Accelerating voltage
   - Beam current
   - Working distance
   - Aperture

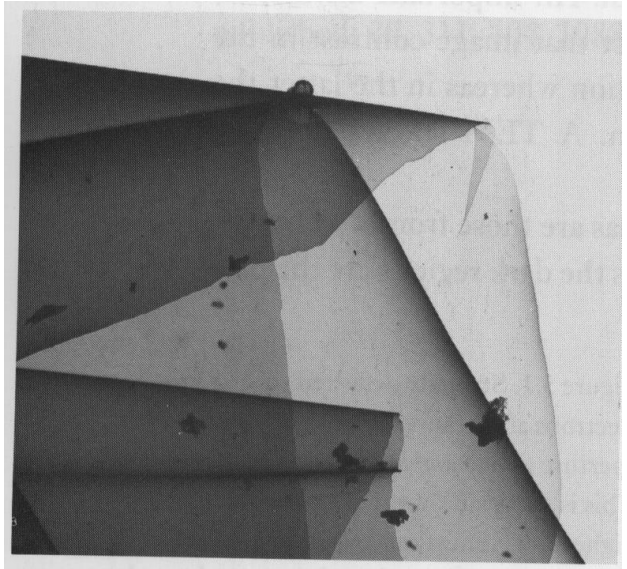
Non-controllable by Operator

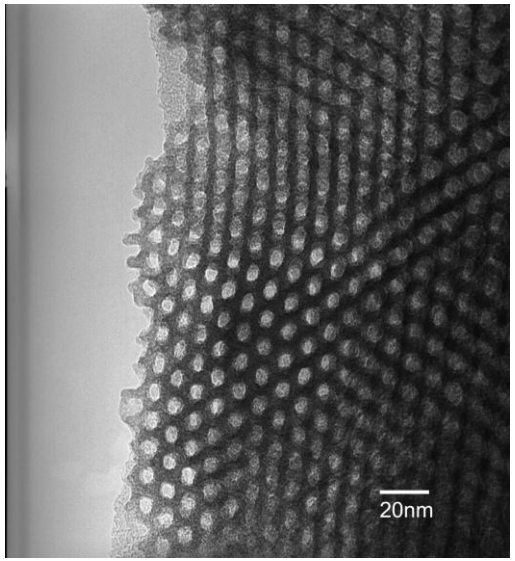
Controllable by Operator

Mass/thickness contrast

- Mass/thickness contrast arises from incoherent elastic scattering at the nucleus (Rutherford cross section).
- The interaction between electrons and heavy atoms is stronger than that between electrons and light atoms . If the thickness is homogeneous, areas in which heavy atoms are concentrated appear with darker contrast than such with light atoms (mass contrast). Of course, more electrons are scattered in thick than in thin areas; thus, thick areas appear dark (thickness contrast). However, a thick area with light elements might have the same contrast as a thinner area with heavier atoms.
- The contrast depends upon atomic number Z and sample thickness t.



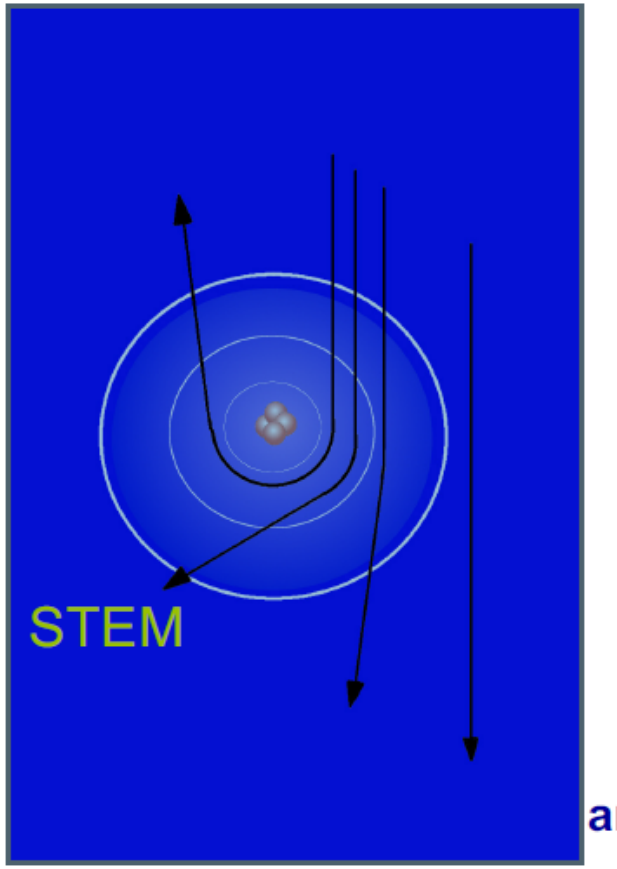




Phase contrast & applications

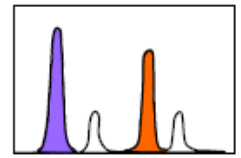
- Phase contrast deals also with crystalline specimens
- Phase contrast is formed **with at least two diffracted electron beams through their interference**.
- Phase contrast is realized with (or without) an aperture at the back focal
- plane of the objective lens
- Phase contrast is mainly used for study of,
    i. Defects
    ii. Precipitates
    iii. Interfacial structures
    iv. Crystal growth
- Phase contrast requires very thin specimens (< 20 nm typically) for the interference.
- Crystals should be orientated to the directions where the zone axes should be parallel to the incidental electron beam.
- Sometimes, image simulations are needed for gain correct interpretation as atoms can be black or white in the image.

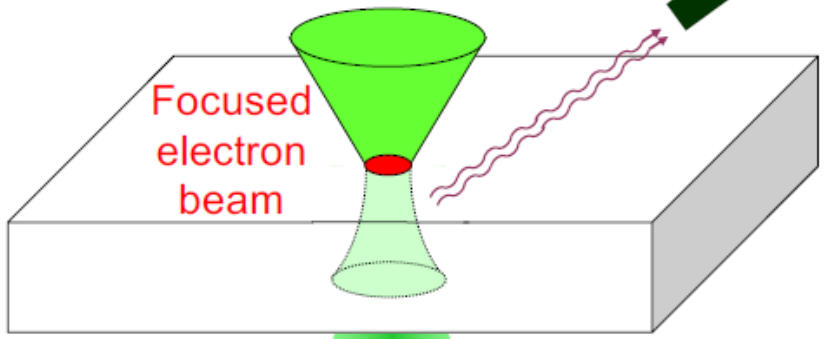
Scanning Transmission Electron Microscopy (STEM)



Small (< 1 nm) electron probe across sample

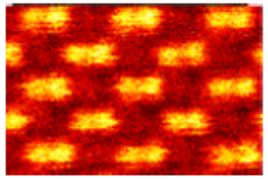
X-ray detector

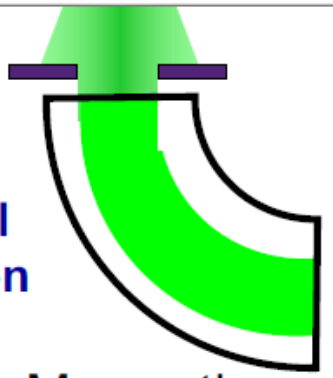


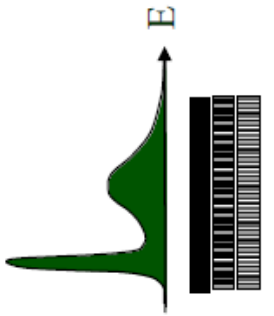


EDS:
Quantitative compositional distributions

STEM Imaging:
Atomic resolution structural and compositional information

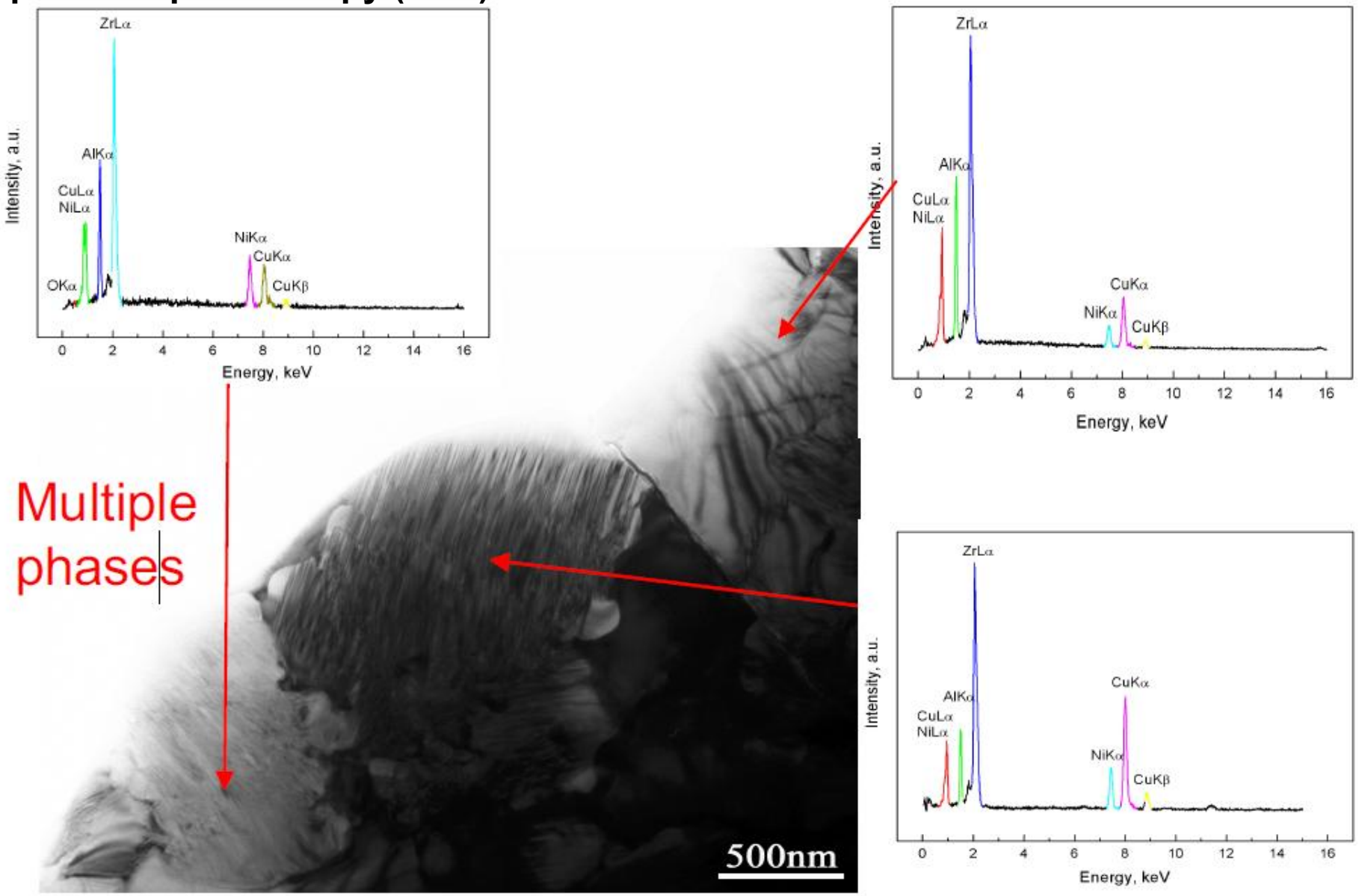




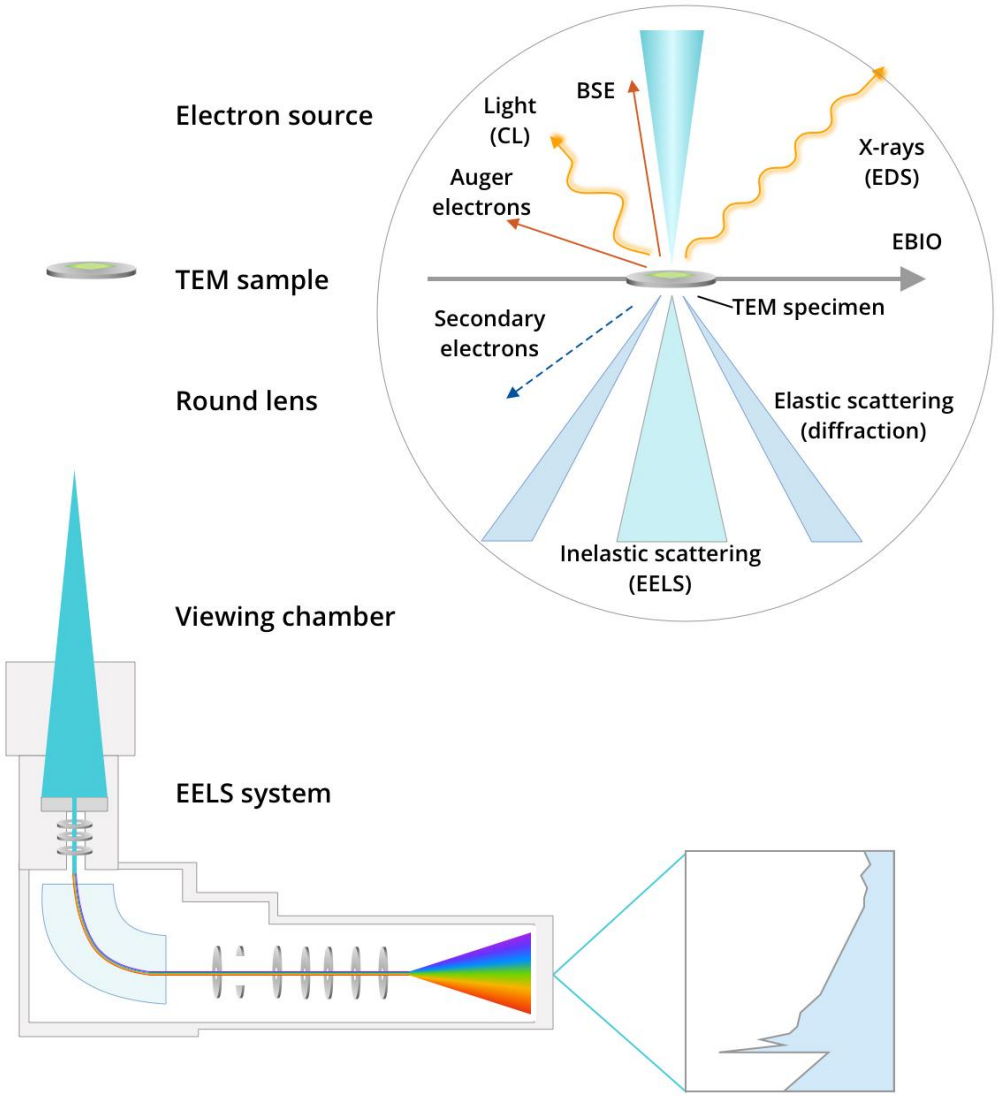


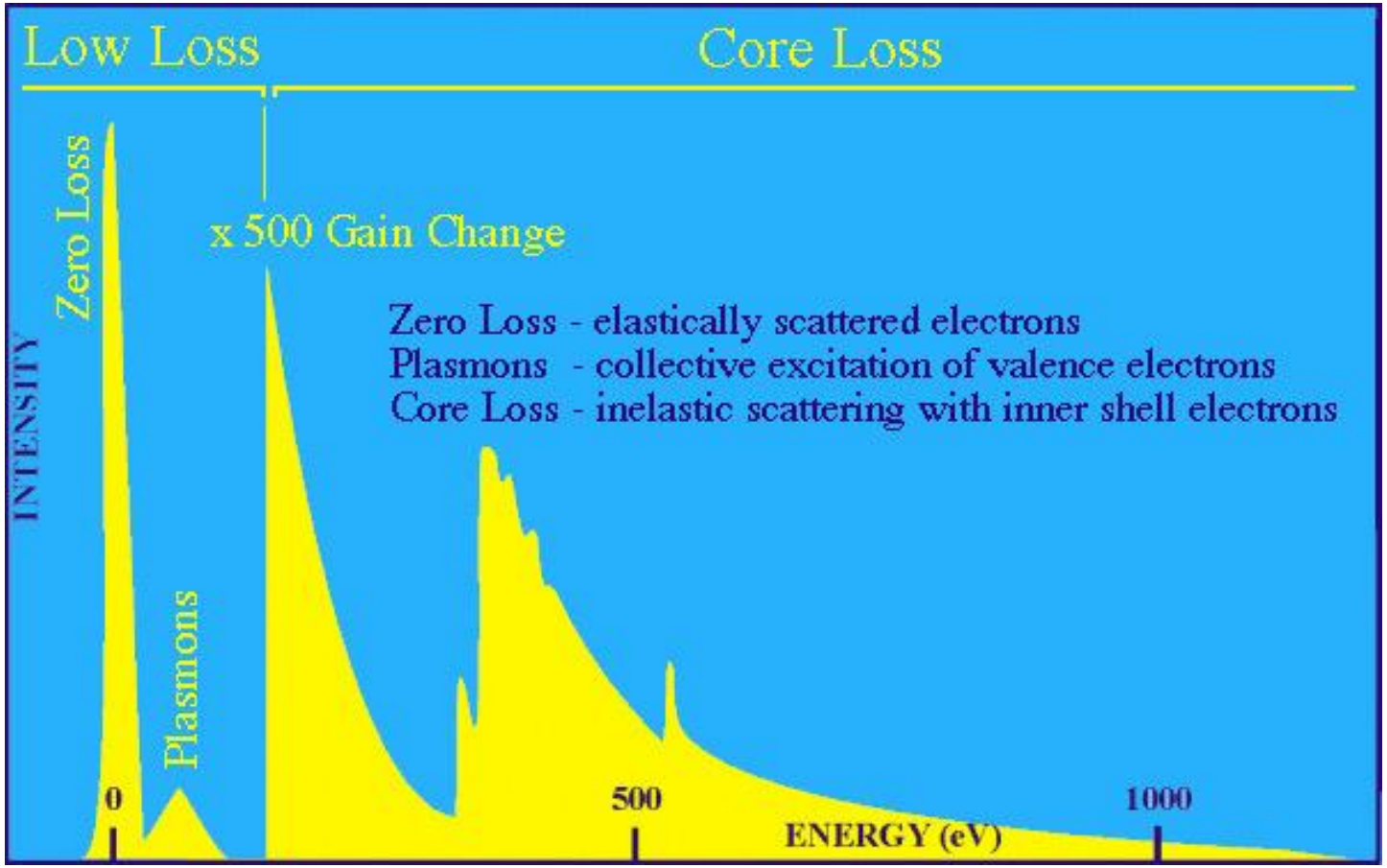
EELS:
Compositional information + electronic structure

Energy Dispersive Spectroscopy (EDS)

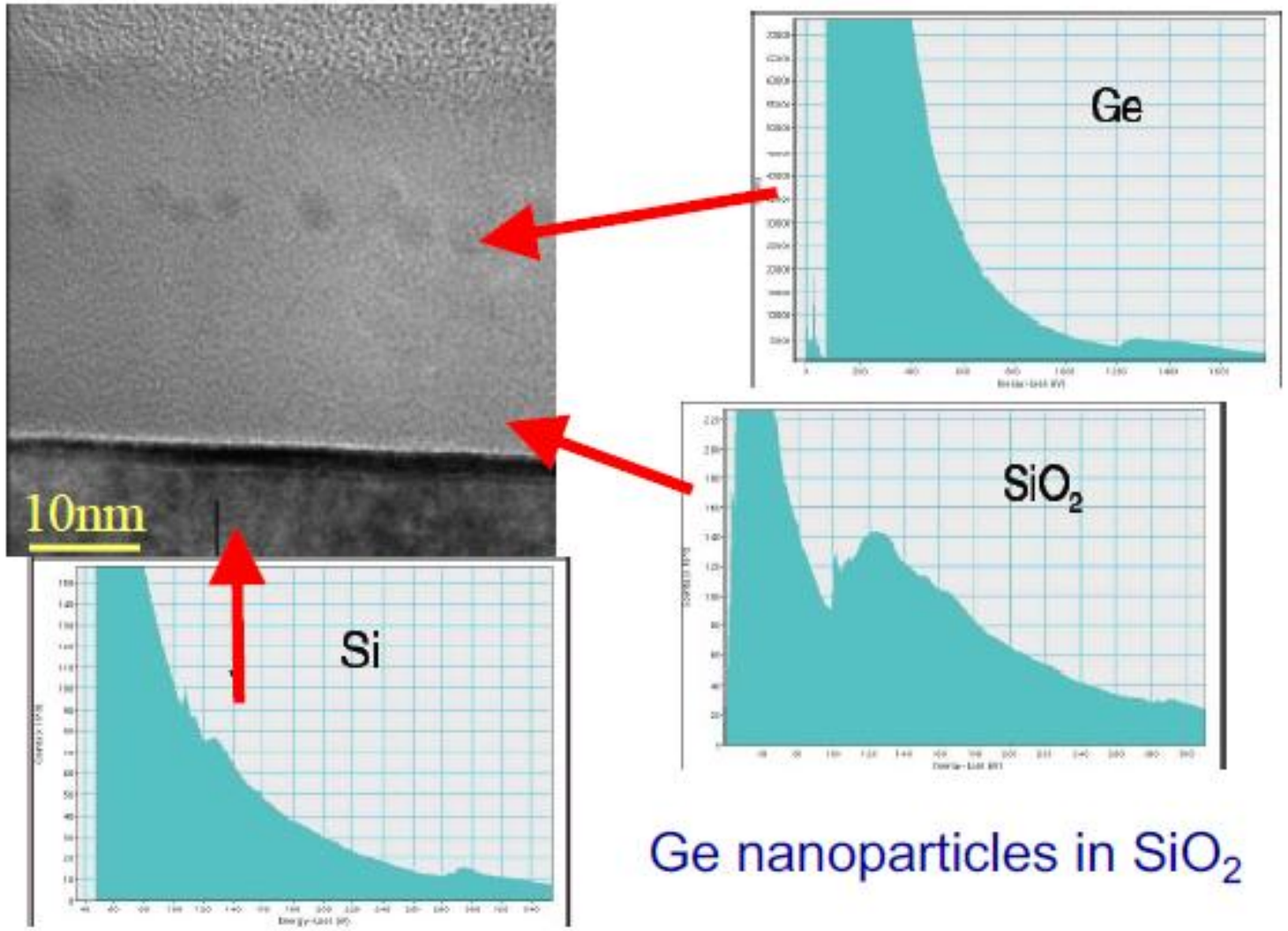


Electron Energy Loss Spectroscopy (EELS)

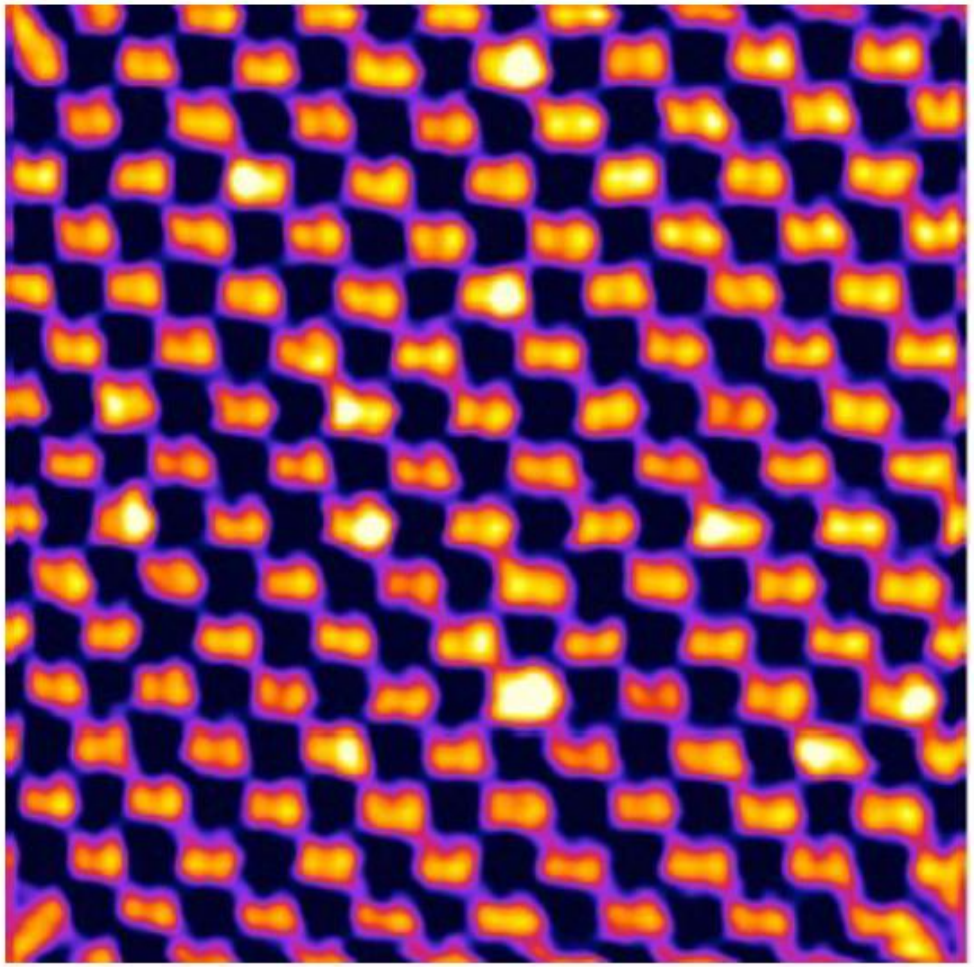




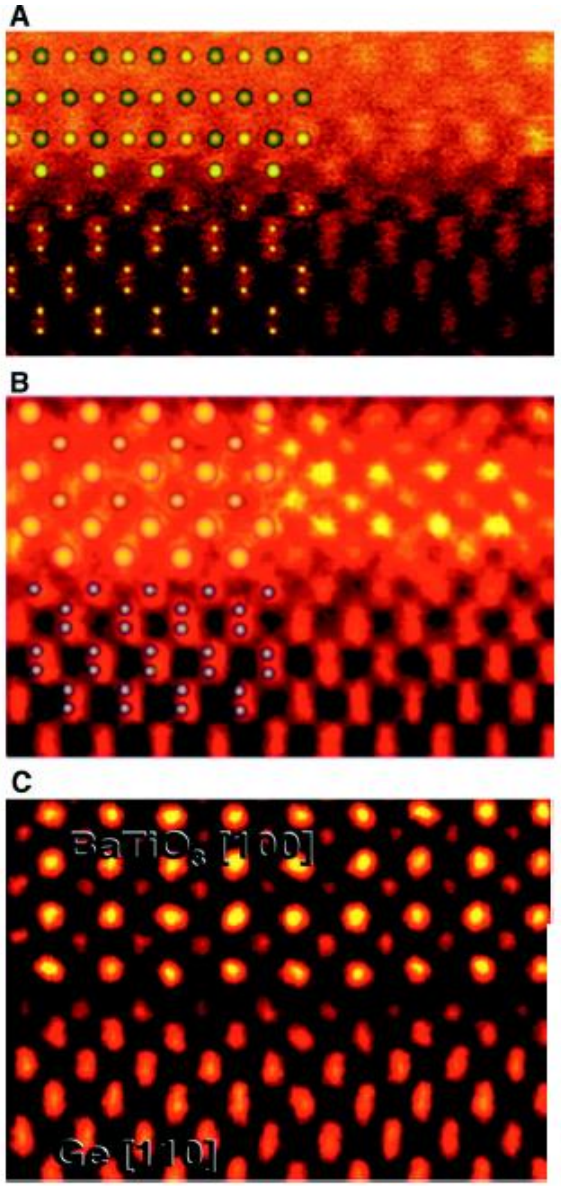
Phase identification using EELS

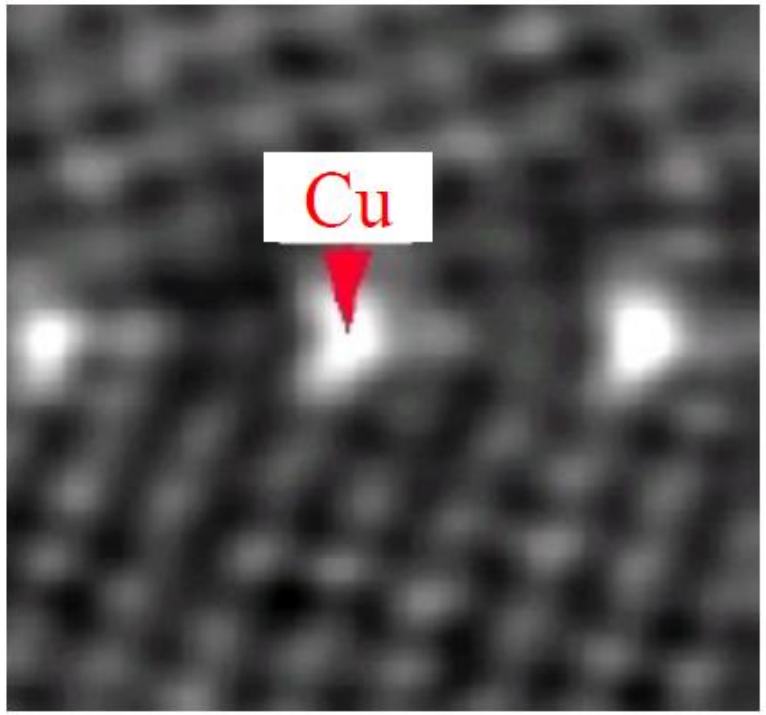


Images from STEM



Bi doped Si





Cu segregated in
Al grain boundary

Perovskite Oxides
grown epitaxially
on Si and Ge

Major applications of TEM

- Microstructural analysis
- Defects identification
- Interfacial analysis
- Crystal structure
- Seeing atomic arrangement of crystals
- Elemental analysis at nanometre scale

Advantages

- Superior resolution.
- Ability to obtain compositional and crystallographic information from specimens with high spatial resolution.

Limitations

- Very small volume of material studied – is it representative?
- Limitations on samples.
- Sample preparation can be quite lengthy and involved.
- Electrons can damage sample and introduce artifacts.

Advantages

- Offers the highest and most powerful magnification of any microscopy technique.
- Versatile imaging modes: dark/bright field and phase contrast (TEM); high-angle annular dark field (STEM).
- Provides the ability to collect electron diffraction patterns (crystallographic information) from nanometer-sized regions by using selected-area electron diffraction (SAED).
- Enables nano-analysis: the ability to collect local information about composition and bonding which can be correlated to high-resolution images.

Disadvantages

- Limited sampling: a typical field of view for an HR-TEM image is no more than 100 nm^2.
- Complex image interpretation: all TEM images (and diffraction patterns) are 2D projections of 3D structures.
- Possibility of electron beam damage, especially for light elements, biological samples, and soft materials. In many cases, this makes it challenging to collect meaningful data before the sample is amorphized.
  - Specialized techniques that utilize advanced equipment like cryo-TEM, low-dose TEM, low-kV TEM can be leveraged to analyze beam-sensitive materials.
- Vacuum environment required – limiting the ability to observe functional materials under realistic “working” conditions.
- Difficult sample preparation to create extremely thin specimens; thin samples often result in imaging artifacts. [6]

The difference between an SEM & TEM:

	Scanning Electron Microscopes (SEM)	Transmission Electron Microscopes (TEM)
Electron stream	Fine, focused beam	Broad beam
Image taken	Topographical/surface	Internal & lattice structures
Resolution	Lower resolution	Higher resolution
Magnification	Up to 2,000,000 times	Up to 50,000,000 times
Image dimension	3D	2D
Sample thickness	Thin and thick samples	Ultrathin samples only
Penetrates sample	No	Yes
Sample restriction	Less restrictive	More restrictive
Sample preparation	Shorter preparation time	Longer preparation time
Cost	Less expensive	More expensive
Speed	Faster	Slower
Operation	Easy to use	More complicated

[1]

Reference

1. Transmission Electron Microscope (TEM)- Definition, Principle, Images (microbenotes.com)
2. SEM vs TEM | Technology Networks
3. Transmission Electron Microscopy: Principle and Components | Soil Mineralogy (soilmanagementindia.com)
4. TEM sample support mesh "grid", with ultramicrotomy sections.@copyright... | Download Scientific Diagram (researchgate.net)
5. Dual-Axis Tomography Holder - Model 2040 - EDEN Instruments | In Situ NanoCharacterization Solutions (eden-instruments.com)
6. Transmission Electron Microscopy - Nanoscience Instruments
7. https://myscope.training/TEM_Electron_energy_loss_spectroscopy__EELS
8. carbon-formvar.jpg (143×115) (tedpella.com)

Other References:

- What if you just keep zooming in? (YouTube)

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AUSTRALIA
CREATE CHANGE

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Thank you

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CRICOS code 00025B